



Boards & Beyond: Renal Slides

Color slides for USMLE Step 1 preparation
from the Boards and Beyond Website

Jason Ryan, MD, MPH

2020 Edition

Boards & Beyond provides a virtual medical school curriculum used by students around the globe to supplement their education and prepare for board exams such as USMLE Step 1.

This book of slides is intended as a companion to the videos for easy reference and note-taking. Videos are subject to change without notice. PDF versions of all color books are available via the website as part of membership.

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Renal Embryology

Renal Embryology

Jason Ryan, MD, MPH



Kidney Development

- Kidneys derive from **mesoderm**
- Three embryonic renal structures form in utero
- **First two degenerate**
- Third becomes adult kidney



Kidney Development

- Pronephros
 - Forms/degenerates week 4
- Mesonephros
 - Interim kidney 1st trimester
 - Contributes to **vas deferens in males**
- **Metanephros**
 - Forms permanent kidney
 - Appears 5th week
 - Develops into kidney through weeks 32-36

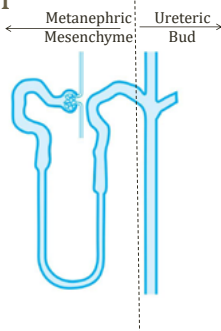


Kidney Formation

- Key Structure #1: **Ureteric bud**
 - Outgrowth of **mesonephric (Wolffian) duct**
 - Gives rise to ureter, pelvis, calyces, collecting ducts
- Key Structure #2: **Metanephric mesenchyme**
 - Mesoderm tissue
 - Also called **metanephric blastema**
 - Interacts with ureteric bud
 - Interaction forms glomerulus to distal tubule
- Aberrant interaction → kidney malformation

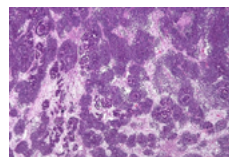


Nephron



Wilms' Tumor

- Most common renal malignancy of young children
- Proliferation of **metanephric blastema**
 - Embryonic glomerular structures
 - Associated with mutations of WT1
 - WT1 expressed in metanephric blastema/mesenchyme



Nephron/Wikipedia



Renal Agenesis

- **Ureteric bud fails to develop**
- **Lack of signals to mesenchyme**
- If single kidney → remaining kidney compensates
 - Hypertrophy
 - **Hyperfiltration**
 - Risk of **focal segmental glomerular sclerosis (FSGS)**
 - Risk of renal failure after decades
- If both kidneys: oligohydramnios, Potter's syndrome

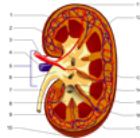
Multicystic Dysplastic Kidney

- **Abnormal ureteric bud-mesenchyme interaction**
- Form of renal dysplasia
- Kidney replaced with cysts
- No/little functioning renal tissue



Ureteropelvic Junction

- Last connection to form
- **Ureteropelvic junction (UPJ) obstruction**
 - Common cause **single kidney** obstruction
 - Narrowing at proximal ureter at junction
 - Hydronephrosis
 - Often detected in utero
 - Poor urinary flow → kidney stones/UTIs
 - Can be treated with surgery after birth



Duplex Collecting System

Duplicated Ureter

- **Two ureteric buds** right or left
- Or bifurcation ureteric bud
- Upper/lower kidneys form
- May lead to poor urine flow
- Hydronephrosis
- Urinary tract infections
- Associated with **vesicoureteral reflux**



Morning2k/Wikipedia

Vesicoureteral Reflux

- **Backward urine flow** from bladder to kidneys
- Leads to recurrent urinary tract infections
- Primary
 - Abnormal closure of ureterovesical junction (UVJ)
 - Occurs in children
 - Associated with **duplex ureters**
- Secondary
 - High bladder pressure → pushes urine backward
 - Seen with posterior urethral valves

Potter's Syndrome

Potter's Sequence

- Fetus exposed to **absent or ↓ amniotic fluid**
- Amniotic fluid = fetal urine
- Severe renal malfunction = ↓ amniotic fluid
- **Loss of fetal cushioning** to external forces
- External compression of the fetus
 - Abnormal face/limb formation
- Alteration in lung liquid movement
 - **Abnormal lung formation**

Potter's Syndrome

Signs

- **Limb deformities**
- **Flat face**
- Pulmonary hypoplasia
- Often fetal death

Oligohydramnios

- First trimester (1-12 weeks): rare
- **Second trimester (13-27 weeks)**
 - Decreased formation of fetal **urine**
- Third trimester (28 weeks to birth)
 - Rupture of membranes

Potter's Syndrome

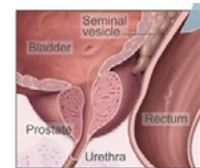
Causes

- **Bilateral renal agenesis**
 - Often detected in utero
 - Fetal kidneys seen on ultrasound at 10 to 12 weeks

Potter's Syndrome

Causes

- **Posterior urethral valves**
 - Occurs in males
 - Tissue (valves) obstruct bladder outflow
 - Ultrasound: dilated bladder, kidneys
 - **Both kidneys effected**



Wikipedia/Public Domain

Potter's Syndrome

Causes

- **Autosomal recessive polycystic kidney disease**
 - Juvenile form of cystic kidney disease
 - Cysts in kidneys and biliary tree
 - Both kidneys affected
 - If severe, may cause oligohydramnios
 - Less severe → renal failure and hypertension in childhood



Horseshoe Kidney

- **Inferior poles fuse**
- Kidney cannot ascend pelvis → retroperitoneum
- Trapped by **inferior mesenteric artery**
- Most patients asymptomatic
- Associated with Turner and Down syndrome
- Associated with vesicoureteral reflux



Image courtesy of Prosyannikov

Horseshoe Kidney



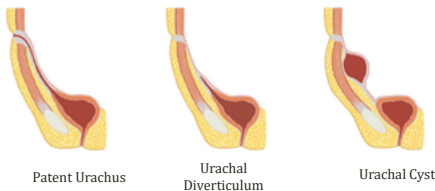
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Image courtesy of Hellerhoff

Urachus

- Connects **dome of bladder to umbilicus**
- Obliterated at birth → **median umbilical ligament**
- Failed/incomplete obliteration can occur
 - Urine can leak from umbilicus
 - Also can form cyst, sinus, diverticulum
 - Can lead to infections

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Urachal Remnants

- Remnant can lead to **adenocarcinoma of bladder**
 - Key feature: Cancer at dome of bladder
- Classic case
 - Adult with painless hematuria
 - Tumor at dome of bladder
 - Path showing adenocarcinoma

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Renal Anatomy

Renal Anatomy

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Kidney Anatomy

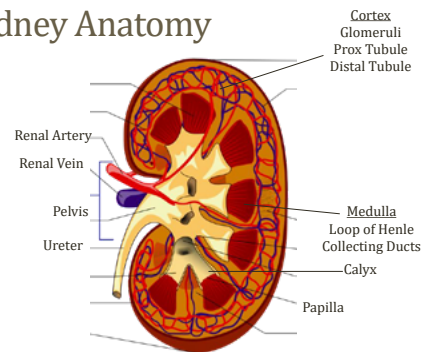
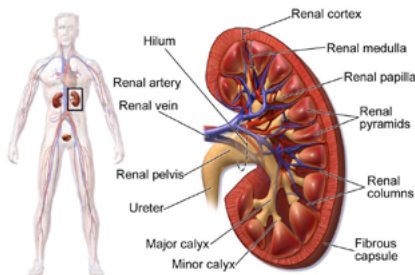


Image courtesy of Piotr Michal Jaworski

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Kidney Anatomy



Kidney Anatomy

Image courtesy of BruceBlais

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Arterial System

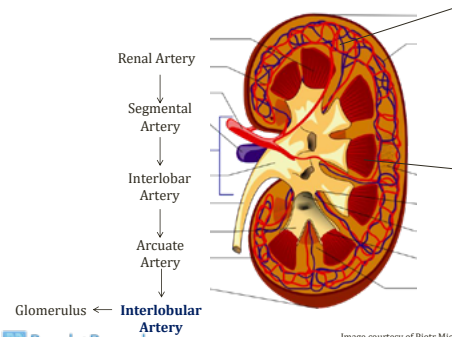
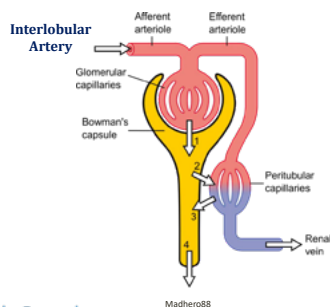


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Glomerular Blood Flow

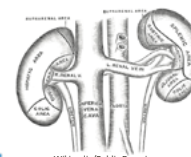


Madher088

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Special Kidney Features

- Right kidney **slightly smaller**
 - Less development in utero due to liver
- Left kidney has **longer renal vein**
 - Often taken for transplant
 - Dead/dying kidney usually not removed in transplant
 - New kidney attached to iliac artery/vein

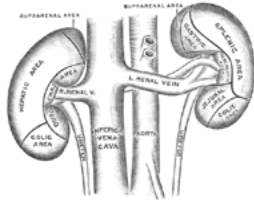


Wikipedia/Public Domain

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Aortic Dissection

- Renal arteries come off abdominal aorta
- Aortic dissection can cause renal ischemia



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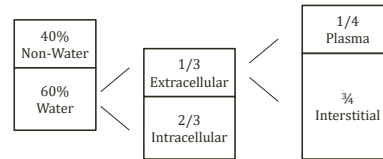
Renal Physiology I

Renal Physiology I

Jason Ryan, MD, MPH

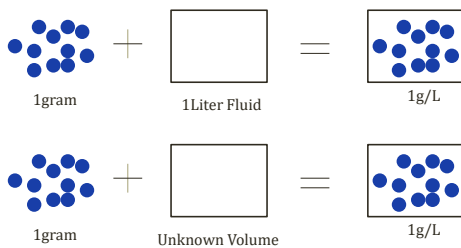
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Fluid Compartments



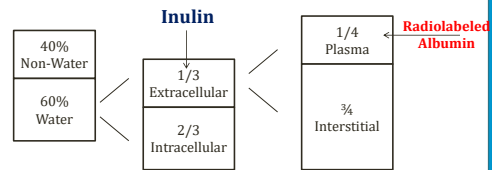
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Determining Fluid Volume



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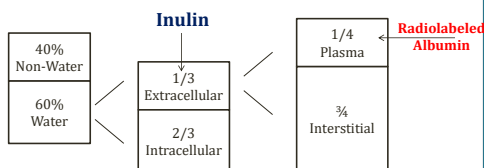
Fluid Compartments



X grams Inulin infused
Equilibrium concentration = Y g/L
ECF = X/Y (Liters)

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Fluid Compartments



10 grams Inulin infused
Equilibrium concentration = 0.25 g/L
ECF = 10/0.25 = 40L

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Sample Question

- A patient is administered 120mg of inulin. An hour later, the patient has excreted 20mg of inulin in the urine. The plasma inulin concentration is 1mg/100ml. What is the extracellular fluid volume for the patient?

Amount of inulin in body = 120mg - 20mg = 100mg
Concentration = 1mg/100ml
ECF = $\frac{100\text{mg}}{0.01\text{mg/ml}}$ = 10000ml = 10L

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Fluid Compartment Shifts

- Plasma osmolarity about 300mosm/kg
- Equilibrium between cells and extracellular fluid
- **Fluid shifts only if difference in osmolarity**

Fluid Compartment Shifts

- Addition/loss of **isotonic fluid**
 - Change in ECF volume
 - No change in ICF volume
- Example: **Hemorrhage**
 - Loss of ECF, no change ICF
- Example: **Infusion of normal saline**
 - Increase ECF, no change ICF

Fluid Compartment Shifts

- Example: **Infusion of 5% dextrose**
 - Hypotonic fluid
 - Increase in ECF volume
 - Increase in ICF volume
- Example: **Mannitol infusion**
 - Raises plasma osmolarity
 - Remains in the vascular system
 - Decrease in ICF volume
 - Increase in ECF volume
 - Reduces volume in interstitial space

Effective Circulating Volume

- Extracellular fluid contained in arterial system
- Maintains tissue perfusion
- Not necessarily correlated with total body water
- Modified by:
 - **Volume**
 - **Cardiac output**
 - **Vascular resistance**

$$BP = CO \times TPR$$

Effective Circulating Volume

- Low ECV leads to low blood pressure
- Low ECV activates:
 - **Sympathetic nervous system**
 - **Renin-angiotensin-aldosterone system**

Condition	ECV	TBW
Volume Depletion	↓	↓
Heart Failure (low CO)	↓	↑
Cirrhosis (low SVR)	↓	↑

Evaluating Kidney Function

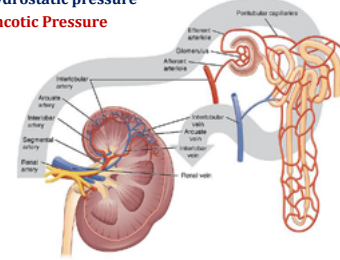
- **Glomerular filtration rate**
 - How much liquid passes through the filter (i.e. glomerulus)?
 - Determined from plasma, urine measurements
 - GFR falls as kidneys fail
- Renal Blood/Plasma Flow
 - How much blood enters kidney
- Filtration Fraction
 - GFR/RPF

Measuring GFR

- Theoretical determination
 - Need to know pressures in capillary, Bowman's capsule
- Clinical determination
 - Need to know plasma concentrations solutes, urine flow

Theoretical Determination GFR

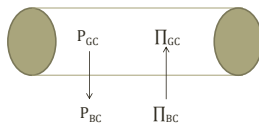
- Filtration Driving Forces
 - **Hydrostatic pressure**
 - **Oncotic Pressure**



OpenStax College/Wikipedia

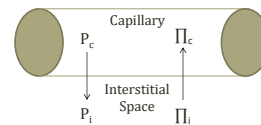
Capillary Fluid Exchange

- **Hydrostatic pressure** – fluid PUSHING against walls
 - High pressure drives fluid TOWARD low pressure
- **Oncotic pressure** – concentrated solution PULLING fluid in
 - High pressure draws fluid AWAY from low pressure

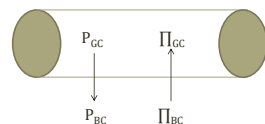


Capillary Fluid Exchange

- Two forces drive fluid into or out of capillaries
- Hydrostatic pressure (P)
 - Molecules against capillaries walls
 - Pushes fluid out
- Oncotic pressure (Π)
 - Solutes (albumin) drawing fluid into capillaries



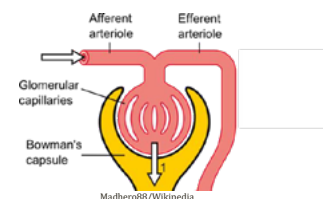
Glomerular Filtration Rate



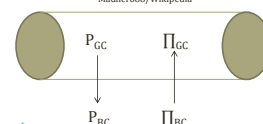
To change GFR:
Change P_{GC} or P_{BC}
Change Π_{GC} or Π_{BC}

Arterioles

Efferent & Afferent



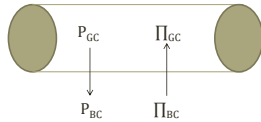
Madhava88/Wikipedia



Raise P_{GC}

Increase GFR

- **Dilate afferent arteriole**
 - More blood IN
 - Increase RPF
 - Increase P_{GC}
 - Increase GFR
 - No change FF

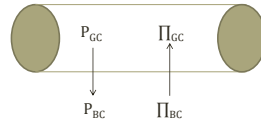


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Raise P_{GC}

Increase GFR

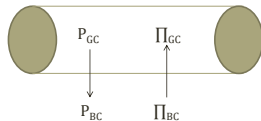
- **Constrict efferent arteriole**
 - Blood backs up behind constricted arteriole
 - Less blood out
 - Decreased RPF
 - Increase P_{GC}
 - Increase GFR
 - Increase FF



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Raise Π_{GC}

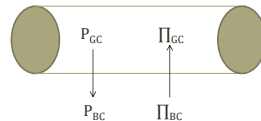
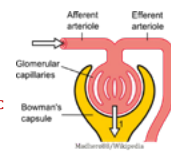
- **Increase protein levels in blood**
 - Less blood drawn into proximal tubule
 - Lower GFR
 - No change RPF
 - Decrease FF



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Change P_{BC}

- **Obstruct ureter $\rightarrow$ Increase P_{BC}**
 - Urine backs up behind obstruction
- Less GFR P_{BC}
- No effect RPF
- Decrease FF



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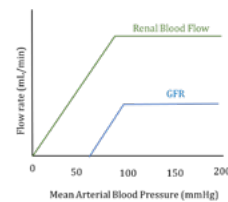
Glomerular Flow Dynamics

	RPF	GFR	FF
Afferent Dilation	↑	↑	--
Efferent Constriction	↓	↑	↑
↑ plasma proteins	--	↓	↓
Ureter obstruction	--	↓	↓

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Autoregulation

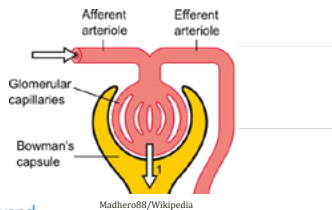
- Constant GFR/RBF over range of blood pressures
- #1: Myogenic mechanism
- #2: Tubuloglomerular feedback



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Myogenic Mechanism

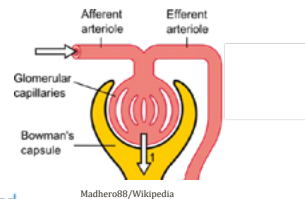
- **Afferent arteriole constricts with high pressure**
 - Responds to changes in stretch
- Result is maintenance of normal GFR/RPF



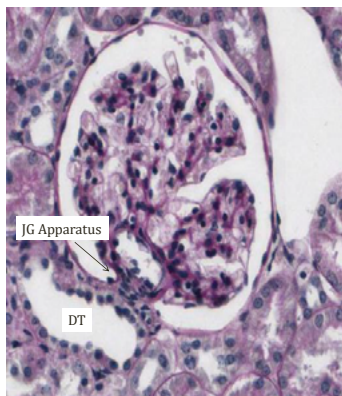
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Tubuloglomerular Feedback

- $\uparrow$ urinary flow in tubule $\rightarrow \uparrow$ NaCl to distal tubule
- NaCl sensed by **macula densa** (part of JG apparatus)
- Macula Densa $\rightarrow$ **vasoconstriction afferent arteriole**



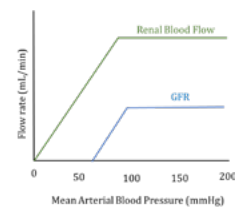
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Severe Volume Loss

- Profound loss of fluid (vomiting, diarrhea, etc.)
- Renal plasma flow falls significantly
- **Auto-regulatory mechanisms overwhelmed**
- $\downarrow$ GFR
- $\uparrow$ BUN/Cr
- Pre-renal failure



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Renal Physiology II

Renal Physiology II

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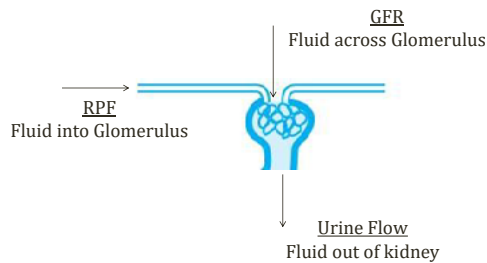
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Renal Function Measurements

- Glomerular filtration rate
 - How much liquid passes through the filter (i.e. glomerulus)?
- Renal plasma flow
 - How much liquid does the kidney handle?
- Filtration fraction
 - Of all substance X entering kidney, what % gets filtered?
- Renal clearance
 - How much of each blood component gets removed?

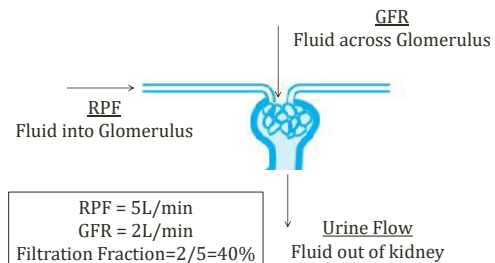
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Renal Function Measurements



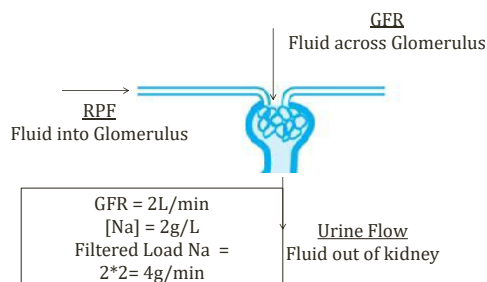
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Renal Function Measurements



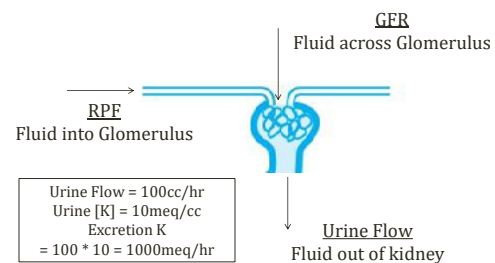
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Renal Function Measurements



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Renal Function Measurements



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Measured Variables

1. Plasma concentration ($P_x = \text{mg/l}$)
 - i.e. Na, Glucose
2. Urine concentration ($U_x = \text{mg/l}$)
3. Urine flow rate ($V = \text{l/min}$)

Use these measured variables to get RPE, GFR, etc.

Renal Clearance

- Number determined for blood substance (Na, Glucose)
- Volume of blood "cleared" of substance X
- Volume of blood that contained amount of X excreted
- Reported liters/min (volume flow)

$$C_x = \frac{U_x * V}{P_x}$$

Determining GFR

- **Inulin clearance** used to determine GFR
- Inulin neither secreted or resorbed
- **All inulin filtered goes out**
- Amount blood "cleared" of inulin is amount of blood filtered by glomerulus
- **Clearance of inulin (liters/min) = GFR**

$$C_{\text{inulin}} = \frac{U_{\text{inulin}} * V}{P_{\text{inulin}}} = \text{GFR}$$

Creatinine

- Breakdown product muscle metabolism
- **Closest naturally occurring substance to inulin**
 - Inulin = All filtered goes out, no secretion/resorption
 - Creatinine = All filtered goes out, small amount secretion
- Using Cr instead of inulin:
 - Secreted Cr will be counted as filtered
 - This will slightly overestimate GFR

Creatinine

$$C_x = \frac{U_x * V}{P_x}$$

- **Special formulas** to convert Cr to GFR
 - Cockcroft-Gault formula
 - Modification of Diet in Renal Disease (MDRD) formula
 - Use age, gender, Cr level to estimate GFR

Cockcroft-Gault

$$\text{CrCl} = (140 - \text{age}) * (\text{Wt in kg}) * (0.85 \text{ if female}) / (72 * \text{Cr})$$

P_{Cr} /GFR Relationship

Amount of Cr out in urine
Equal to amount produced

$$C_{\text{Cr}} = \frac{U_{\text{Cr}} * V}{P_{\text{Cr}}} = \text{GFR}$$

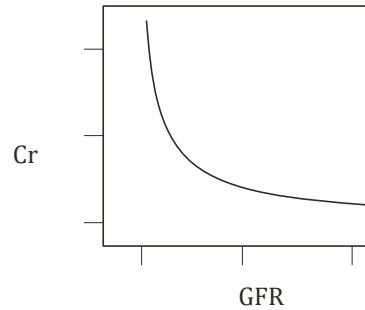
Creatinine Clearance

$$C_{Cr} = \frac{\text{Constant} \approx \text{GFR}}{P_{Cr}}$$

Double [Cr] (1.0 to 2.0) → Half the GFR

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Creatinine



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Creatinine

- Worsening renal function = high blood Cr level
- Some sample values:
 - Normal kidney function → Cr = 0.8 mg/dl
 - Chronic kidney disease → Cr = 2.0 mg/dl
 - End stage renal disease (dialysis) → Cr = 4.0mg/dl

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Creatinine

- **GFR declines with age**
 - Not always accompanied by rise in Cr
 - Use of formulas is key
 - Must adjust some medication dosages



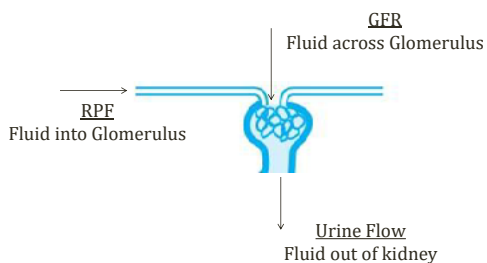
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Cockcroft-Gault

$$CrCl = (140 - \text{age}) * (\text{Wt in kg}) * (0.85 \text{ if female}) / (72 * Cr)$$

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Renal Function Measurements

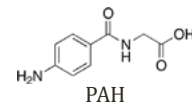


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Renal Plasma Flow (RPF)

- Use **para-aminohippuric acid (PAH)** to estimate RPF
- PAH is filtered and secreted
- 100% of PAH that enters kidney leaves blood in urine
- Clearance PAH (l/min) = Plasma to kidney (l/min)

$$C_{PAH} = \frac{U_{PAH} * V}{P_{PAH}} = RPF$$



*PAH clearance underestimates RPF by 10%
Not all renal plasma/blood to glomeruli

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Plasma versus Blood

- Blood = Plasma + cells/proteins
- Renal Blood Flow > Renal Plasma Flow
- Separate calculations RBF vs. RPF

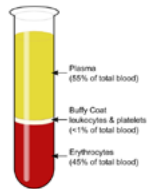


Image courtesy of KnuteKnudsen

Renal Blood Flow (RBF)

- RBF determined from RPF
- Blood = Plasma + Cells/Proteins
- Cells/Proteins (%) $\approx$ Hct (%)

$$RPF = RBF (1 - Hct)$$

$$RBF = 10 \text{ cc/min}$$

$$40\% \text{ if cells (Hct)}$$

$$60\% \text{ RBF is plasma}$$

$$RPF = 10 (1 - 0.4) = 10 (0.6)$$

$$= 6 \text{ cc/min}$$

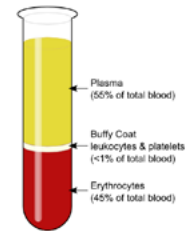


Image courtesy of KnuteKnudsen

Renal Blood Flow (RBF)

- RBF determined from RPF
- Blood = Plasma + Cells/Proteins
- Cells/Proteins (%) $\approx$ Hct (%)

$$RPF = RBF (1 - Hct)$$

$$RBF = \frac{RPF}{1 - Hct}$$

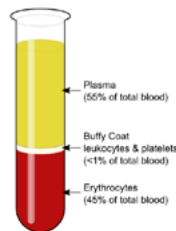


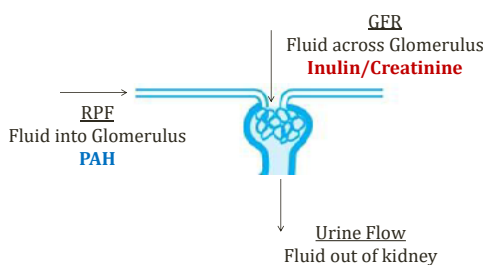
Image courtesy of KnuteKnudsen

Renal Blood Flow (RBF)

- RPF = 1 liter/min
- Hct = 40%

$$RBF = \frac{1}{1 - Hct} = \frac{1}{0.6} = 1.6 \text{ l/min}$$

Renal Function Measurements



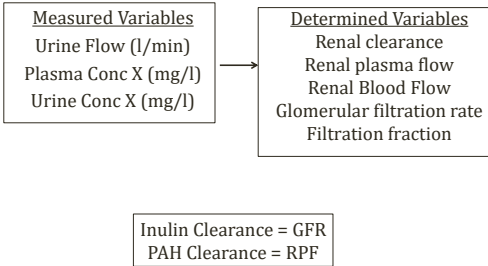
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Other Renal Function Variables

- Filtration Fraction
 - How much of plasma to kidney gets filtered?
 - GFR/RPF
 - Normal = 20%
- Filtered Load X
 - How much of substance X gets filtered?
 - $P_x \times GFR$
 - Amount of X delivered to proximal tubule

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Quantifying Kidney Function



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Prostaglandins and NSAIDs

- Prostaglandins **dilate afferent arteriole** → ↑ RPF
- NSAIDs (ibuprofen) block PG production
- Afferent arteriole constricts
- ↓ RPF ↓ GFR -- FF
- Clinical effects:
 - Acute renal failure
 - Acute heart failure

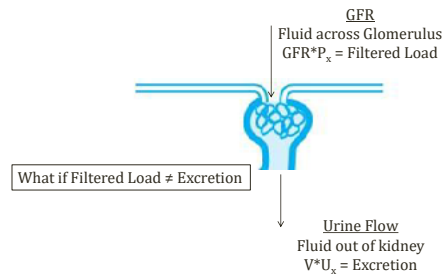
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ACE Inhibitors

- Angiotensin II constricts most blood vessels
- Angiotensin II **constricts efferent arteriole** preferentially
- ACE inhibitors blunt Angiotensin II effects
- ↓ GFR ↑ RPF ↓ FF

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Secretion and Absorption



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Secretion and Absorption

- Excreted = Filtered - Reabsorbed + Secreted
- Amount filtered (X) = $GFR \times P_x$
- Amount excreted (X) = $V \times U_x$

Example:
10mgX/min filtered, 20mgX/min excreted
Additional 10mgX/min must be secreted

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Secretion and Absorption

- Filtered = Excreted if no secretion/resorption
- Filtered < Excreted if some secreted
- Filtered > Excreted if some resorbed

Example #1:
Filtered = 100mg/min
Excreted = 120mg/min
Additional 20mg/min must be secreted

Example #2:
Filtered = 100mg/min
Excreted = 80mg/min
20mg/min must be resorbed

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Secretion and Absorption

- If clearance (x) = GFR → no secretion/resorption
- GFR < C_x → secretion
- GFR > C_x → resorption

Example #1:

GFR = 100ml/min

C_x = 120ml/min

Additional 20ml/min "cleared" by secretion

Example #2:

GFR = 100ml/min

C_x = 80ml/min

Additional 20ml/min "uncleared" by resorption



Intake and Output

- Amount of any substance in must equal amount out
- When insults occur (renal failure, diarrhea), there is a transient imbalance that alters plasma levels
- Steady state returns
- Eat 10grams per day salt → excrete 10grams per day



Solutes in Renal Failure

- **Regulated solutes (Na/K):** No concentration change
- **Unregulated solutes (Cr/Urea):** ↑ plasma level

	Normal GFR=100	GFR=50	Filtered Load	Excretion	Fractional Excretion
K	4.5	4.5	↓	--	↑
Na	145	145	↓	--	↑
Urea	20	40	--	--	--



Question 1

- A patient has a urine output of 4800cc/day (200cc/hr). Plasma concentration of substance X is 4mg/dL. Urine concentration of X is 8mg/dL. What is the clearance of substance X?

$$C_x = \frac{U_x * V}{P_x} = \frac{8 * 200}{4} = 400\text{cc/hr}$$



Question 2

- A patient is infused with inulin. At steady state, plasma concentration of inulin is 3mg/dl and urine concentration is 6mg/dl. If the GFR is 200ml/hr, what is the urine flow rate?

$$GFR = \frac{U_{\text{inulin}} * V}{P_{\text{inulin}}} = C_{\text{inulin}}$$

$$V = \frac{GFR * P_{\text{inulin}}}{U_{\text{inulin}}} = \frac{200 * 3}{6} = 100\text{ml/hr}$$



Question 3

- A patient is infused with PAH. At steady state, plasma concentration of PAH is 5mg/dl. Urine concentration is 10mg/dl. If the urine flow rate is 200ml/hr and the hematocrit is 0.50, what is the renal blood flow?

$$C_{\text{PAH}} = \frac{U_{\text{PAH}} * V}{P_{\text{PAH}}} = \text{RPF} \quad \text{RBF} = \frac{\text{RPF}}{1 - \text{Hct}}$$

$$\text{RPF} = \frac{10 * 200}{5} = 400 \quad \text{RBF} = \frac{400}{1 - 0.5} = 800\text{ml/hr}$$



Question 4

- A lab animal has an inulin clearance of 100cc/hr. Plasma concentration of substance X is 4mg/mL. It is known that substance X is not reabsorbed, but is secreted at a rate of 25mg/hr. What is the excretion rate of substance X?

$$\begin{aligned}\text{Amount filtered (X)} &= \text{GFR} * P_x \\ \text{Excreted} &= \text{Filtered} - \text{Reabsorbed} + \text{Secreted}\end{aligned}$$

Question 4

- A lab animal has an inulin clearance of 100cc/hr. Plasma concentration of substance X is 4mg/mL. It is known that substance X is not reabsorbed, but is secreted at a rate of 25mg/hr. What is the excretion rate of substance X?

$$\begin{aligned}\text{Amount filtered (X)} &= \text{GFR} * P_x = 100 * 4 = 400\text{mg/hr} \\ \text{Excreted} &= \text{Filtered} - \text{Reabsorbed} + \text{Secreted} \\ \text{Excreted} &= 400 - 0 + 25 \\ &= \underline{425\text{mg/hr}}\end{aligned}$$

Key Points

- If given inulin clearance, that is GFR
- GFR used to calculate filtered load of other substances
- Just need plasma concentration (P_x)

Key Points

- Amount filtered = $\text{GFR} * P_x$
- Amount excreted = $V * U_x$
- Excreted = Filtered + Secreted - Reabsorbed
- For Inulin Filtered = Excreted

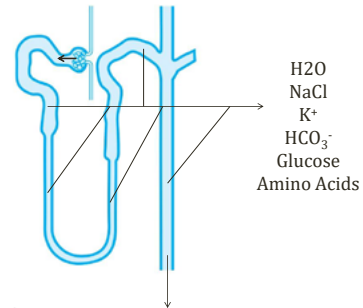
Nephron Physiology

Nephron Physiology

Jason Ryan, MD, MPH

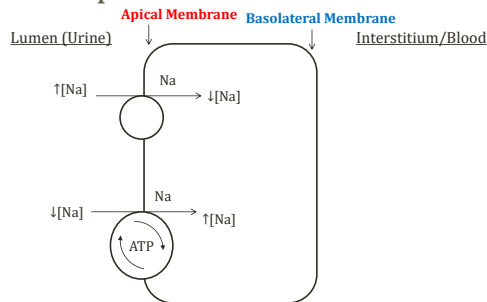
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Nephron



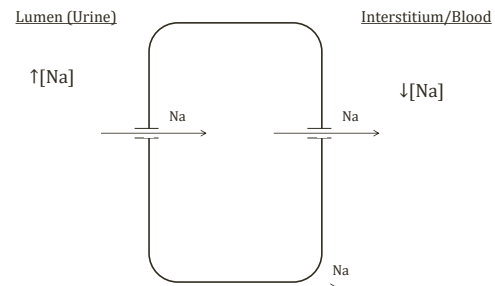
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Transport



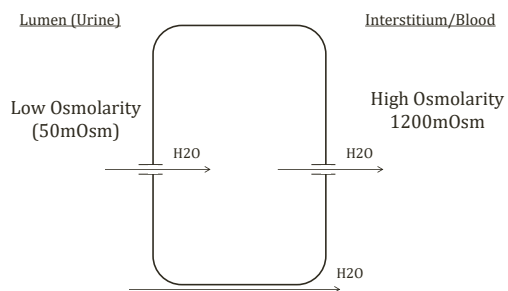
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Diffusion



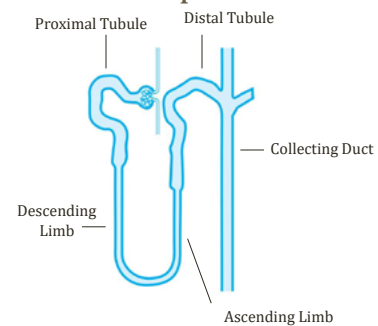
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Osmotic Diffusion



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Segments of Nephron

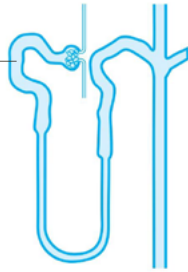


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Proximal Tubule

100%
Glucose
Amino Acids

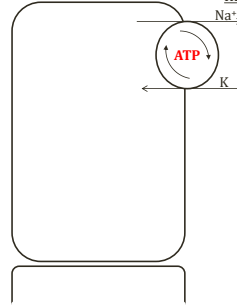
67%
Water
Bicarb
NaCl
Potassium
Phosphate



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Proximal Tubule

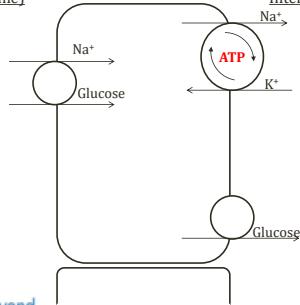
Lumen (Urine) Interstitium/Blood



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Proximal Tubule

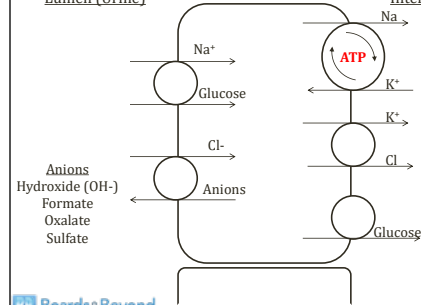
Lumen (Urine) Interstitium/Blood



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Proximal Tubule

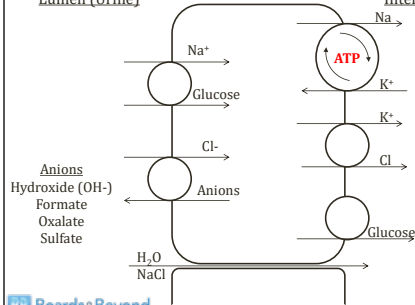
Lumen (Urine) Interstitium/Blood



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Proximal Tubule

Lumen (Urine) Interstitium/Blood



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Glucose

- Completely reabsorbed proximal tubule
- **Na/Glucose co-transport**
- At glucose ~160mg/dl → glucose appears in urine
- Glucose ~350mg/dl → all transporters saturated
- Diabetes mellitus = "sweet" diabetes



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Pregnancy

- Pregnancy: **some glycosuria normal**
- $\uparrow$ GFR
- $\downarrow$ glucose reabsorption
- Serum glucose testing for diabetes

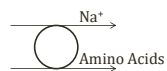


glyvind Holmstad/Wikipedia

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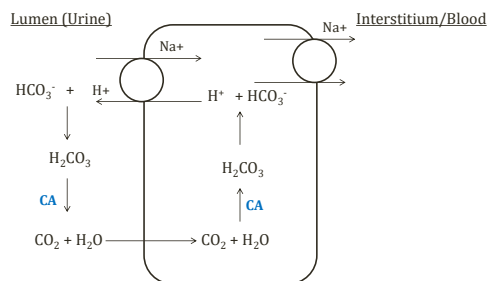
Amino Acids

- All amino acids reabsorbed in proximal tubule
 - Na/AA transporters
- **Hartnup disease**
 - No tryptophan transporter in proximal tubule
 - Tryptophan deficiency
 - Skin rash resembling pellagra (plaques, desquamation)
 - **Amino acids in urine**



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Bicarbonate



CA = Carbonic Anhydrase

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Proximal Tubule Bicarbonate

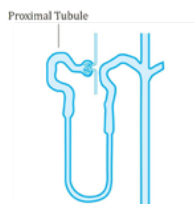
Clinical Correlations

- Carbonic anhydrase inhibitors
 - Weak diuretics
 - Result in bicarb loss in urine
- Type II Renal Tubular Acidosis
 - Ion defect
 - Inability to absorb bicarb
 - Metabolic acidosis

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Fanconi Syndrome

- Loss of **proximal tubule functions**
 - Impaired resorption of solutes
 - HCO_3^- , glucose, amino acids, phosphate
 - Low molecular weight proteins



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Fanconi Syndrome

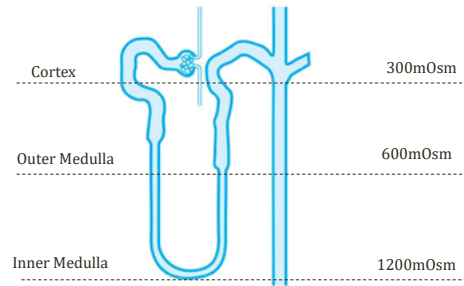
- **Polyuria, polydipsia** (diuresis from glucose)
 - Normal serum glucose (contrast with diabetes)
- Non anion gap acidosis (loss of HCO_3^-)
- Hypokalemia ($\uparrow$ nephron flow)
- Hypophosphatemia (loss of phosphate)
- **Amino acids in urine**

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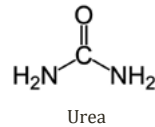
Osmolarity of Nephron



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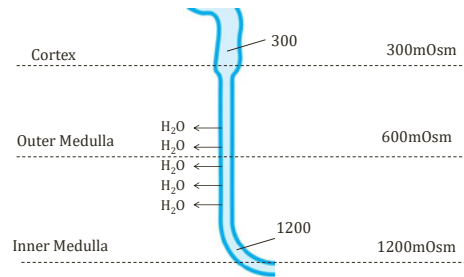
Osmolarity of Nephron

- Created by **Na, Cl, and Urea**
- Urea generated by liver and protein metabolism
- Reabsorbed by **collecting duct**
 - High permeability to urea
- Essential to maintaining gradients



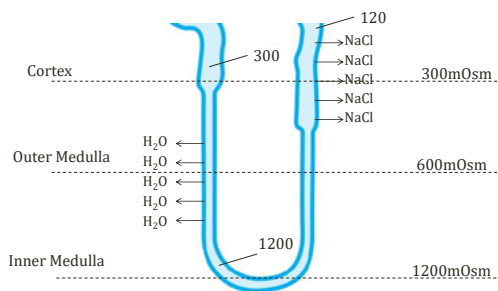
Boards&Beyond

Thin Descending Loop Henle



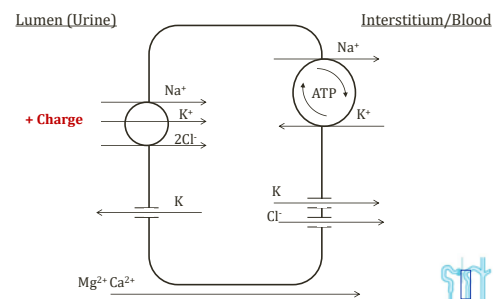
Boards&Beyond

Thick Ascending Loop Henle



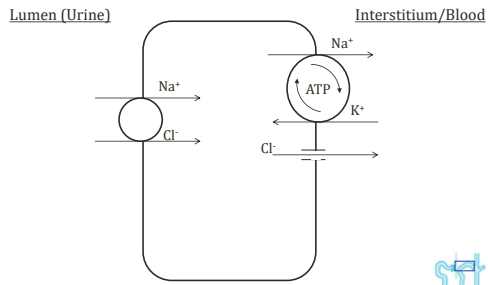
Boards&Beyond

Thick Ascending Loop Henle



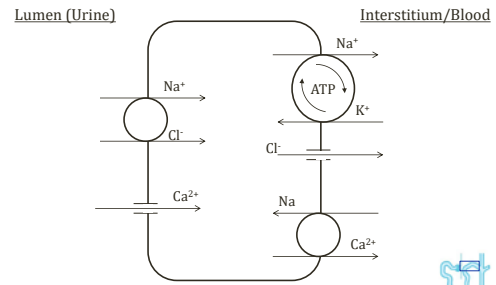
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Distal Tubule



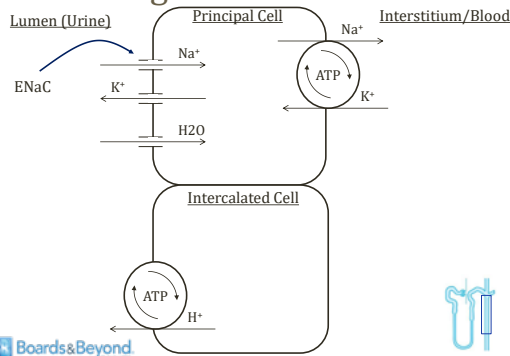
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Distal Tubule



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Collecting Duct



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Key Points

- Collecting duct functions
 - Reabsorb Na^+ / H_2O
 - Secrete K^+ / H^+
- Increased Na^+ delivery to CD $\rightarrow$ increased K^+ excretion
 - Contributes to hypokalemia with loops/thiazides

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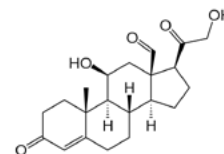
Collecting Duct Hormones

- Aldosterone**
- Antidiuretic hormone (ADH)**
- Regulate collecting duct function

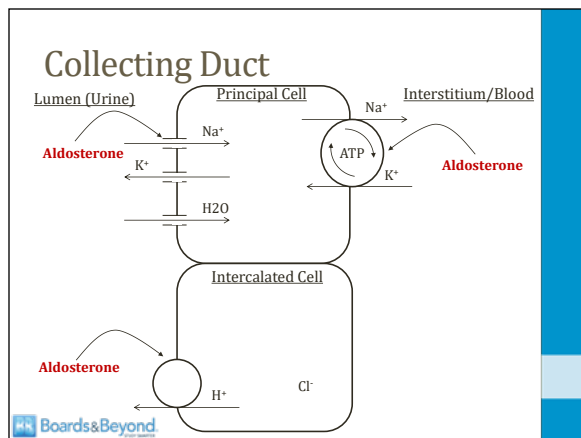
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Aldosterone

- Steroid (mineralocorticoid) hormone
- Increases **Na^+ / K^+ -ATPase proteins**
- Increases **Na^+ channels (ENaC)** of principal cells
- Promotes K^+ secretion** principal cells
- Promotes H^+ secretion** intercalated cells

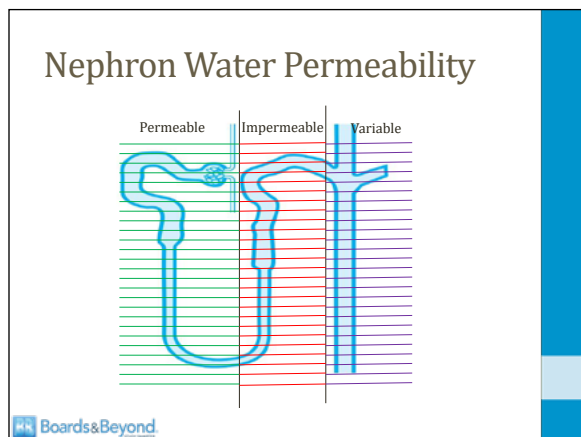


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Aldosterone

- Overall effect:
 - $\uparrow$ sodium/water resorption ($\uparrow$ effective circulating volume)
 - $\uparrow$ K excretion
 - $\uparrow$ H⁺ excretion
- Release stimulated by:
 - Angiotensin II
 - High potassium
 - ACTH (minor effect)



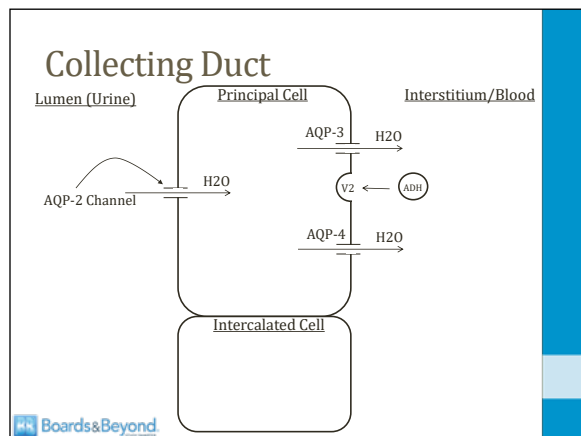
Antidiuretic Hormone (ADH)

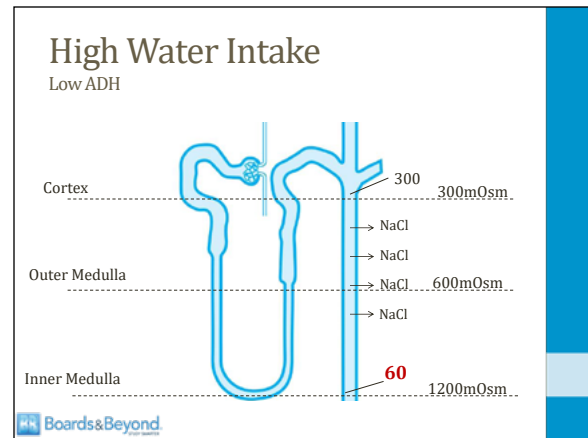
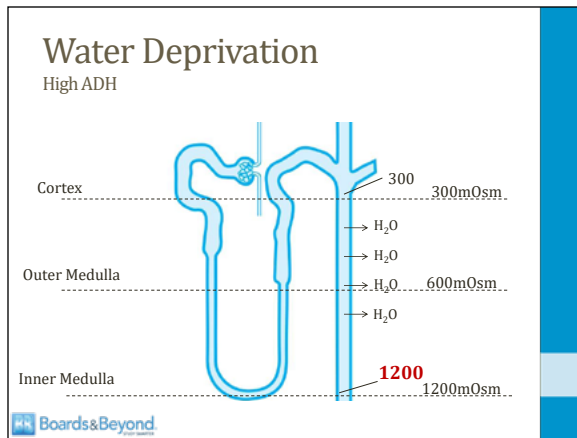
Vasopressin

- Promotes free water retention (inhibits secretion)
- Two receptors: V1, V2
 - V1: Vasoconstriction
 - V2: Antidiuretic response
- Released by **posterior pituitary**
- Two stimuli for release
 - Major physiologic stimuli: **hyperosmolarity**
 - Volume loss: non-osmotic release**

ADH Water Resorption

- V2 receptors: **principal cells collecting duct**
- G-protein, cAMP second messenger system
- Endosome insertion into cell membrane
- Endosomes contain **aquaporin 2**
 - Water channel
- Result is $\uparrow$ permeability of cells to water

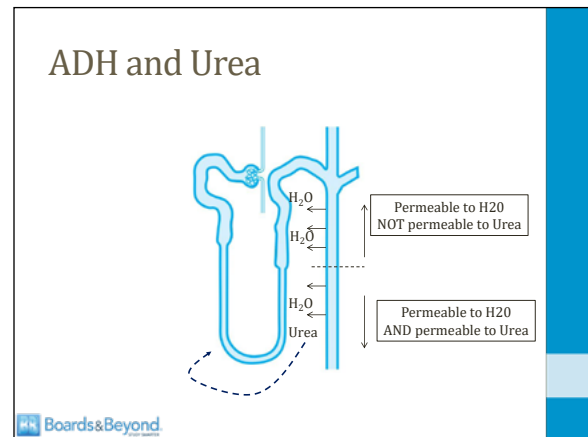




ADH and Urea

- Key osmole in kidney (with Na, Cl)
- **Medullary collecting duct** permeable to urea
- **ADH increases urea reabsorption**
- Urea enters medullary interstitium
- **Thin descending limb transporters “recycle” urea**

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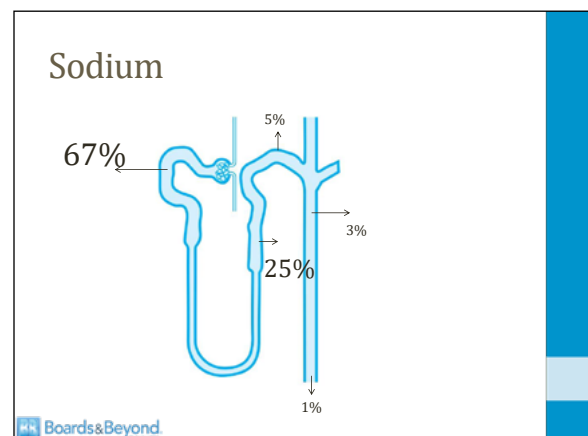


Collecting Duct

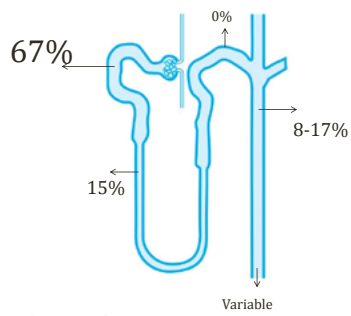
Major Functions

- Resorption Na/H₂O
 - Depends on ADH (H₂O) and Aldosterone (Na)
- Secretion of K⁺ and H⁺
 - Depends on Aldosterone
- Urea resorption

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Water



Renal Endocrinology

Renal Endocrinology

Jason Ryan, MD, MPH

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Renal Hormones

- Released by kidney
 - Erythropoietin
 - Renin (enzyme)
 - 1,25 Vitamin D
- Act on kidney
 - Angiotensin II
 - Atrial Natriuretic Peptide (ANP)
 - Antidiuretic hormone (ADH)
 - Aldosterone
 - Parathyroid hormone (PTH)

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JG Apparatus

Juxtaglomerular Apparatus

- JG Cells
 - Modified smooth muscle of afferent arteriole
- Macula densa
 - Part of distal convoluted tubule
- JG cells secrete renin**

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Glomerulus

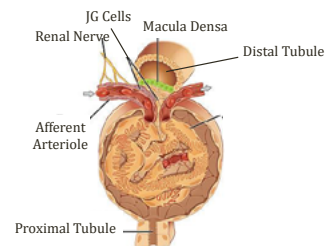
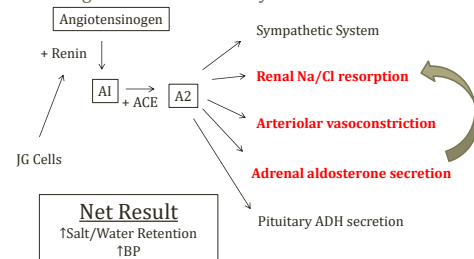


Image courtesy of OpenStax College

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RAAS

Renin-Angiotensin-Aldosterone System



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Stimulation Renin Release

- Low perfusion pressure**
 - Low blood pressure or low circulating volume
 - Sensed by afferent arteriole → JG cell renin release
- Low NaCl delivery**
 - Sensed by macula densa → JG cell renin release
 - Also constricts afferent arteriole: "tubuloglomerular feedback"
- Sympathetic activation**
 - β_1 receptors
 - Also constricts (α) afferent/efferent arterioles
 - Decreases GFR to limit sodium/water excretion

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RAAS

Renin-Angiotensin-Aldosterone System

- **Renin**
 - Converts angiotensinogen to angiotensin I
- **Angiotensin II**
 - Multiple effects
- **Aldosterone**
 - Collecting duct effects
 - Resorption of Na
 - Excretion of K, H⁺

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Angiotensin II

- **Efferent arteriole constriction**
- ↓ RPF
- ↑ GFR
- Less renal blood flow
- More Na/H₂O filtration

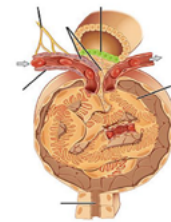


Image courtesy of OpenStax College

Angiotensin II

- **Increased Na/H₂O reabsorption**
 - Several mechanisms
 - Increased proximal tubule resorption via **capillary effect**
 - Direct proximal tubule resorption through **Na/H⁺ exchange**
 - Stimulates aldosterone release

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Capillary Effect

- Altered by efferent arteriole constriction
- ↓ hydrostatic pressure from less blood flow
- ↑ oncotic pressure from more H₂O filtered
- Net result is that efferent arteriole constriction by Angiotensin II leads to increased NaCl resorption

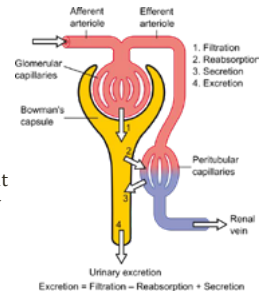
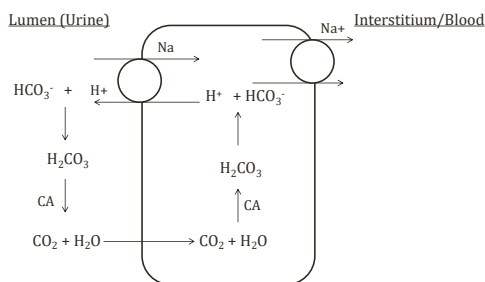


Image courtesy of Madhoo88

Na/H⁺ Exchange

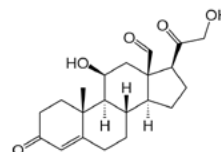
Proximal Tubule



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Aldosterone

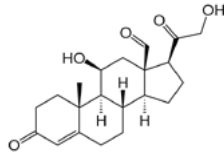
- Synthesized/released by **adrenal cortex**
 - Zona glomerulosa cells
- Freely crosses cell membrane (steroid)
- Binds to cytosolic protein receptor
- Activated receptor **modifies gene expression**



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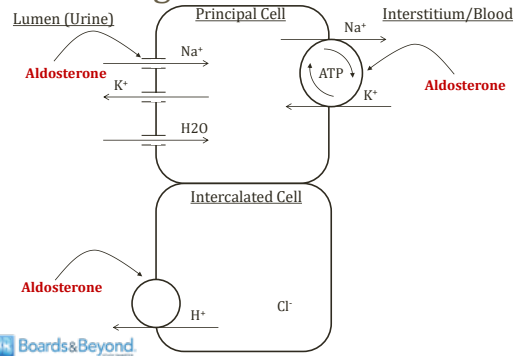
Aldosterone

- Increases **Na/K-ATPase proteins**
- Increases **Na channels (ENaC)** of principal cells
- Promotes K secretion** principal cells
- Promotes H⁺ secretion** intercalated cells



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Collecting Duct



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Aldosterone

- Overall effect:
 - ↑ sodium/water resorption (↑ effective circulating volume)
 - ↑ K excretion
 - ↑ H⁺ excretion
- Release stimulated by:
 - Angiotensin II
 - High potassium
 - ACTH (minor effect)

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RAAS Drugs

- ACE-inhibitors
 - Block conversion AI to AII
 - Lower blood pressure
- Angiotensin receptor blockers (ARBs)
 - Block effects of angiotensin II
 - Lower blood pressure

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RAA System Drugs

- Beta Blockers
 - Block sympathetic stim of JG apparatus
 - Block renin release
 - Lower blood pressure

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RAA System Drugs

- Aldosterone antagonists
 - Spironolactone, eplerenone
 - Lower blood pressure
 - Will ↑ K, ↑ H⁺ (↓ pH)
- Potassium-sparing diuretics
 - Triamterene/amiloride
 - Inhibit ENaC

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Natriuretic Peptides

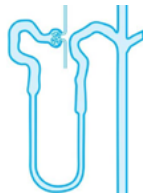
- Atrial natriuretic peptide (ANP)
- Brain natriuretic peptide (BNP)
- Release in response to volume (myocyte stretch)
- **Oppose actions of RAAS**
 - Relax vascular smooth muscle via cGMP
 - Vasodilator ($\downarrow$ SVR)
 - $\uparrow$ diuresis

Parathyroid Hormone

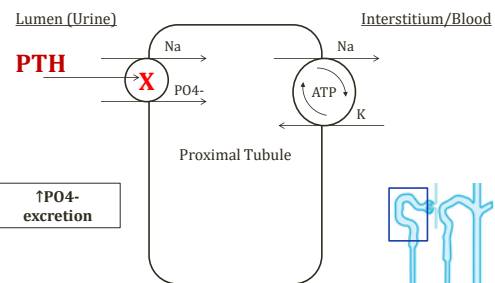
- Maintains **calcium** levels
- Released by **chief cells of parathyroid gland**
- Main stimulus is $\downarrow$ $[Ca^{2+}]$
- Net Effects:
 - $\uparrow$ $[Ca^{2+}]$ plasma
 - $\downarrow$ $[PO_4^{3-}]$ plasma
 - $\uparrow$ $[PO_4^{3-}]$ urine

Parathyroid Hormone Effects

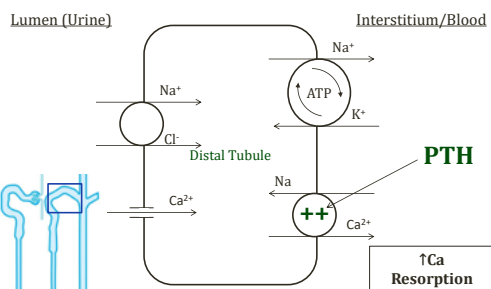
- Kidney:
 - $\uparrow$ Ca^{2+} resorption (DCT)
 - $\downarrow$ PO_4^{3-} resorption (PCT)
 - $\uparrow$ $1,25-(OH)_2$ vitamin D production
- Also has effects on GI tract and bone



Parathyroid Hormone

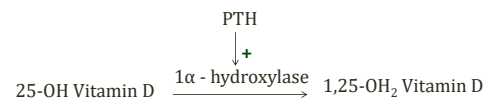


Parathyroid Hormone



Vitamin D and the Kidney

- Proximal Tubule converts vitamin D to active form



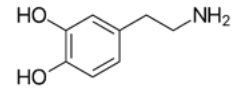
EPO

Erythropoietin

- Stimulates red blood cell production in bone marrow
- Made by interstitial cells peritubular capillary
- Released in response to hypoxia
- Decreased production in renal failure
- Normocytic anemia

Dopamine

- Synthesized in the proximal tubule
- Dilates afferent and efferent arterioles
 - Increased RPF
 - Little change in GFR
- Promotes sodium/water excretion (natriuretic)
- Unclear physiologic significance



Acid Excretion

Acid Excretion

Jason Ryan, MD, MPH



Types of Acids

- Two types of acids produced via metabolism
 - Volatile acids
 - Non-volatile acids



Volatile Acids

- CO_2
- Combines with water to form carbonic acid and H^+
- Eliminated by lungs (not kidneys)



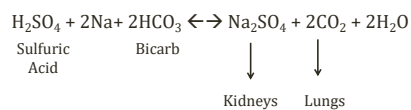
Non-volatile Acids

- Not from CO_2
- Derived from amino acids, fatty acids, nucleic acids



Non-volatile Acids

- Example: Sulfuric acid $\text{H}_2\text{SO}_4 \leftrightarrow \text{H}^+ + \text{SO}_4^-$

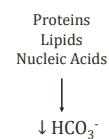


Key Points

Acid buffered by bicarbonate (no change pH)
Bicarbonate must be replenished by kidneys



Non-volatile Acids



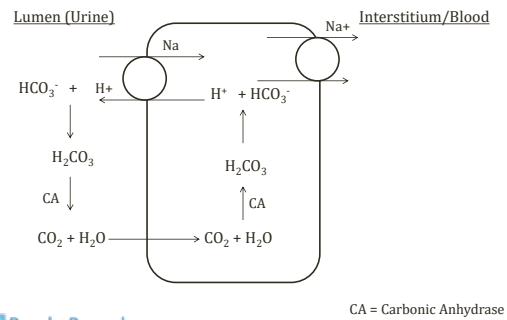
Renal Acid-Base Regulation

- #1: Reabsorb/Generate bicarb
- #2: Excrete H^+

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Bicarb Reabsorption

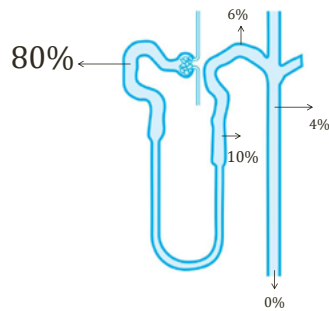
Proximal Tubule



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Bicarb Reabsorption

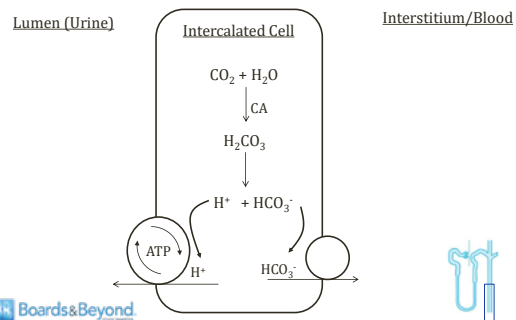
Nephron



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HCO_3^- Generation

Collecting Duct



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HCO_3^- Generation

Collecting Duct

- High H^+ $\rightarrow$ low pH $\rightarrow$ damage to nephron
- **Buffers** soak up H^+
- Protect from low pH
- Problem: Bicarbonate reabsorbed
- Need other buffers

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Urinary Buffers

- **Titrateable acids**
- **Ammonia**

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Titratable Acids

- Urinary substances that absorb H^+
- Acids
- Measured by titration ("titratable")

Titratable Acids

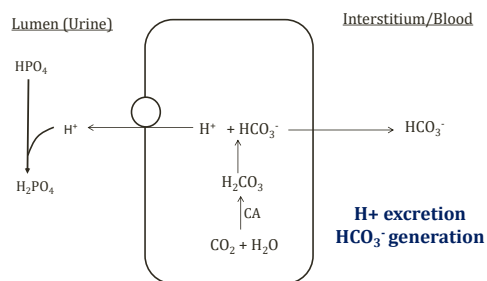
- Mostly phosphate
- Exists in multiple states
- HPO_4 (one hydrogen)
- H_2PO_4 (two hydrogens)

Titratable Acids

- HPO_4 filtered by glomerulus
- Form H_2PO_4 with addition of H^+
- H_2PO_4 excreted in urine = excretion of H^+

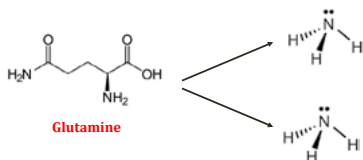
$\uparrow H_2PO_4$ excretion = $\uparrow H^+$ excretion

Titratable Acids



Ammonia

- Limited supply of titratable acids
 - Varies with dietary intake (especially phosphate)
- Supply of ammonia (NH_3) is **adaptable**
- More NH_3 generated by kidneys when $\uparrow H^+$
- Synthesized from **glutamine** (amino acid)

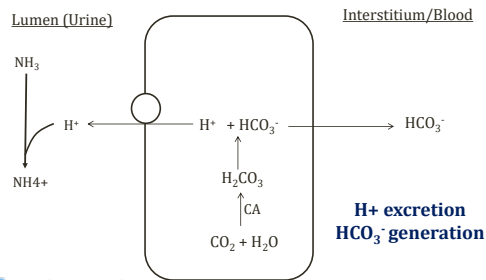


Terminology

- Ammonia = NH_3
- Ammonium = NH_4^+

Ammonia

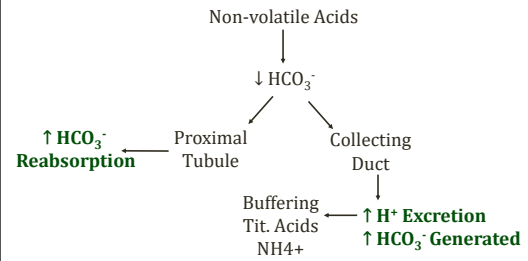
$\uparrow \text{NH}_4^+$ excretion = $\uparrow \text{H}^+$ excretion



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Renal Acid-Base

Summary



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Net Acid Excretion

- Urinary Acid – Urinary Base
- Positive value indicates acid being excreted

$$\text{Net Acid Excretion} = \text{Titratable Acids} + \text{NH}_4^+ - \text{HCO}_3^-$$

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Net Acid Excretion

- Acidosis: Increased net acid excretion
- Alkalosis: Decreased net acid excretion

$$\text{Net Acid Excretion} = \text{Titratable Acids} + \text{NH}_4^+ - \text{HCO}_3^-$$

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Acid Base Principles

Acid Base Principles

Jason Ryan, MD, MPH

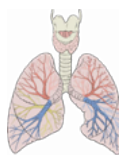
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Acid Base Balance

- Normal arterial pH: 7.37 to 7.42
- Tightly controlled
- **Lungs**: excrete carbon dioxide
- **Kidneys**: excrete acid & produce bicarbonate



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Patrick J. Lynch, medical illustrator

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Henderson-Hasselbalch Equation

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Maintained by kidneys

Maintained by lungs

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CO₂ Acid Dissociation Equation



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Arterial Blood Gas

- Normal HCO₃⁻ = 22 – 26 mEq/L
- Normal pCO₂ = 35 – 45 mmHg
- Normal pH = 7.37-7.42

HCO₃⁻: 24
pCO₂: 40
pH: 7.4



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Acid Base Disorders

- Acidosis/alkalosis
 - Disorder-altering H⁺ levels
- Acidemia/alkalemia
 - Presence of high or low pH in bloodstream
- Acidosis without acidemia occurs in mixed disorders
 - i.e. acidosis + alkalosis at same time

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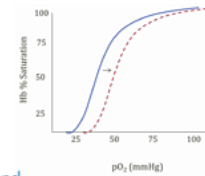
Acidosis Effects

- Myocardial depression (↓ contractility)
- ↑ CO₂: cerebral vasodilation
 - CO₂: major cerebral autoregulator
 - Increased cerebral blood flow
 - Increased intracranial pressure → headaches

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Acidosis Effects

- **Hyperkalemia**
 - High H⁺ shifts into cells in exchange for K⁺
- Shift in **oxyhemoglobin dissociation curve**
 - Bohr effect
 - ↓ pH leads to hemoglobin releasing more oxygen



Alkalosis Effects

- Cerebral vasoconstriction
 - Decrease in cerebral blood flow
- Hypokalemia
- Shift in oxyhemoglobin dissociation curve

Acid-Base Disorders

- **Metabolic Disorders**
 - Excess or insufficient HCO₃⁻
 - Metabolic acidosis (↓ HCO₃⁻)
 - Metabolic alkalosis (↑ HCO₃⁻)
- **Respiratory disorders**
 - Excess or insufficient CO₂
 - Respiratory acidosis (↑ CO₂)
 - Respiratory alkalosis (↓ CO₂)

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Acid-Base Problems

- Given pH, CO₂, HCO₃⁻
- What is the disorder?



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Acid-Base Problems

1. Check the **pH**
 - pH < 7.37 = acidosis
 - pH > 7.42 = alkalosis
2. Check the **HCO₃⁻** and **pCO₂**
 - Increased or decreased?
 - HCO₃⁻: normal 22-28 mEq/L
 - pCO₂ from ABG; normal 35-45mmHg

Acid-Base Problems

3. Determine acid-base disorder

- Acidosis + ↓ HCO_3^- = metabolic acidosis
- Acidosis + ↑ pCO_2 = respiratory acidosis
- Alkalosis + ↑ HCO_3^- = metabolic alkalosis
- Alkalosis + ↓ pCO_2 = respiratory alkalosis

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Acid-Base Problems

4. Calculate anion gap (metabolic acidosis)

5. Use special formulas to check for mixed disorders

- Combined respiratory/metabolic
- Two metabolic disorders

Compensatory Changes

- HCO_3^- and CO_2 are not independent
- Abnormal $\text{HCO}_3^- \rightarrow$ Abnormal CO_2
- Abnormal $\text{CO}_2 \rightarrow$ Abnormal HCO_3^-
- This is called **compensation**

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Compensatory Changes

- Respiratory disorders $\rightarrow$ abnormal CO_2
 - Compensation: HCO_3^- (renal)
- Metabolic disorders $\rightarrow$ Abnormal HCO_3^-
 - Compensation CO_2 (respiratory)

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Compensatory Changes

Acid-Base Disorder	Primary Abnormality	Compensation
Metabolic Acidosis	↓ HCO_3^-	↓ CO_2
Metabolic Alkalosis	↑ HCO_3^-	↑ CO_2
Respiratory Acidosis	↑ CO_2	↑ HCO_3^-
Respiratory Alkalosis	↓ CO_2	↓ HCO_3^-

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Compensatory Changes

- Most acid-base disorders: HCO_3^- and CO_2 abnormal
- One is "culprit" causing disorder
- Other is compensatory change

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Compensatory Changes

- Simple disorders
 - Culprit and compensatory change: same direction
 - HCO_3^- and pCO_2 both increased or both decreased

Example 1
 $\text{pH} = 7.30$ (acidosis)
 HCO_3^- = low
 pCO_2 = low
 Metabolic acidosis with respiratory compensation

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

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Compensatory Changes

- Simple disorders
 - Culprit and compensatory change: same direction
 - Both increased or both decreased

Example 2
 $\text{pH} = 7.30$ (acidosis)
 HCO_3^- = high
 pCO_2 = high
 Respiratory acidosis with metabolic compensation

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Respiratory Compensation

- Hyperventilation or hypoventilation
- Alters CO_2
- Compensates for metabolic disorders (HCO_3^-)
- Hyperventilation**
 - Physiologic response to metabolic acidosis
 - Kussmaul breathing = deep, labored breathing
 - Trying to blow off CO_2



Patrick J. Lynch, medical illustrator

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Respiratory Compensation

- Hyperventilation
 - Blows off CO_2
 - Plasma CO_2 level falls
 - Less H^+ in blood
 - pH rises
- Hypoventilation
 - Retains CO_2
 - Plasma CO_2 level rises
 - More H^+ in blood
 - pH falls

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$



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Renal Compensation

- Acidosis
 - Excess H^+ filtered/secreted
 - Bicarbonate** reabsorbed
 - Bicarbonate generated
- Alkalosis
 - Reverse of acidosis

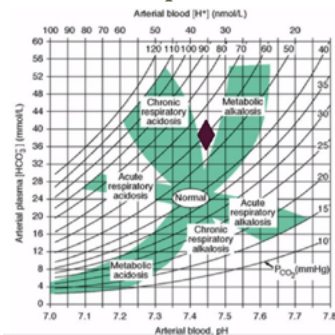
$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$



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Acid-Base Map



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Mixed Disorders

- **Two disorders** at same time
 - Metabolic acidosis AND respiratory acidosis
 - Metabolic acidosis AND metabolic alkalosis
 - Two metabolic acidoses
 - Occurs in many pathologic states
 - Example: vomiting and diarrhea

Mixed Disorder Recognition

- Determine **"expected"** response
 - Expected HCO_3^- for respiratory disorder
 - Expected CO_2 for metabolic disorder
- If actual $\neq$ expected $\rightarrow$ 2nd disorder present
- Compensation back to normal pH **very rare**
 - Normal pH usually implies a mixed disorder

Mixed Disorder Recognition

- If actual $\neq$ expected, determine abnormality
 - Example: CO_2 higher than expected
 - Example: HCO_3^- lower than expected
- Usual rules then apply for determining 2^o disorders:
 - $\uparrow \text{CO}_2$ = respiratory acidosis
 - $\downarrow \text{CO}_2$ = respiratory alkalosis
 - $\downarrow \text{HCO}_3^-$ = metabolic acidosis
 - $\uparrow \text{HCO}_3^-$ = metabolic alkalosis

Compensation Formulas

- **Winter's Formula**
- Metabolic Alkalosis Formula
- Acute/Chronic Respiratory Equations
- Delta-Delta

Metabolic Acidosis

- Compensatory respiratory alkalosis ($\downarrow \text{CO}_2$)
- Hyperventilation
- **Winter's Formula:**
 - Calculates expected CO_2
 - If actual $\text{CO}_2 \neq$ expected, mixed disorder

$$p\text{CO}_2 = 1.5 (\text{HCO}_3^-) + 8 \pm 2$$

Metabolic Acidosis

- Compensatory respiratory alkalosis ($\downarrow \text{CO}_2$)
- Hyperventilation
- **Winter's Formula:** $p\text{CO}_2 = 1.5 (\text{HCO}_3^-) + 8 \pm 2$
 - Calculates expected CO_2
 - If actual $\text{CO}_2 \neq$ expected, mixed disorder

Example 1
 pH = 7.23 (acidosis)
 $\text{HCO}_3^- = 9 \text{ mEq/L}$ (nl = 24)
 $p\text{CO}_2 = 22 \text{ mmHg}$ (nl = 40)
 Expected $p\text{CO}_2 = 1.5 (9) + 8 = 22 \pm 2$

Metabolic Acidosis

- Compensatory respiratory alkalosis ($\downarrow$ CO_2)
- Hyperventilation
- **Winter's Formula:** $\text{pCO}_2 = 1.5 (\text{HCO}_3^-) + 8 \pm 2$
 - Calculates expected CO_2
 - If actual $\text{CO}_2 \neq$ expected, mixed disorder

Example 2
 $\text{pH} = 7.10$ (acidosis)
 $\text{HCO}_3^- = 12 \text{ mEq/L}$ (nl = 24)
 $\text{pCO}_2 = 40 \text{ mmHg}$ (nl=40)
 Expected $\text{pCO}_2 = 1.5 (12) + 8 = 26 \pm 2$
 $\text{pCO}_2 > \text{expected}$
 Concomitant Respiratory Acidosis

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Metabolic Alkalosis

- Compensatory respiratory acidosis ($\uparrow$ CO_2)
- Hypoventilation
- $\uparrow$ pCO_2 0.7 mmHg per 1.0 meq/L $\uparrow$ $[\text{HCO}_3^-]$
- If actual $\text{pCO}_2 \neq$ expected, mixed disorder

$$\Delta \text{pCO}_2 = 0.7 * (\Delta [\text{HCO}_3^-])$$

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Metabolic Alkalosis

- Compensatory respiratory acidosis ($\uparrow$ CO_2)
 - Hypoventilation
 - $\uparrow$ pCO_2 0.7 mmHg per 1.0 meq/L $\uparrow$ $[\text{HCO}_3^-]$
 - If actual $\text{pCO}_2 \neq$ expected, mixed disorder
- $$\Delta \text{pCO}_2 = 0.7 * (\Delta [\text{HCO}_3^-])$$

Example 1
 $\text{pH} = 7.48$ (alkalosis)
 $\text{HCO}_3^- = 34 \text{ mEq/L}$ (nl = 24)
 $\text{pCO}_2 = 47 \text{ mmHg}$ (nl=40)
 $\Delta [\text{HCO}_3^-] = (34 - 24) = 10$
 Expected $\Delta \text{pCO}_2 = 0.7 (10) = 7$
 Actual $\Delta \text{pCO}_2 = 47 - 40 = 7$
 No Secondary Disorder

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Respiratory Acidosis

- Acute compensation
 - Minutes
 - Intracellular buffers raise $[\text{HCO}_3^-]$
 - Hemoglobin and other proteins
 - Small $\uparrow$ pH
- Chronic compensation
 - Days
 - Renal generation of $\uparrow$ $[\text{HCO}_3^-]$
 - Larger $\uparrow$ pH

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Respiratory Acidosis

- Acute compensation
 - 1 meq/L $\uparrow$ $[\text{HCO}_3^-]$ for every 10 mmHg $\uparrow$ pCO_2
 - $\Delta [\text{HCO}_3^-] = \Delta \text{pCO}_2 / 10$
- Chronic compensation
 - 3.5 meq/L $\uparrow$ $[\text{HCO}_3^-]$ for every 10 mmHg $\uparrow$ pCO_2
 - $\Delta [\text{HCO}_3^-] = 3.5 * \Delta \text{pCO}_2 / 10$

Example 1
 $\text{pH} = 7.10$ (acidosis)
 $\text{pCO}_2 = 80 \text{ mmHg}$ (nl=40)
 Acute $\Delta [\text{HCO}_3^-] = 40 / 10 = 4$
 $[\text{HCO}_3^-] = 28$; $\text{pH} = 7.17$
 Chronic $\Delta [\text{HCO}_3^-] = 3.5 * 40 / 10 = 14$
 $[\text{HCO}_3^-] = 38$; $\text{pH} = 7.30$

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Respiratory Alkalosis

- Acute compensation
 - 2 meq/L $\downarrow$ $[\text{HCO}_3^-]$ for every 10 mmHg $\downarrow$ pCO_2
 - $\Delta [\text{HCO}_3^-] = 2 * \Delta \text{pCO}_2 / 10$
- Chronic compensation
 - 4 meq/L $\downarrow$ $[\text{HCO}_3^-]$ for every 10 mmHg $\downarrow$ pCO_2
 - $\Delta [\text{HCO}_3^-] = 4 * \Delta \text{pCO}_2 / 10$

Example 1
 $\text{pH} = 7.7$ (alkalosis)
 $\text{pCO}_2 = 20 \text{ mmHg}$ (nl=40)
 Acute $\Delta [\text{HCO}_3^-] = 2 * 20 / 10 = 4$
 $[\text{HCO}_3^-] = 20$; $\text{pH} = 7.62$
 Chronic $\Delta [\text{HCO}_3^-] = 4 * 20 / 10 = 8$
 $[\text{HCO}_3^-] = 16$; $\text{pH} = 7.53$

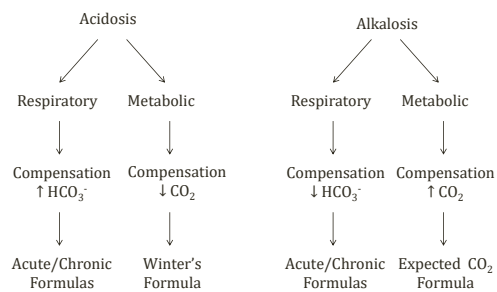
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Compensation Timeframe

- Respiratory compensation to metabolic disorders
 - Occurs in **minutes**
 - Rapid change in respiratory rate
- Metabolic compensation to respiratory disorders
 - Acute, mild compensation in minutes from buffers
 - Chronic, significant compensation in **days** from kidneys



Summary



Respiratory Acid-Base Disorders

Respiratory Acid-Base Disorders

Jason Ryan, MD, MPH

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Acid-Base Disorders

1. Respiratory alkalosis
2. Respiratory acidosis
3. Metabolic alkalosis
4. Metabolic acidosis

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Respiratory Alkalosis

pH > 7.42
↓ pCO₂

Increased **pH** = $6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 * \text{pCO}_2}$ ↓ = compensation
↓ = primary abnormality

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Respiratory Alkalosis

- Caused by **hyperventilation**
 - Pain
 - Anxiety
 - Early high-altitude exposure
 - Early aspirin overdose
 - Mechanical ventilation
- Normal respiratory rate < 25/min

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High Altitude

- **Lower atmospheric pressure**
 - Sea level: 760 mmHg
 - Machu Picchu = 560 mmHg
- **Lower pO₂**
 - P_{A02} sea level = 100 mmHg
 - P_{A02} Machu Picchu = 75mmHg



Wikipedia/Public Domain

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High Altitude

- Hypoxia → **hyperventilation**
- ↓ pCO₂ → respiratory alkalosis (pH rises)

Parameter	Change
P _{O2} (alveoli, artery)	Decrease
Respiratory rate	Increase
Carbon dioxide	Decrease
pH	Increase

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High Altitude

Acclimatization

- Renal response
 - After 24-48 hours: kidneys **excrete HCO_3^-**
 - pH will fall back toward normal
 - **Acetazolamide** increases HCO_3^- excretion
- Red cell response to hypoxemia
 - Synthesis of **2,3-Bisphosphoglycerate**
 - Unloading of oxygen from hemoglobin



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Aspirin Overdose



- Two acid-base disorders
- Shortly after ingestion: **respiratory alkalosis**
 - Salicylates stimulate medulla
 - Hyperventilation
- Hours after ingestion: **AG metabolic acidosis**
 - Salicylates ↓ lipolysis, uncouple oxidative phosphorylation
 - Inhibits citric acid cycle
 - Accumulation of pyruvate, lactate, ketoacids

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Aspirin Overdose



- pH
 - Variable due to mixed disorder
 - Acidotic, alkalotic, normal
- pCO_2
 - Low due to hyperventilation
- HCO_3^-
 - Low due to acidosis
- Winter's formula predicts CO_2 higher than actual
- CO_2 lower than expected for compensation

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Aspirin Overdose

- Sample case: pH 7.36, pCO_2 20, HCO_3^- 11
- Metabolic acidosis
- Winter's formula
 - $\text{pCO}_2 = 1.5 (\text{HCO}_3^-) + 8 \pm 2$
 - $\text{pCO}_2 = 1.5 (11) + 8 \pm 2 = 25$
- $\text{pCO}_2 < \text{expected}$
- Concomitant respiratory alkalosis



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Respiratory Acidosis

pH < 7.37
↑ pCO_2

$$\text{Decreased pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

↑ = compensation
↑ = primary abnormality

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Respiratory Acidosis

- Caused by **hypoventilation**
- Decreased CNS respiratory drive
 - Opiates
 - Barbiturates
 - CNS disease
- Respiratory muscle diseases
 - Guillain-Barre
 - Polio
 - Multiple sclerosis
 - Amyotrophic lateral sclerosis

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Respiratory Acidosis

- Impaired gas exchange
 - Pneumonia
 - Pulmonary edema
 - Acute respiratory distress syndrome
 - COPD
- Airway obstruction
 - Aspiration
 - Obstructive sleep apnea

ARDS



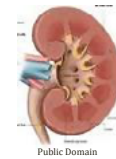
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Respiratory Acidosis

Renal Compensation

- Excess H⁺ filtered/secreted
- **Bicarbonate** reabsorbed
- Bicarbonate generated
- Classic example: COPD
 - Chronic increase pCO₂
 - Chronic respiratory acidosis
 - Compensation: **increased HCO₃⁻**

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 * \text{pCO}_2}$$

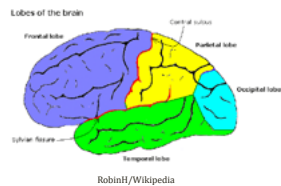


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Acute hypercapnia

- Hypercapnia can affect **CNS system**
- Mild to moderate: anxiety, headaches
- Severe: delirium, eventually coma



CO₂

RobinH/Wikipedia

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Metabolic Alkalosis

Metabolic Alkalosis

Jason Ryan, MD, MPH



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Acid-Base Disorders

1. Respiratory alkalosis
2. Respiratory acidosis
3. Metabolic alkalosis
4. Metabolic acidosis

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Metabolic Alkalosis

pH > 7.42
↑ HCO₃⁻

Increased pH = $6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 * \text{pCO}_2}$

↑ = primary abnormality

↑ = respiratory compensation



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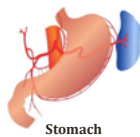
Metabolic Alkalosis

1. Gastrointestinal acid loss
2. Hypokalemia
3. Contraction alkalosis
4. Hyperaldosteronism
5. Milk-alkali syndrome
6. Rare renal syndromes

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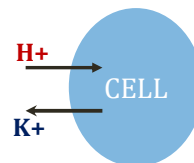
Gastrointestinal Acid Loss

- Vomiting
 - Gastric suction
 - Parietal cells secrete **HCl**
 - Loss of stomach contents → **alkalosis**
 - **Hypochloremia** (Cl^- loss)
 - **Hypokalemia** (K^+ loss; RAAS activation)
 - Hypochloremic, hypokalemic metabolic alkalosis
- 
- Stomach



Hypokalemia

- K^+ exchanges with H^+
- **Shifts in and out of cells**
- $\downarrow K^+ \rightarrow$ shift K^+ out of cells $\rightarrow H^+$ into cells
- Hypokalemia $\rightarrow$ alkalosis (vice versa)



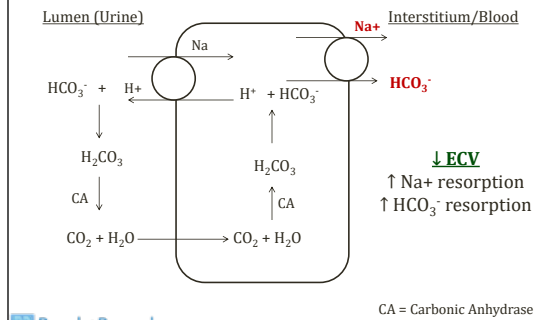
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Contraction Alkalosis

- Loss of fluid with Na and Cl
- ↓ Effective Circulating Volume (ECV)
- Renin-Angiotensin-Aldosterone (RAAS) activation
- Sympathetic nervous system (SNS) activation
- ↑ HCO_3^- resorption **proximal tubule**
- ↑ H^+ secretion **collecting duct**

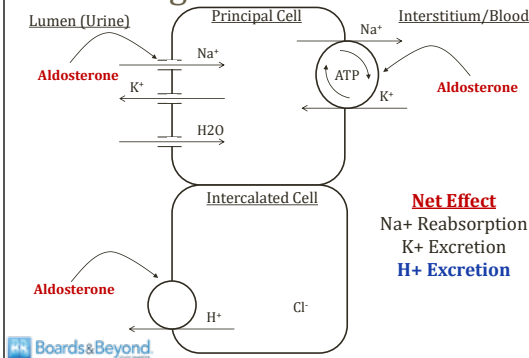
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Proximal Tubule



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Collecting Duct



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Contraction Alkalosis

- Vomiting (contraction plus gastric losses)
- Diuretics (contraction plus ↓ K)
- Heart failure (treated with diuretics)
- Cirrhosis (treated with diuretics)
- New data suggest **chloride depletion** is true cause
 - Low volume → low NaCl
 - Chloride repletion resolves alkalosis

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Diuretics

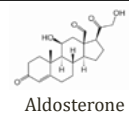
- Metabolic alkalosis: **loop and thiazide diuretics**
- Increased Na delivery to distal tubule
 - Increased K excretion → hypokalemia
 - Increased H^+ excretion
- Volume contraction → contraction alkalosis



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Hyperaldosteronism

Primary Aldosteronism



- Adrenal overproduction of aldosterone
- Adrenal hyperplasia
- Adrenal adenoma (Conn's syndrome)
- Increased secretion H^+
- **Metabolic alkalosis**
- Hypokalemia
- **Hypertension**



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Milk-alkali Syndrome

- Hypercalcemia, metabolic alkalosis, renal failure
- Excessive intake:
 - Calcium
 - Alkali (base)
- Usually **calcium carbonate** and/or milk
 - Alkaline CaCO_3
- Often taken for dyspepsia



Midnightcomm

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Milk-alkali Syndrome

- Increased Ca intake → hypercalcemia
- Hypercalcemia interferes with kidney function
 - Inhibition Na-K-2Cl in TAL
 - Blockade (ADH)-dependent water reabsorption collecting duct
- Results in volume contraction
- Contraction + alkali = **metabolic alkalosis**
- ↓ GFR from volume contraction
 - ↑ BUN, Cr

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Bartter and Gitelman

- Congenital disorders
- Occur in children
- Impaired sodium resorption in nephron
- Polyuria
- Hypokalemia (muscle cramps)
- Metabolic alkalosis

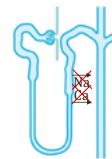


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Bartter Syndrome

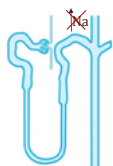
- Defective sodium resorption thick ascending limb
- Similar to administration of **loop diuretic**
- Presents in childhood
- Polyuria, polydipsia or nocturia
- Activation RAAS → metabolic alkalosis
- Hypokalemia
- **High urinary calcium**
 - TAL cannot absorb Ca



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Gitelman Syndrome

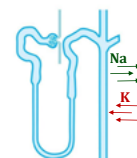
- Defective sodium resorption distal tubule
- Similar to administration of **thiazide diuretic**
- Presents in childhood
- Polyuria, polydipsia or nocturia
- Activation RAAS → metabolic alkalosis
- Hypokalemia (cramps)
- **Low urinary calcium**
 - Distal tubule cannot secrete Ca



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Liddle Syndrome

- Genetic disorder of **increased activity of ENaC**
 - Epithelial sodium channel
 - Activity increased by aldosterone
- Presents in juveniles
- Similar clinical syndrome to hyperaldosteronism
 - Hypertension
 - Hypokalemia
 - Metabolic alkalosis



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Liddle Syndrome

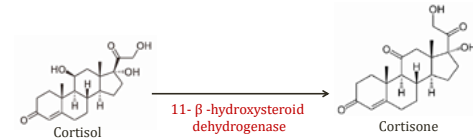
- Young patient with hypertension, hypokalemia
- **Aldosterone levels low**
- **Treatment: amiloride**
 - Potassium-sparing diuretic
 - Inhibits ENaC activity

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SAME

Syndrome of Apparent Mineralocorticoid Excess

- Renal aldosterone receptors bind cortisol
- Cortisol → cortisone by renal cells
- Enzyme: **11-β-hydroxysteroid dehydrogenase**
- SAME: deficiency 11-β-hydroxysteroid dehydrogenase
- Cortisol produces **aldosterone effects**



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SAME

Syndrome of Apparent Mineralocorticoid Excess

- Presents in children/adolescents
- Similar clinical syndrome to hyperaldosteronism
 - Hypertension
 - Hypokalemia
 - Metabolic alkalosis
- **Aldosterone levels low**
- Treatment:
 - Potassium-sparing diuretics (amiloride, spironolactone)
 - Dexamethasone

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Licorice

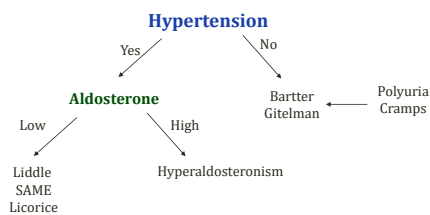
- Contains **glycyrrhetic acid** (a steroid)
 - Weak mineralocorticoid effect
 - Inhibits renal 11-beta-hydroxysteroid dehydrogenase
- Large amounts may cause disease
- Hypertension, hypokalemia, metabolic alkalosis
- Plasma aldosterone level low



Pikaluk/Flickr

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Metabolic Alkalosis Causes



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Urinary Chloride

- Rarely used test in metabolic alkalosis
- Low (<20 mEq/L) with **gastric acid loss (HCl)**
 - Vomiting
 - Nasogastric suction
- Variable findings with other causes
- High immediately after diuretic administration



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NaCl Fluid Administration

- Contains sodium, chloride and water
- Resolves many forms of metabolic alkalosis
 - **"Chloride responsive"**
 - Diuretics
 - Vomiting
- Some exceptions
 - Hyperaldosteronism
 - Bartter, Gitelman
 - Heart failure, cirrhosis



Hypoventilation

- Respiratory compensation in metabolic alkalosis
- Increased $p\text{CO}_2$
- Lowers pH

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 * p\text{CO}_2}$$



Patrick J. Lynch, medical illustrator

Renal Tubular Acidosis

Renal Tubular Acidosis

Jason Ryan, MD, MPH

Boards&Beyond

Non-AG Metabolic Acidosis

- Diarrhea
- Acetazolamide
- Spironolactone/Addison's disease
- Saline infusion
- Hyperalimentation
- **Renal tubular acidosis**

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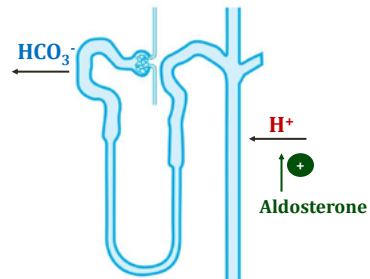
Renal Tubular Acidosis

- Rare disorders of nephron ion channels
- All cause **non-anion-gap metabolic acidosis**
- Often present with **low $[\text{HCO}_3^-]$** or **abnormal K^+**

$\text{pH} < 7.37$
 $\downarrow \text{HCO}_3^-$

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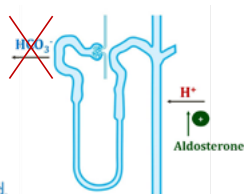
Renal Acid Handling



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Type II (proximal) RTA

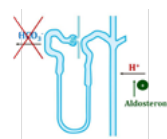
- Defect in **proximal tubule HCO_3^- resorption**
- Non-anion gap metabolic acidosis
- Often asymptomatic



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Type II (proximal) RTA

- **Hypokalemia**
 - Loss of HCO_3^- resorption $\rightarrow$ diuresis
 - Volume contraction
 - $\uparrow$ aldosterone $\rightarrow \uparrow \text{K}^+$ excretion $\rightarrow$ hypokalemia
- Urine $\text{pH} < 5.5$
 - Distal tubule excretes H^+ as acidosis becomes established
 - Urine becomes acidic
 - Negative urine anion gap



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Type II (proximal) RTA

- Milder than type I: $[\text{HCO}_3^-]$ 12-20
 - Distal intercalated cells function normally
 - Secrete acid to compensate

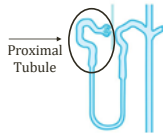
	Normal	Proximal (II)	Distal (I)	Type IV
HCO_3^- mg/dL	24	12-20	<10	>17

Type II (proximal) RTA

- Sample Case
 - No symptoms : routine blood work
 - Mild weakness (low K)
 - Mildly reduced HCO_3^- (12 – 20)
 - Hypokalemia
 - Urine pH is low (<5.3)
- Treatment: Sodium bicarbonate

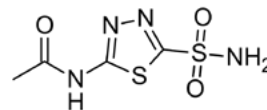
Type II (proximal) RTA

- Can be seen with **Fanconi syndrome**
 - Generalized failure of proximal tubule
 - Urine loss of phosphate, glucose, amino acids, urate, protein
- Can be seen in **multiple myeloma**
 - Some forms of light chains toxic to proximal tubule
 - Causes Fanconi syndrome



Acetazolamide

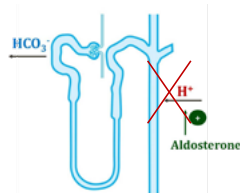
- Carbonic anhydrase inhibitor
- Weak diuretic effect
 - Block some Na resorption
- Causes a **non-anion gap metabolic acidosis**
 - Increased elimination of HCO_3^-



Acetazolamide

Type I (distal) RTA

- Impaired **acidification of urine** by distal nephron
- Non-anion gap metabolic acidosis
- ↓ excretion H^+ (acidemia)
- ↓ resorption K^+ (hypokalemia)



Type I (distal) RTA

- Very low HCO_3^- (often <10meq/L)**
- Urine pH is high
 - Distal tubule cannot "acidify" the urine
 - Urine is alkaline
- Diagnosis established if alkaline urine (pH > 5.5) despite a metabolic acidosis (with normal kidneys)

Type I (distal) RTA

- Key symptoms: chronic kidney stones
 - Alkaline urine precipitates stones (sometimes bilateral)
 - Acidosis → ↑Ca from bones
 - Acidosis suppresses calcium resorption (↑Ca in urine)
- Rickets
- Growth failure in children

Type I (distal) RTA

- Many etiologies
- Associated with **autoimmune diseases**
 - Sjögren's syndrome
 - Rheumatoid arthritis
- Medications
 - Amphotericin B
- Rare genetic forms

Urine Anion Gap

- Used for diagnosis of metabolic acidosis
- Evaluation of **renal acid excretion**
- In acidosis, excess NH_4 excreted (removes H^+)
- NH_4 not measured directly
- Surrogate: **urinary anion gap**
- NH_4 leaves with Cl
- **Negative** UAG when acid (H^+) being excreted
- UAG should be negative in acidosis

$$\text{UAG} = \text{Na} + \text{K} - \text{Cl}$$

Urine Anion Gap

- In GI metabolic acidosis (diarrhea):
 - UAG becomes **negative**
 - Excretion of NH_4 with Cl increases
 - Urine Cl concentration goes up
- Also negative in proximal (type II) RTA
 - Intact distal H^+ secretion intact

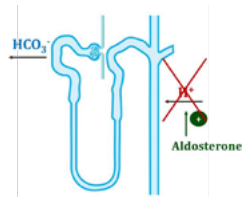
$$\text{UAG} = \text{Na} + \text{K} - \text{Cl}$$

neGUTtive in GI

Urine Anion Gap

$$\text{UAG} = \text{Na} + \text{K} - \text{Cl}$$

- In distal RTA and type IV RTA UAG is **positive**
 - Kidneys can't excrete H^+
 - NH_4 and Cl don't increase
 - UAG ($\text{Na} + \text{K} - \text{Cl}$) does not become negative



Ammonium Chloride Challenge

- Used for diagnosis of metabolic acidosis
- "Challenge" patient with NH_4Cl
- Gives an acid load
- Should lower urine pH
- In distal RTA, **urine pH remains >5.3**



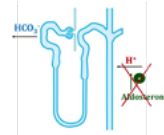
Type I (distal) RTA

- Classic case
 - Patient with Sjogren's disease
 - Recurrent bilateral kidney stones
 - Very low bicarb on blood work (<10)
 - Hypokalemia
 - Urine pH is high (>5.5)
 - UAG is positive
 - If given NH_4Cl urine remains with high pH
- Treatment: Sodium bicarbonate

Type IV RTA

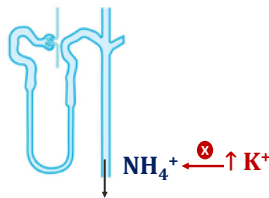
Hypoaldosteronism

- Distal tubule failure to respond to **aldosterone**
 - Aldosterone deficiency/resistance
- Decreased excretion K^+
- Major feature: **hyperkalemia**
- Mild non-anion gap metabolic acidosis
 - $\text{HCO}_3^- > 17$ (normal = 24)



Type IV RTA

- Major pathologic defect: **decreased NH_4^+ excretion**
 - Loss of urinary buffering $\rightarrow$ low urinary pH (<5.3)
- **Hyperkalemia** $\rightarrow$ $\downarrow$ ammonium



Type IV RTA

- **Hyporeninemic hypoaldosteronism**
 - Low renin activity
 - Decreased aldosterone production
 - Diabetes (associated with low renin production)
 - NSAIDs (impair renin release)
 - Other drugs

Type IV RTA

- **RAAS drugs** ($\downarrow$ aldosterone)
 - Angiotensin-converting enzyme (ACE) inhibitors
 - Angiotensin II receptor blockers (ARBs)
 - Direct renin inhibitors (Aliskiren)

Type IV RTA

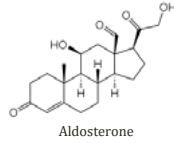
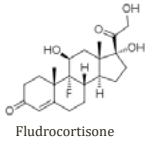
- **Aldosterone resistance**
 - Usually caused by drugs that inhibit tubular function
 - Potassium sparing diuretics
 - TMP/SMX



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Type IV RTA

- Classic case:
 - Diabetic with renal insufficiency
 - Unexplained hyperkalemia
- Treatment: **fludrocortisone**
 - Mineralocorticoid



Renal Tubular Acidosis

Type	Key Features
I	Distal; High urine pH; kidney stones; very low HCO ₃ ⁻
II	Proximal; mild acidosis; Fanconi's
IV	Aldosterone; hyperkalemia; ammonium

Type	Plasma K ⁺	Urine pH
I	Low (<3.5)	High (>5.4)
II	Low (<3.5)	Low (<5.4)
IV	High (>5.0)	Low (<5.4)

Metabolic Acidosis

Metabolic Acidosis

Jason Ryan, MD, MPH

Boards&Beyond

Acid-Base Disorders

1. Respiratory alkalosis
2. Respiratory acidosis
3. Metabolic alkalosis
4. Metabolic acidosis

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Metabolic Acidosis

- Most complex set of acid-base disorders
- Reduced pH (acidosis)
- Reduced HCO_3^- (metabolic acidosis)
- Causes categorized by **anion gap**

pH < 7.37
↓ HCO_3^-

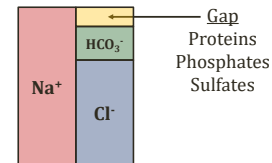
Decreased **pH** = $6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$

↓ = primary abnormality
↓ = respiratory compensation

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The Anion Gap

- Sodium (Na^+): major serum cation
- Balanced by anions like Cl^- and HCO_3^-
- Anion "gap": unmeasured anions
 - Proteins (albumin)
 - Phosphates
 - Sulfates



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The Anion Gap

- Anion Gap = $\text{Na} - (\text{Cl}^- + \text{HCO}_3^-)$
- Normal: **8 to 12**
- Metabolic acidosis subtypes:
 - Normal anion gap
 - Increased anion gap

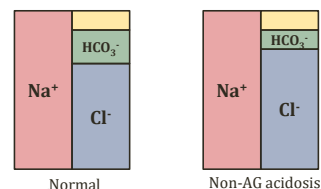
Measurement	Value
Sodium (Na^+)	140 mEq/L
Chloride (Cl^-)	103 mEq/L
Bicarbonate (HCO_3^-)	26 mEq/L

Anion Gap = $140 - (103 + 26) = 11$

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Why the Anion Gap Matters

- Acidosis from **primary loss of HCO_3^-**
 - Body compensates with retention of chloride (Cl^-)
 - $\text{AG} = \text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$
 - **Normal anion gap**

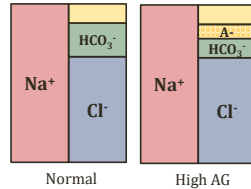


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Why the Anion Gap Matters

- Acidosis from **primary retention of acid**

- Examples: ketoacids, lactic acid
- HCO_3^- falls
- A^- to compensate for fall in HCO_3^-
- No change in Cl^-
- $\text{AG} = \text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$
- **Increased anion gap**



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Hyperchloremia

- Increased serum chloride
- Seen in all **non-anion gap metabolic acidoses**
- Negatively-charged HCO_3^- decreases
- No increase in acid-related anions
- Cl^- rises to compensate for lost negative charge

↑Cl⁻

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Two Cases

Measurement	Value	Normal
pH	7.31	7.37-7.42
Sodium (Na^+)	134 mEq/L	135-145
Chloride (Cl^-)	108 mEq/L	96-106
Bicarbonate (HCO_3^-)	16 mEq/L	22-26
Anion gap	10	8-16

Measurement	Value	Normal
pH	7.27	7.37-7.42
Sodium (Na^+)	132 mEq/L	135-145
Chloride (Cl^-)	93 mEq/L	96-106
Bicarbonate (HCO_3^-)	11 mEq/L	22-26
Anion gap	28	8-16

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Non-Anion Gap Metabolic Acidosis

- **Diarrhea**
 - Most common
 - Loss of HCO_3^- in stool
- **Saline infusion**
 - NaCl: lots of Cl^-
 - "Chloride toxicity"
 - Cl^- drives HCO_3^- from plasma
- Renal tubular acidosis



Non-Anion Gap Metabolic Acidosis

- Hyperalimentation
 - Acid
 - Lowers pH
- Acetazolamide
 - Blocks formation and resorption HCO_3^-
- Spironolactone/Addison's disease
 - Loss of aldosterone effects
 - Cannot excrete H^+ effectively
 - Body retains H^+



Anion Gap Metabolic Acidosis

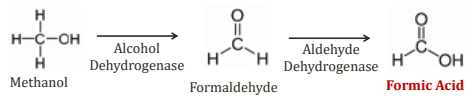
- Methanol
- Uremia
- Diabetic ketoacidosis
- Propylene glycol
- Iron tablets or INH
- Lactic acidosis
- Ethylene glycol
- Salicylates

MUD PILES

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Methanol

- Antifreeze, industrial cleaners, windshield wiper fluid
- Metabolized to **formic acid**
- Central nervous system poison
- Visual loss, coma



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Methanol

- Classic scenario:
 - Suspected ingestion (accidental, suicide, alcoholic)
 - Confusion (may appear inebriated)
 - **Visual symptoms**
 - High AG metabolic acidosis
- Treatment:
 - Inhibit alcohol dehydrogenase
 - Fomepizole (Antizol)
 - Ethanol

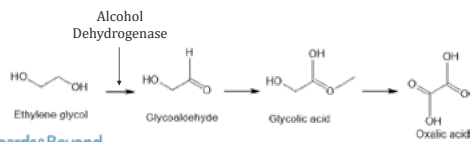


Petr Novák, Wikipedia

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Ethylene Glycol

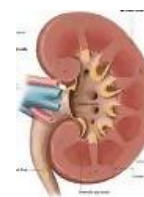
- Antifreeze, industrial cleaners, windshield wiper fluid
- Metabolized to glycolate and oxalate
- **Kidney toxins**
- Glycolate: toxic to renal tubules
- Oxalate: precipitates calcium oxalate crystals



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Ethylene Glycol

- Classic scenario:
 - Suspected ingestion (accidental, suicide, alcoholic)
 - Flank pain, oliguria, anorexia (acute renal failure)
 - High AG metabolic acidosis
- Treatment:
 - Inhibit alcohol dehydrogenase
 - Fomepizole (Antizol)
 - Ethanol

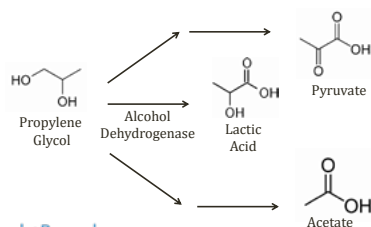


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Propylene Glycol

- Antifreeze (lowers freezing point of water)
- Solvent for **IV benzodiazepines**
- Metabolized to pyruvic acid, acetic acid, lactic acid



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Propylene Glycol

- Main clinical feature of overdose is **CNS depression**
- High AG metabolic acidosis from lactate & other acids
- Many other adverse effects:
 - Hemolysis
 - Seizure, coma, and multisystem organ failure
- No visual symptoms or nephrotoxicity



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Isopropyl Alcohol

- Also found in many industrial products
- Effects similar to ethanol
 - Key scenario: ingestion by alcoholic
- Less toxic than methanol or ethylene glycol
- **Does NOT cause anion gap metabolic acidosis**
 - Absence of high AG acidosis suggest IA ingestion
- No role for fomepizole or ethanol
- Main symptom of ingestion is coma



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Uremia

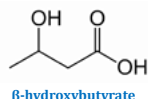
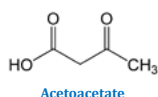
- **Advanced** kidney disease
 - Early kidney disease can have non-AG acidosis
 - Reduction in H^+ excretion (loss of tubule function)
- Kidneys cannot excrete **organic acids**
- Retention of phosphates, sulfates, urate, others
- Increased anion gap acidosis

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Diabetic Ketoacidosis

DKA

- Usually occurs in **type I diabetics**
- Insulin requirements rise → cannot be met
 - Often triggered by infection
- Fatty acid metabolism → **ketone bodies**
 - β -hydroxybutyrate
 - Acetoacetate



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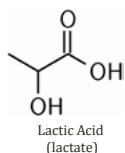
Diabetic Ketoacidosis (DKA)

- Polyuria, polydipsia ($\uparrow$ glucose → diuresis)
- Abdominal pain, nausea, vomiting
- **Kussmaul respirations**
 - Deep, rapid breathing
 - From acidosis
- High AG metabolic acidosis from ketones

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Lactic Acidosis

- Low tissue oxygen delivery
- Pyruvate converted to lactate
- High levels ($>4.0\text{mmol/L}$) → lactic acidosis
- Anion gap metabolic acidosis
- Clinical scenarios:
 - Shock ($\downarrow$ tissue perfusion)
 - Ischemic bowel
 - Metformin therapy (especially with renal failure)
 - Seizures



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Iron

- Acute iron poisoning
- Initial gastrointestinal phase (0 to 6 hours)
 - Iron toxic to GI mucosal cells
 - **Abdominal pain**
- Weeks later: bowel obstruction (scarring)



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Iron



Tomihahndorf

- Later (24 hours)
 - Cardiovascular toxicity: **shock**
 - Coagulopathy: iron inhibits thrombin formation/action
 - Hepatic dysfunction: worsening coagulopathy
 - Acute lung injury
- Anion-gap metabolic acidosis
 - From **ferric irons (Fe³⁺)**
 - Also lactate (hypoperfusion)



Isoniazid

INH

- Tuberculosis antibiotic
- Overdose causes **seizures**
 - Often severe, refractory (status epilepticus)
- Seizures cause **lactic acidosis**
- Anion gap metabolic acidosis

M. tuberculosis



Filix/Public Domain

Aspirin Overdose



- Two acid-base disorders
- Shortly after ingestion: **respiratory alkalosis**
 - Salicylates stimulate medulla
 - Hyperventilation
- Hours after ingestion: **AG metabolic acidosis**
 - Salicylates uncouple oxidative phosphorylation
 - Accumulation of pyruvate, lactate, ketoacids



Salicylic Acid
(salicylate)



Acetylsalicylic Acid
(aspirin)

Aspirin Overdose



- pH
 - Variable due to mixed disorder
 - Acidotic, alkalotic, normal
- pCO₂
 - Low due to hyperventilation
- HCO₃⁻
 - Low due to acidosis
- Winter's formula predicts CO₂ higher than actual
- CO₂ lower than expected for compensation

Aspirin Overdose

- Sample case: pH 7.36, pCO₂ 20, HCO₃⁻ 11
- Metabolic acidosis
- Winter's formula
 - pCO₂ = 1.5 (HCO₃⁻) + 8 +/- 2
 - pCO₂ = 1.5 (11) + 8 +/- 2 = 25
- pCO₂ < expected
- Concomitant respiratory alkalosis



Hyperventilation

- **Respiratory compensation** to metabolic acidosis
- Lowers pCO₂
- Increases pH

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \cdot \text{pCO}_2}$$

Winter's Formula

- Acidosis: compensatory **respiratory alkalosis**
 - ↓ pCO₂
 - Hyperventilation
- Winter's Formula gives expected ↓ pCO₂
- If actual CO₂ ≠ expected → mixed disorder
- Check Winter's formula for all metabolic acidoses

$$pCO_2 = 1.5 (HCO_3^-) + 8 +/- 2$$

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The Delta-Delta

Delta Ratio

- Anion gap ↑ should be similar to HCO₃⁻ ↓
- Ratio ΔAG / Δ HCO₃⁻ assesses for 2° acid-base disorder
 - Only used in AG metabolic acidosis

$$\Delta\Delta = \frac{\Delta AG}{\Delta HCO_3^-}$$

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The Delta-Delta

Delta Ratio

- ΔΔ 1-2 = normal
- ΔΔ <1
 - HCO₃⁻ too low
 - 2° non-AG metabolic acidosis
- ΔΔ >2
 - HCO₃⁻ too high
 - 2° metabolic alkalosis
 - Or pre-existing chronic respiratory acidosis

$$\Delta\Delta = \frac{\Delta AG}{\Delta HCO_3^-}$$

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The Delta-Delta

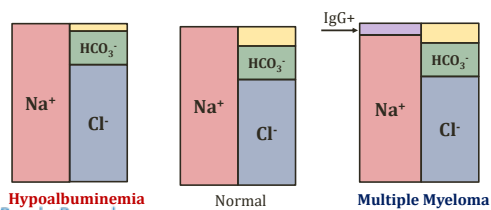
Example

- pH=7.21 (acidosis)
- HCO₃⁻ = 12
- Na⁺ = 150, Cl⁻ = 96
- Anion gap: 42
- Delta AG = 42-12 = 30
- Delta HCO₃⁻ = 24 - 12 = 12
- Delta-Delta = 30/12 = 2.5
- HCO₃⁻ is too high
- Concurrent metabolic alkalosis
- Or prior compensated respiratory acidosis

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Low Anion Gap

- Hypoalbuminemia
- Multiple myeloma
 - IgG is cationic (+)
 - Will lower measured (+) ions



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Acid Base Problems

Acid Base Problems

Jason Ryan, MD, MPH

Boards&Beyond

Case 1

A 40-year-old man presents to the emergency room with a three day history of severe diarrhea. Several coworkers have been ill with similar symptoms. An arterial blood gas is drawn showing:

pH 7.30, pCO₂ 33mmHg.

Electrolytes are:

Na 134, K 2.9, Cl 108, HCO₃⁻ 16

What is the acid-base disorder?

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Case 1

134	108	
2.9	16	

- Diarrhea → non-AG metabolic acidosis
- No other clues to suggest a 2nd disorder
- pH = 7.30 → acidosis
- HCO₃⁻ = 16 (low) → metabolic acidosis
- pCO₂ = 33 (low) → respiratory compensation
- Abnormal same direction → mixed disorder less likely
- Anion gap = 134 - 108 - 16 = 10 (normal)
- Winter's formula

$$\begin{aligned} \text{pCO}_2 &= 1.5 (\text{HCO}_3^-) + 8 \pm 2 \\ &= 1.5 (16) + 8 \\ &= 32 \pm 2 \end{aligned}$$

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Non-AG Metabolic Acidosis

Case 2

An 80-year-old man with a severe cardiomyopathy presents with shortness of breath and edema for the past two days.

An arterial blood gas is drawn showing:

pH 7.25, pCO₂ 62mmHg

Electrolytes show:

HCO₃⁻ 27

What is the acid-base disorder?

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Case 2

- CHF exacerbation → acute respiratory acidosis
- pH = 7.25 (acidosis)
- pCO₂ = 62 (high) → respiratory acidosis
- HCO₃⁻ = 27 (high) → metabolic compensation
- Abnormal same direction → mixed disorder less likely
- Acute respiratory acidosis ↑ HCO₃⁻ 1/10 ↑ CO₂
- Expected ↑ HCO₃⁻ = 2 (HCO₃⁻ of 26)
- No concurrent disorder

Acute respiratory acidosis

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Case 3

A 40-year-old woman with rheumatoid arthritis presents for a routine exam. She has normal vitals and a normal physical exam. She was hospitalized for a kidney stone six months ago which has since resolved. Serum electrolytes show:

Na 140, K 3.4, Cl 110, HCO₃⁻ 16

Because of the low HCO₃, an ABG is done:

pH 7.25, pCO₂ 32mmHg

What is the acid-base disorder?

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Case 3

140	110	
3.4	16	

- pH 7.25 (acidosis)
- HCO_3^- 16 (low) → metabolic acidosis
- pCO_2 32 (low) → respiratory compensation
- Expected $\text{PCO}_2 = 32$
- $\text{AG} = 140 - 110 - 16 = 14$
- Non-AG metabolic acidosis

$$\begin{aligned}\text{pCO}_2 &= 1.5 (\text{HCO}_3^-) + 8 \pm 2 \\ &= 1.5 (16) + 8 \\ &= 32 \pm 2\end{aligned}$$

Non-AG metabolic acidosis

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Case 3

- Must consider RTA given RA history/kidney stones
- UAG should be checked
 - Urine Na + K - Cl
 - Should be negative due to acidosis
 - If positive, suggests RTA
- Acid challenge with NH_4Cl should be done
 - Urine pH will remain >5.3 after NH_4Cl
 - Type I RTA cannot acidify urine

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Case 4

75-year-old man has a long-standing history of severe COPD for which he requires chronic oxygen therapy.

Serum electrolytes show:

Na 140, K 4.0, Cl 94, HCO_3^- 34

An ABG is done:

pH 7.32, pCO_2 69mmHg

What is the acid-base disorder?

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Case 4

140	94	
4.0	34	

- pH = 7.32 (acidosis)
- $\text{PCO}_2 = 69 \rightarrow$ respiratory acidosis
- $\text{HCO}_3^- = 34 \rightarrow$ metabolic compensation
- This is chronic
- Expected $\Delta[\text{HCO}_3^-] = 3.5 * \Delta\text{pCO}_2/10$
- $\Delta\text{pCO}_2 = 69 - 40 = 29$
- Expected $\Delta[\text{HCO}_3^-] = 3.5 * 29/10 = 10$
- Actual $\Delta[\text{HCO}_3^-] = 34 - 24 = 10$

Chronic respiratory acidosis

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Case 5

A 50-year-old man is found obtunded and poorly responsive.

An arterial blood gas is drawn showing:

pH 7.52, pCO_2 47mmHg

Electrolytes show:

Na 140, Cl- 96; HCO_3^- 34

What is the acid-base disorder?

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Case 5

- pH = 7.52 (alkalosis)
- $\text{HCO}_3^- = 34$ (high) → metabolic alkalosis
- $\text{PCO}_2 = 47$ (high) → respiratory compensation
- $\Delta\text{pCO}_2 = 0.7 * (\Delta[\text{HCO}_3^-])$
- $\Delta\text{PCO}_2 = 47 - 40 = 7$
- $\Delta[\text{HCO}_3^-] = 34 - 24 = 10$
- Expected $\Delta\text{PCO}_2 = 0.7 * (10) = 7$

Pure metabolic alkalosis

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Case 5

- Cause?
 - Contraction alkalosis, hypokalemia, diuretics, vomiting, hyperaldosteronism, antacid use
- Need to know volume status
 - Reduced in contraction, diuretics, vomiting
- Need to know urinary chloride
 - Low with GI losses (vomiting)
- This disorder often fluid (saline) responsive

Case 6

A 59-year-old man with a history of alcoholism and depression presents with altered mental status. He was found by his ex-wife sleeping in his tool shed. He reports blurry vision and black spots.

An arterial blood gas is drawn showing:

pH 7.30, pCO₂ 28mmHg

Electrolytes show:

Na 141, Cl⁻ 102; HCO₃⁻ 14

What is the acid-base disorder?

Case 6

141	102	
	14	

- pH = 7.30 (acidosis)
- HCO₃⁻ = 14 → metabolic acidosis
- pCO₂ = 28mmHg → respiratory compensation
- Expected PCO₂ = 29 +/- 2
- No secondary respiratory disorder
- AG = 141 - 102 - 14 = 25 (high)
- ΔAG = 25 - 12 = 13; ΔHCO₃⁻ = 24 - 14 = 10
- ΔΔ = 13/10 = 1.3
- No secondary metabolic disorder

$$\begin{aligned} \text{pCO}_2 &= 1.5 (\text{HCO}_3^-) + 8 \pm 2 \\ &= 1.5 (14) + 8 \\ &= 29 \pm 2 \end{aligned}$$

AG metabolic acidosis

Case 7

A 50-year-old man with diabetes presents to the emergency room with confusion. His wife says he has been thirsty and urinating frequently. In addition, he takes narcotics for back pain and she believes he has been taking more pills than usual lately for abdominal pain.

An arterial blood gas is drawn showing:

pH 7.28, pCO₂ 40mmHg.

Electrolytes are:

Na 134, K 3.5, Cl 94, HCO₃⁻ 12

What is the acid-base disorder?

Case 7

134	94	
3.5	12	

- Diabetic, polyuria, polydipsia, abd pain → DKA
- Expect AG metabolic acidosis
- Narcotic use
 - Possible respiratory depression
 - Respiratory acidosis
- pH = 7.28 → acidosis
- HCO₃⁻ = 12 (low) → metabolic acidosis
- pCO₂ = 40 (normal) → NO respiratory compensation
- Anion gap = 134 - 94 - 12 = 28 (high)

Case 7

134	94	
3.5	12	

- Winter's formula pCO₂ = 26
- pCO₂ higher than expected at 40
- Concomitant respiratory acidosis

$$\begin{aligned} \text{pCO}_2 &= 1.5 (\text{HCO}_3^-) + 8 \pm 2 \\ &= 1.5 (12) + 8 \\ &= 26 \pm 2 \end{aligned}$$

AG metabolic acidosis with respiratory acidosis

Case 8

A 60-year-old woman presents to the emergency room with a massive vomiting for 3 days. On exam, she is hypotensive and tachycardic. Skin turgor is diminished.

An arterial blood gas is drawn showing:

pH 7.24, pCO₂ 24 mmHg.

Electrolytes are:

Na 140, K 3.2, Cl 79, HCO₃⁻ 10

What is the acid-base disorder?

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Case 8

140	79	
3.2	10	

- Vomiting → metabolic alkalosis
- Dehydration → Possible lactic acidosis
- pH = 7.24 → acidosis
- HCO₃⁻ = 10 (low) → metabolic acidosis
- pCO₂ = 24 (low) → respiratory compensation
- Abnormal same direction → mixed disorder less likely
- Anion gap = 140 - 79 - 10 = 51 (high)

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Case 8

140	79	
3.2	10	

- Winter's Formula pCO₂ = 23 +/- 2
- Actual pCO₂ = 24
- Normal respiratory compensation $pCO_2 = 1.5 (HCO_3^-) + 8 \pm 2$
= 1.5 (10) + 8
= 23 +/- 2
- ΔAG = 51 - 12 = 39
- ΔHCO₃⁻ = 24 - 10 = 14
- ΔΔ = 39/14 = 2.8
- Concurrent metabolic alkalosis

AG Metabolic Acidosis with metabolic alkalosis

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Summary

- Diarrhea non-AG metabolic acidosis
- Acute respiratory acidosis
- Renal tubular acidosis - Urine anion gap
- Chronic respiratory acidosis
- Metabolic alkalosis - Volume status/urine chloride
- Methanol toxicity
- AG metabolic acidosis with respiratory acidosis
 - Winter's formula doesn't match compensation
- AG Metabolic acidosis with metabolic alkalosis
 - Delta-delta abnormal

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Electrolyte Disorders

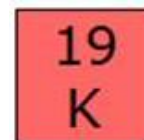
Electrolytes

Jason Ryan, MD, MPH

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Potassium

- Needed for **HEART** and **SKELETAL MUSCLES**
- Hypo/hyper effects:
 - EKG changes
 - Arrhythmias
 - Weakness

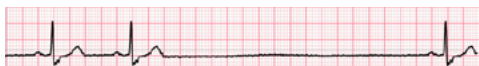


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Hyperkalemia

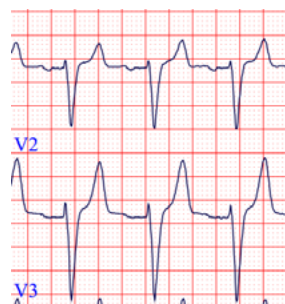
Signs/Symptoms

- **Arrhythmias**
 - Sinus arrest
 - AV block
- **Muscle weakness** → paralysis
 - Lower extremities → trunk → upper extremities
- EKG changes
 - Peaked T waves
 - QRS widening



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Peak T waves



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QRS Widening



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Hyperkalemia

Causes

- Most cases: ↓ K excretion in urine
 - **Acute and chronic kidney disease**
- Need **aldosterone** → renal secretion K⁺
 - Type IV RTA (aldosterone resistance)

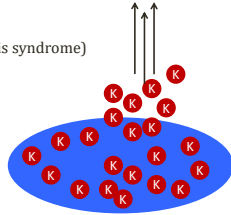


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Hyperkalemia

Causes

- **Increased K release from cells**
 - Acidosis
 - Insulin deficiency
 - Beta blockers
 - Digoxin
 - Lysis of cells (tumor lysis syndrome)
 - Hyperosmolarity



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Hypokalemia

Signs/Symptoms

- Arrhythmias
 - PACs, PVCs
 - Bradycardia
- **Muscle weakness** → paralysis
 - Lower extremities → trunk → upper extremities
- EKG changes
 - U waves
 - Flattened T waves

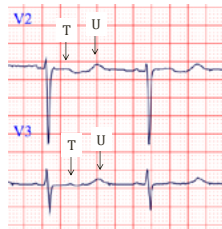
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U Waves/Flat T waves

U Wave: Origin unclear

May represent repolarization of Purkinje fibers

Can be normal

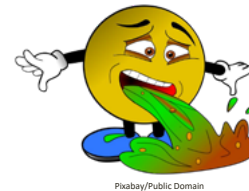


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Hypokalemia

Selected Causes

- Increased renal losses
 - **Diuretics**
 - Type I and II RTAs
- Increased GI losses
 - **Vomiting/diarrhea**



Pixabay/Public Domain

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Hypokalemia

Selected Causes

- Increased K entry into cells
 - Hyperinsulinemic states
 - Beta agonists: albuterol, terbutaline, dobutamine
 - Alkalosis
- **Hypomagnesemia**
 - Promotes urinary K loss
 - Cannot correct K until Mg is corrected!!

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Hypercalcemia

Symptoms

- Often asymptomatic
- May cause **recurrent kidney stones**
- Acute hypercalcemia → **polyuria**
 - Nephrogenic diabetes insipidus
 - Loss of ability to concentrate urine
 - Downregulation of aquaporin channels
 - Excessive free water excretion
 - ↓ GFR → acute renal failure

20
Ca

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Hypercalcemia

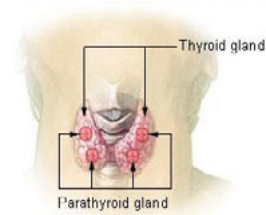
Symptoms

- Stones (kidney)
 - Polyuria
 - Kidney stones
 - Renal failure
- Bones (bone pain)
- Groans (abdominal pain)
 - Constipation, anorexia, nausea
- Psychiatric overtones
 - Anxiety, altered mental status

Hypercalcemia

Selected Causes

- **Hyperparathyroidism**
- **Malignancy**



Wikipedia/Public Domain

Hypercalcemia

Selected Causes

- **Hypervitaminosis D**
 - Massive consumption calcitriol supplements
 - **Sarcoidosis**: Granulomatous macrophages 1 α -hydroxylase
- **Milk alkali syndrome**
 - Largely historical (milk/bicarb)
 - High intake calcium carbonate (ulcers)
 - Excess calcium and alkali intake
 - Hypercalcemia
 - Metabolic alkalosis
 - Renal failure



Wikimediacommons

Hypocalcemia

Signs/Symptoms

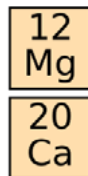
↓ Ca

- **Tetany**
 - Muscle twitches
 - Calcium blocks Na channels in neurons
 - Low Ca $\rightarrow$ easy depolarization $\rightarrow$ spontaneous contractions
 - High Ca $\rightarrow$ difficult depolarization $\rightarrow$ weakness
 - Hyper-excitability of neurons and motor endplates
 - Trousseau's sign: Hand spasm with BP cuff inflation
 - Chvostek's sign: Facial contraction with tapping on nerve
- **Seizures**

Hypocalcemia

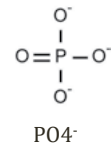
Selected Causes

- Hypoparathyroidism ($\downarrow$ PTH)
- Renal failure ($\downarrow$ active vitamin D)
- Pancreatitis (saponification of Mg/Ca in necrotic fat)
- Drugs (Foscarnet)
- **Magnesium**: Hypo/Hypermagnesemia



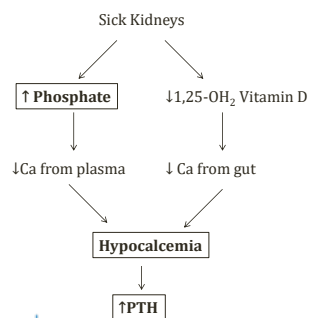
Hyperphosphatemia

Selected Causes



- **Acute and chronic kidney disease**
- Hypoparathyroidism
- Huge phosphate load
 - Tumor lysis syndrome
 - Rhabdomyolysis
 - Large amount of phosphate laxatives (Fleet's Phospho-soda)

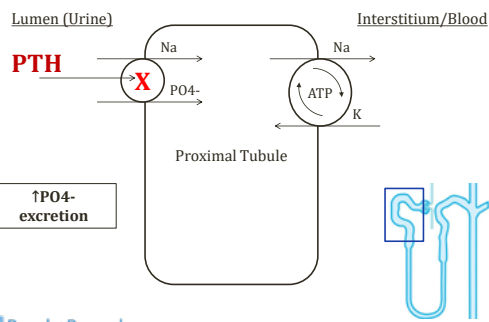
Calcium-Phosphate in Renal Failure



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Hyperphosphatemia

Hypoparathyroidism



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Hyperphosphatemia

Symptoms

- Most patients asymptomatic
- Signs and symptoms usually from **hypocalcemia**
- Phosphate precipitates serum calcium

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Hyperphosphatemia

Symptoms

- **Metastatic calcifications**
 - "Calciophylaxis"
 - Seen in chronic hyperphosphatemia in CKD
 - Excess phosphate taken up by vascular smooth muscle
 - Smooth muscle osteogenesis
- **Vascular wall calcification**
 - Increased systolic blood pressure
 - Small vessel thrombosis
 - Painful nodules, skin necrosis



Niels Olsen/Wikipedia

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Hypophosphatemia

Symptoms

- Main acute symptom is **weakness**
 - ATP depletion
 - Often presents are **respiratory muscle weakness**
- If chronic: bone loss, osteomalacia

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Hypophosphatemia

Selected Causes

- Primary hyperparathyroidism
- **Diabetic ketoacidosis (DKA)**
 - Glucose induced diuresis → ↑ P₀₄ excretion
- **Refeeding syndrome in alcoholics**
 - Low P₀₄ from poor nutrition
 - Food intake → metabolism → further ↓ P₀₄
- Antacids
 - Ammonium hydroxide
- Urinary wasting
 - Fanconi Syndrome

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Hypermagnesemia

Signs/Symptoms

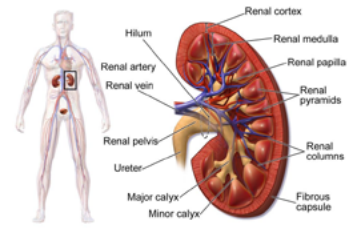
- **Mg blocks Ca and K channels**
- Neuromuscular toxicity
 - ↓ reflexes
 - Paralysis
- Bradycardia, hypotension, cardiac arrest
- Hypocalcemia (inhibits PTH secretion)

$\uparrow \text{Mg} \rightarrow \downarrow \text{PTH} \rightarrow \downarrow \text{Ca}$

Hypermagnesemia

Selected Causes

- **Renal insufficiency**



Kidney Anatomy

Image courtesy of BruceBlais

Hypomagnesemia

Symptoms

- Neuromuscular excitability
 - Tetany, tremor
- Cardiac arrhythmias
- **Hypocalcemia**
- **Hypokalemia**

Hypomagnesemia

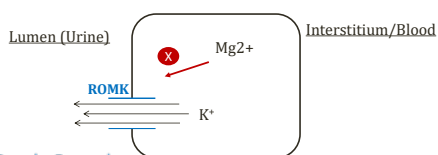
Parathyroid Gland

- Low Mg
 - ↑ PTH release (same effect as calcium)
 - ↑ GI and renal magnesium along with calcium
- Very low Mg → inhibits PTH release
 - Some Mg required for normal CaSR function
 - Abnormal function → suppression of PTH release
 - **Hypocalcemia often seen in severe hypomagnesemia**

Hypomagnesemia

Potassium

- Magnesium inhibits potassium excretion
- **ROMK**
 - Renal outer medullary potassium channel
 - Found in cortical collecting duct
- **K+ won't correct until Mg2+ corrected**



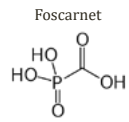
Hypomagnesemia

Selected Causes

- GI losses (secretions contain Mg)
 - Diarrhea
- Pancreatitis
 - Saponification of Mg/Ca in necrotic fat
- Renal losses
 - Loop and thiazide diuretics
 - **Alcohol abuse** (alcohol-induced tubular dysfunction)
- Drugs
 - **Omeprazole (impaired absorption)**
 - Fosarnet

Foscarnet

- Antiviral pyrophosphate analog
- Binds/inhibits viral DNA polymerase
- Adverse effects
 - Nephrotoxicity (limiting side effect)
 - Seizures (often related to electrolytes)
 - Hypocalcemia (chelates calcium)
 - Hypomagnesemia (induces renal wasting of magnesium)
 - Hypokalemia
 - Hypophosphatemia
 - Hypercalcemia
 - Hyperphosphatemia



Sodium and Water Balance

Sodium and Water Balance

Jason Ryan, MD, MPH



Balance

- Water in = water out → “water balance”
- Sodium in = sodium out → “sodium balance”
- Major regulators:
 - Antidiuretic hormone (ADH)
 - Sympathetic nervous system (SNS)
 - Renin-angiotensin-aldosterone system (RAAS)



Effective Circulating Volume

- Portion of extracellular fluid
- Contained in arterial system
- Maintains tissue perfusion
- Not necessarily correlated with total body water



Effective Circulating Volume

- Modified by:
 - **Volume**
 - **Cardiac output**
 - **Vascular resistance**
- Major determinant: **sodium**
 - Excess sodium → ↑ ECV
 - Restricted sodium → ↓ ECV



Effective Circulating Volume

- Low ECV can lead to **low blood pressure**
- May cause **orthostatic hypotension**
- Dizziness/fainting on standing



Effective Circulating Volume

- Low ECV activates:
 - **Sympathetic nervous system**
 - **Renin-angiotensin-aldosterone system**
- Retention of sodium/water



Effective Circulating Volume

- Some disease states have chronically ↓ ECV
- Chronic activation of SNS and RAAS
- Chronic retention of sodium/water by kidneys
- May or may not lead to increased total body water

Condition	ECV	TBW
Volume Depletion	↓	↓
Heart Failure (low CO)	↓	↑
Cirrhosis (low SVR)	↓	↑

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Antidiuretic Hormone

ADH; Vasopressin

- Retention of free water
- Major physiologic trigger is **plasma osmolality**
 - Sensed by hypothalamus
 - ADH released by posterior pituitary gland
 - ADH → free water resorption by kidneys
 - Water retention adjusted to maintain normal osmolality

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Antidiuretic Hormone

ADH; Vasopressin

- Also released with **low ECV**
 - "Non-osmotic release" of ADH
 - Second trigger in addition to serum osmolality
 - Only activated with very low ECV

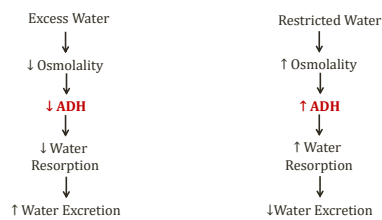
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Water Balance

- Plasma sodium maintained at ~ 140meq/L
- Water intake → water excretion → normal sodium
- Water balance maintained by ADH
- ADH → retention of excess free water
- Water balance reflected by **plasma sodium**
 - Normal sodium: In = Out (in balance)
 - Hyponatremia: In > Out
 - Hypernatremia: In < Out

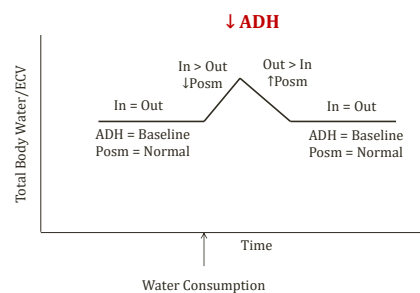
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Water Balance



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Water Balance

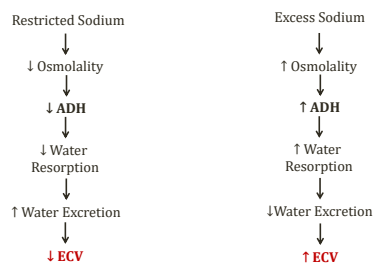


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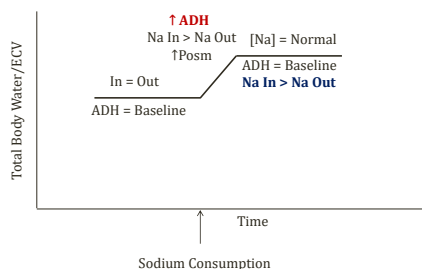
Sodium Balance

- Plasma sodium maintained at $\sim 140\text{meq/L}$
- Excess sodium $\rightarrow \uparrow$ osmolality
- $\uparrow$ osmolality $\rightarrow$ water retention $\rightarrow$ normal sodium
- Water retention $\rightarrow \uparrow$ ECV
- **Sodium intake expands ECV**

Sodium Balance



Sodium Balance



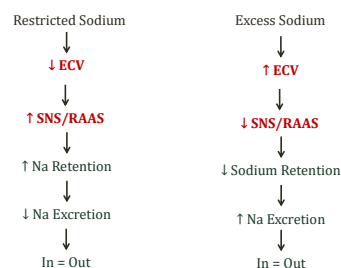
Sodium Balance

- ECV controlled by SNS and RAAS
 - Sympathetic nervous system
 - Renin-angiotensin-aldosterone system
- Activated when ECV is low
- Inhibited when ECV is high
- Sodium alters ECV $\rightarrow$ alters SNS/RAAS

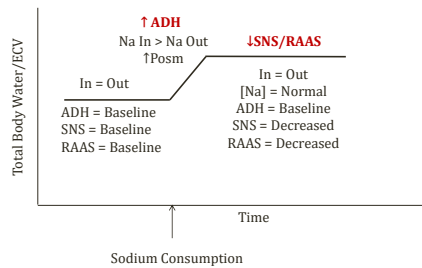
Sodium Balance

- Sodium intake $\rightarrow$ Expanded ECV
- Expanded ECV $\rightarrow \downarrow$ SNS and $\downarrow$ RAAS
- Result: Increased sodium excretion
- $\text{Out} = \text{In} \rightarrow$ balance restored

Sodium Balance



Sodium Balance



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Sodium Balance

Key Points

- High sodium intake expands ECV
 - Weight gain
 - May cause hypertension
- Low sodium intake contracts ECV
 - Weight loss
 - May improve hypertension

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Out of Balance

- Lack of water balance
 - Alters **plasma sodium level**
 - Hypo or hypernatremia
- Lack of sodium balance
 - Alters **total body volume/ECV**
 - Hypo or hypervolemia

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GI Losses

- Nausea, vomiting, diarrhea
- Activation of SNS/RAAS
- **Volume loss → ↑ ADH release**
 - Non osmotic release of ADH
 - Driven by volume sensors
 - No longer controlled by plasma sodium level
- Water balance control by ADH lost
- Free water always retained by kidneys
- Plasma sodium determined by relative intake/losses

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GI Losses

- **Hyponatremia** often occur
 - Drinking free water
 - Not eating (no sodium)
- Hypernatremia can occur
 - Not taking enough free water

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Heart Failure

- Chronically ↓ ECV (low cardiac output)
- Chronic activation of SNS and RAAS
- Sodium chronically retained
- Free water also retained to balance sodium

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Heart Failure

- **Sodium balance disrupted**
- Sodium excretion always reduced
- High sodium intake → intake > excretion
- **Hypervolemia** often occurs

Heart Failure

- **ECV does not ↑ normally with fluid retention**
 - Failing heart unable to increase CO
 - Heart failure patients always have low ECV
- Result: **Congestion**
 - Pulmonary edema
 - Elevated jugular venous pressure
 - Pitting edema

Heart Failure

- **Water balance disrupted**
- **↓ ECV → ↑ ADH release**
 - ADH always high
 - Driven by volume sensors ("non-osmotic")
 - No longer controlled by plasma sodium level
- Water balance control by ADH lost
- Free water always retained by kidneys
- Plasma sodium determined by relative intake/losses
- **Hyponatremia** often occurs

SIADH

Syndrome of Inappropriate ADH Secretion

- Excessive ADH release
- Excess water retention → hyponatremia
- Normal **volume status**
 - Water retention → ↑ ECV → ↓ SNS/RAAS
 - Sodium excretion → ↓ ECV (back to normal)
- Key findings
 - Hyponatremia
 - Normal volume status
 - Concentrated urine

Sodium Disorders

Sodium Disorders

Jason Ryan, MD, MPH

Boards&Beyond

Sodium Disorders

- In general, these are disorders of **WATER** not sodium
- Hyponatremia
 - Too much water
- Hypernatremia
 - Too little water



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Sodium Symptoms

- Hypo and hypernatremia effect **brain**
- Low sodium = low plasma osmotic pressure
 - Fluid into tissues
 - Brain swells
- High sodium = high plasma osmotic pressure
 - Fluid out of tissues
 - Brain shrinks

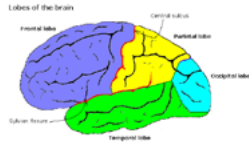


Image courtesy of RobinH

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Hyponatremia

Symptoms

- **Malaise, stupor, coma**
- Nausea

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Hyponatremia

Key Diagnostic Tests

- Plasma osmolality
- Urinary sodium
- Urinary osmolality

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Plasma Osmolality

- Amount of solutes present in plasma
- Key solute: **Sodium**
- Osmolality should be **LOW** in **HYPONATREMIA**
- 1st step in hyponatremia is to make sure it's low

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Plasma Osmolality

$$\text{Serum Osmolality} = 2 * [\text{Na}] + \frac{\text{Glucose}}{18} + \frac{\text{BUN}}{2.8}$$

Normal = 285 (275 to 295)

Plasma Osmolality

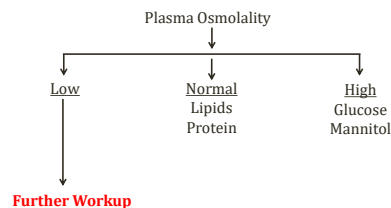
- Hyponatremia with **HIGH** osmolality
 - Hyperglycemia or mannitol
 - Glucose or mannitol = osmoles
 - Raise plasma osmolality
 - Water out of cells → hyponatremia

Plasma Osmolality

- Hyponatremia with **NORMAL** osmolality
 - Artifact in serum Na measurement
 - Hyperlipidemia
 - Hyperproteinemia (multiple myeloma)
 - "Pseudohyponatremia"

Plasma Osmolality

- 1st step in evaluation of hyponatremia unknown cause



Urinary Osmolality

- Concentrations of all osmoles in urine (Na, Cl, K, Urea)
- Varies with water ingestion and urinary concentration
- Low Uosm = dilute urine (lots of free water in urine)
- High Uosm = concentrated urine (little free water)

Urinary Sodium

- Usually > 20meq/L
- Varies with dietary sodium and free water in urine
- Usually high when urine osmolality is high
- Key exception:
 - **Sympathetic nervous system (SNS)**
 - **Renin-angiotensin-aldosterone system (RAAS)**
 - Activation → low urinary sodium with ↑Uosm
 - Hemorrhage, heart failure, cirrhosis

Antidiuretic Hormone

ADH; Vasopressin

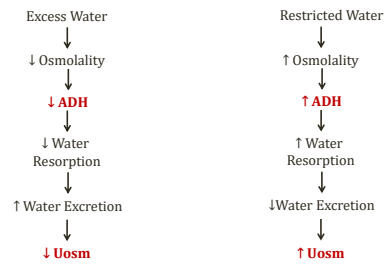
- Osmolality sensed by hypothalamus
- ADH released by posterior pituitary gland
- ADH → free water resorption by kidneys

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Antidiuretic Hormone

ADH; Vasopressin

- Responds to water intake to maintain sodium levels



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Antidiuretic Hormone

ADH; Vasopressin

- Any cause of high ADH can cause hyponatremia
- Sodium no longer controlled by ADH (always high)
- Plasma free water varies with intake
- Increased intake → hyponatremia

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Hyponatremia

General Points

- Urine should be diluted
 - More free water than solutes
 - Low urine osmolality (<100mosm/kg)
 - Low urinary sodium (<30meq/L)

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Hyponatremia

General Points

- If urine is diluted
 - Kidneys responding appropriately
 - ADH level is low (as it should be)
 - Problem is outside the kidneys
- If urine is not diluted
 - Kidneys are NOT responding appropriately
 - Too much ADH
 - Or drugs/pathology interfering with kidney function

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Hyponatremia

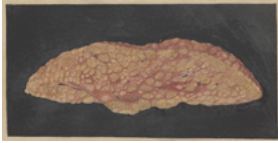
Causes

1. Heart failure and Cirrhosis
2. Kidneys ineffective
3. High ADH
4. Psychogenic polydipsia/Dietary

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Heart failure and Cirrhosis

- Perceived hypovolemia → ADH levels high
- Urine not diluted ($U_{osm} > 100$)
- **Clinical signs of hypervolemia**



Wellcome Images



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Kidneys ineffective

- Advanced **renal failure**
 - Kidneys cannot excrete free water normally
 - Urine cannot be diluted
 - Minimum U_{osm} rises even with low ADH
 - Normal < 100
 - Greater than 200 to 250 mosmol/kg with renal failure
 - Key point: **↑ U_{osm}** indicates abnormal response to ↓ Na
- May occur with euvolemia or hypervolemia

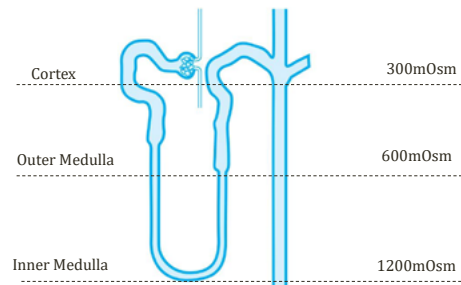
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Kidneys ineffective

- **Diuretics**
 - Cause sodium and water loss
 - Most commonly **thiazides**
 - Can occur with loop diuretics
- Highly variable urinary findings
 - ↑ sodium and water excretion
 - Dehydration → ↑ ADH
 - Water/Na in urine vary by dose, dietary intake
 - Key test: **Response to discontinuation of drugs**

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Diuretics



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Diuretics

- Loop diuretics
 - Medullary gradients **diminished**
 - Difficult to reabsorb free water (loops = powerful diuretic)
 - Low likelihood of excess water → hyponatremia
- Thiazide diuretics
 - Medullary gradients **intact**
 - Intact ability to absorb free water
 - More sodium out in urine (diuretic effect)
 - Higher likelihood of excess water → hyponatremia

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High ADH

- Any cause of dehydration → ↑ ADH
 - **Vomiting, diarrhea**
 - **Sweat**
- Sodium level varies with water intake
- Free water intake → hyponatremia

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High ADH

- **Adrenal insufficiency**
 - Cortisol normally suppresses ADH release
 - Loss of cortisol (primary/secondary) → ↑ ADH
 - Loss of aldosterone (primary) → loss of salt/water → ↑ ADH
- **Hypothyroidism**
- SIADH

SIADH

Syndrome of Inappropriate Antidiuretic Hormone Secretion

- Too much ADH released (inappropriate)
- Causes hyponatremia
- High urinary Na ($>40\text{meq/L}$)
- High urinary osmolality ($>100\text{ mOsm/kg}$)
- No other cause for high ADH
 - Heart failure
 - Cirrhosis
 - Dehydration
 - Thyroid/adrenal disease

SIADH

Causes

- Drug induced (carbamazepine, cyclophosphamide)
- Paraneoplastic (small cell lung cancer)
- CNS
- Pulmonary disease

Volume Status SIADH

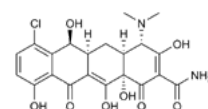
- Fluid retention due to ADH
- Body responds with **↓RAAS**
- ↓ aldosterone → ↑Na in urine (worsens hyponatremia)
- ↓ aldosterone → ↓ water resorption by kidneys
- Result: **normal volume status**

SIADH

- Diagnostic Criteria
 - Hypotonic hyponatremia ($\downarrow P_{\text{osm}}, \downarrow \text{Na}$)
 - Normal liver, renal, cardiac function
 - Clinical euvolemia
 - Normal thyroid, adrenal function
 - Urine osmolality $> 100\text{ mOsm/kg}$

SIADH

- Common treatment: fluid restriction
- Special treatment option:
 - **Demeclocycline**
 - Tetracycline antibiotic
 - ADH antagonist



Demeclocycline

Psychogenic Polydipsia

- Need to drink >18L/day to get hyponatremia
- Occurs in **psychiatric patients** (compulsive)
- Hyponatremia
- **Low urine osmolality** (<100mosm/kg)
 - Indicates kidneys working
 - Kidneys trying to eliminate free water
- Water restriction resolves hyponatremia

Special Diets

- Tea and toast
- Beer drinkers ("beer potomania")
- Very little sodium ingestion
- Minimum urine osmolality ~60 mosmol/kg
- Minimal sodium intake may limit free water excretion
- **Free water intake > output**
- Result: hyponatremia

Special Diets

- Salt consumed must equal salt excreted
- Imagine highest water to salt ratio is 10:1



Normal Diet
3 salt
30 water



Restricted Diet
1 salt
10 water

Special Diets

- Normal diet
 - 1000mOsm/day solute
 - Most dilute urine = 50mOsm/L
 - Max free water output = $1000/50 = 20\text{L/day}$
- Special diet
 - 250mOsm/day solute
 - Most dilute urine = 50mOsm/L
 - Max free water output = $250/50 = 5\text{L/day}$
 - Water intake >5L/day → hyponatremia

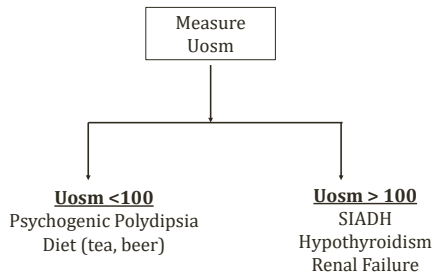
Special Diets

- **Low urine osmolality** (<100mosm/kg)
 - Indicates kidneys working
 - Kidneys trying to eliminate free water
- Free water excretion limited by solute availability

Hyponatremia

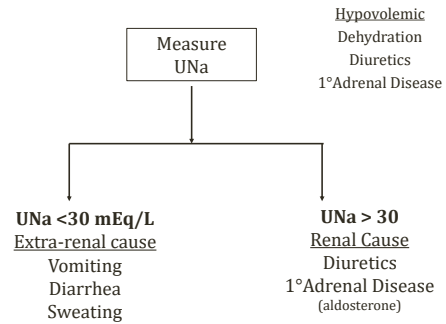
Hypervolemic	Euvolemic	Hypovolemic
Cirrhosis	SIADH	Dehydration
CHF	Hypothyroid	Diuretics
Renal failure	2° Adrenal Disease	1° Adrenal Disease
	Renal failure	
	Polydipsia	
	Dietary	

Euvolemic Hyponatremia



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Hypovolemic Hyponatremia



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↑ ADH and ↑ Uosm

Hypervolemic	Euvolemic	Hypovolemic
Cirrhosis	SIADH	Dehydration
CHF	Hypothyroid	Diuretics
Renal failure	2° Adrenal Disease	1° Adrenal Disease
	Renal failure	
	Polydipsia	
	Dietary	

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↓ ADH and ↓ Uosm

Hypervolemic	Euvolemic	Hypovolemic
Cirrhosis	SIADH	Dehydration
CHF	Hypothyroid	Diuretics
Renal failure	2° Adrenal Disease	1° Adrenal Disease
	Renal failure	
	Polydipsia	
	Dietary	

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↓ ADH and ↑ Uosm

Hypervolemic	Euvolemic	Hypovolemic
Cirrhosis	SIADH	Dehydration
CHF	Hypothyroid	Diuretics
Renal failure	2° Adrenal Disease	1° Adrenal Disease
	Renal failure	
	Polydipsia	
	Dietary	

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Hyponatremia

Treatment

- **Fluid restriction**
- 3% saline
- Vaptan drugs (tolvaptan, lixivaptan, and conivaptan)
 - Block ADH
 - Main use is in severe hyponatremia of heart failure

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Central Pontine Myelinolysis

"Osmotic demyelination syndrome"

- Associated with overly rapid correction ↓Na
 - Usually >10meq per 24 hours
- Demyelination of central pontine axons
- Lesion at base of pons
- Loss of corticospinal and corticobulbar tracts
- Quadriplegia
- Can be similar to locked-in syndrome

Hypernatremia

Symptoms

- **Irritability, stupor, coma**

Hypernatremia

Causes

1. **Water loss**
 - Skin and lungs (more H₂O than Na)
 - ADH will be high
 - Uosm will high
2. Diabetes insipidus
 - **Loss of ADH activity**
 - Central: trauma, tumors
 - Congenital nephrogenic (rare)
 - Acquired (nephrogenic): Many causes

Acquired Diabetes Insipidus

- Hypercalcemia
- Hypokalemia
- Drugs
 - **Lithium**
 - **Amphotericin B**

Diabetes Insipidus

Symptoms

- Polyuria and polydipsia
- Similar to diabetes mellitus via different mechanism

Diagnosis Diabetes Insipidus

- Suspected with polyuria and polydipsia
- Often normal [Na]
 - Water loss stimulates thirst
 - Hypernatremia occurs if not enough water
 - Central lesion (central DI) can impair thirst
- Urine osmolality low (50-200mOsm/kg)

Diagnosis Diabetes Insipidus

Diagnosis

- Fluid restriction
 - After 8 hours of no fluid, urine should be concentrated
 - If urine is dilute → absent/ineffective ADH
- Administration of vasopressin or desmopressin
 - Should concentrate urine if kidneys work
 - If no concentration → nephrogenic DI
 - If concentration → central DI

Hypernatremia

Treatment

- Water (ideally PO)
- IV Fluids (D5W)



Diabetes Insipidus Treatments

- Central DI: **Desmopressin**
 - ADH analog
 - No vasopressor effect (contrast with vasopressin)

Diabetes Insipidus Treatments

- Nephrogenic DI: **Thiazides and NSAIDs**
- Thiazides
 - Increase in proximal Na/H₂O reabsorption
 - Less H₂O delivery to collecting tubules
 - Paradoxical antidiuretic effect
- NSAIDs
 - Inhibit renal synthesis of prostaglandins (ADH antagonists)

Glomerular Disease Principles

Glomerular Disease Principles

Jason Ryan, MD, MPH

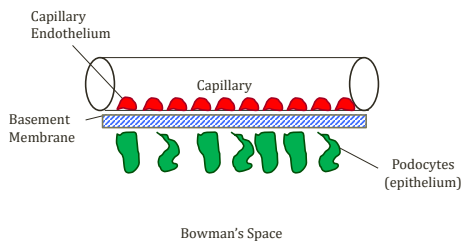


Glomerulus Functions

- Allow “ultrafiltrate” into Bowman’s space
 - Water, electrolytes, glucose, amino acids
- Prevent filtration of most proteins
- Prevent filtration of red blood cells
- Glomerular pathology
 - **Proteinuria**
 - **Hematuria**



Glomerular Filtration Barrier



Capillary Endothelium

- Fenestrated (i.e. has openings)
- Only small (~40nm) molecules pass through
- Repels red cells, white cells, platelets
- First barrier to filtration
- Capillary damage → RBC in urine → **hematuria**
- Capillary damage → inflammation → **nephritis**



Basement Membrane

- Negatively charged molecules
 - Type IV collagen
 - Heparan sulfate
- Repels (-) molecules like albumin
- Also size barrier



Podocytes

- Also called **epithelial cells**
- Long “processes” called “foot processes”
- Wrap capillaries
- Slits between foot processes filter blood
- Further size barrier small molecules
- Damage → loss of protein barrier



Albumin

- Small (~3.6nm)
- Can fit through all size barriers
- Negatively charged
- Repelled by GBM charge barrier
- Podocyte/GBM disease → albumin in urine

Glomerular Diseases

- Breakdown of components of filtration barrier
- Things in urine that shouldn't be there:
 - Red blood cells
 - Protein (especially albumin)

Hematuria

- Urinalysis
- Dipstick: tests for the presence of heme
 - Heme has peroxidase activity → reacts with strip
 - Heme positive: hemoglobin or myoglobin
- Microscopy: red cells visualized

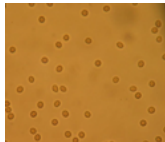


Image courtesy of Bobgalindo



Image courtesy of J3D3

Hematuria

- Many, many causes
- Gross: abnormal color to urine from blood
- Microscopic: Incidental finding on urinalysis
- Can occur after exercise
- Common causes:
 - UTI
 - Kidney stones
- Feared cause: bladder cancer
- Glomerular disease is rare cause

Glomerular Bleeding

- Red cell casts
- Dysmorphic red blood cells
- Acanthocytes
- Proteinuria
- Red, smoky brown or "coca cola"
- Clots generally not seen



Image courtesy of Anwar Siddiqui,

Proteinuria

- Urine dipstick
 - Color change indicates amount of protein
 - Primarily detects albumin (good for glomerular disease!)
 - 1+, 2+, 3+, 4+
 - Affected by urine concentration

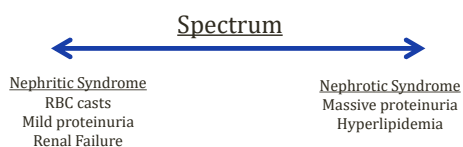
Proteinuria

- Urine protein-to-creatinine ratio
 - "Spot urine"
 - 1st or 2nd morning urine sample after avoiding exercise
 - Normal ratio less than 0.2 mg/mg

Proteinuria

- 24-hour urine collection
 - Gold standard for protein evaluation
 - Gives you grams/day or protein excretion
 - Normal is less than 150 mg/day
 - Cumbersome for patients
 - Errors in collection common

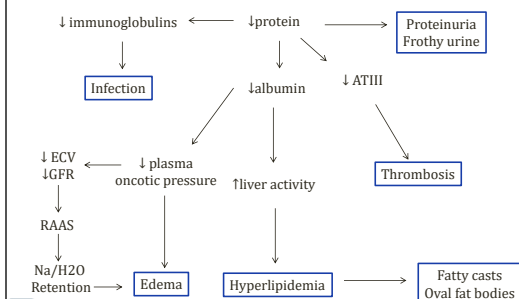
Glomerular Diseases



Nephrotic Syndrome

- Filtration barrier to protein is lost
- RBC filtration barrier remains intact
- Massive proteinuria
 - 4+ on dipstick
 - >3.5g/day
- Triggers cascade of pathology

Nephrotic Syndrome



Urine in Nephrotic Syndrome

- Urinary lipid may be present
- Trapped in casts (fatty casts)
- Enclosed by plasma membrane of degenerative epithelial cells (oval fat bodies)
- Under polarized light fat droplets have appearance of Maltese cross

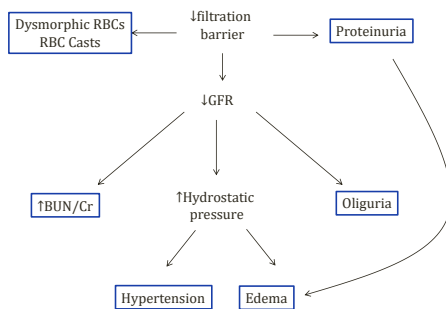
Nephrotic Syndrome

- Classic presentation
 - Frothy urine
 - Swelling of ankles
 - Swelling around eyes (periorbital)
 - Often mistaken for allergic reaction
 - Serum total cholesterol >300mg/dl
 - Proteinuria (>3.5g/day)

Nephritic Syndrome

- Inflammatory process damages entire glomeruli
- Filtration barrier to RBCs and protein lost
- Glomerular damage: ↓GFR
- RBC in urine
 - Dysmorphic
 - RBC Casts
- Protein in urine
 - Less than nephrotic syndrome due to lower GFR
 - <3.5g/day

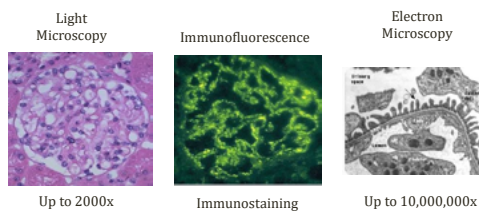
Nephritic Syndrome



Nephritic Syndrome

- Classic presentation
 - Dark urine (RBCs)
 - Swelling
 - Fatigue (uremia)
 - Proteinuria (<3.5g/day)

Microscopy



Microscopy



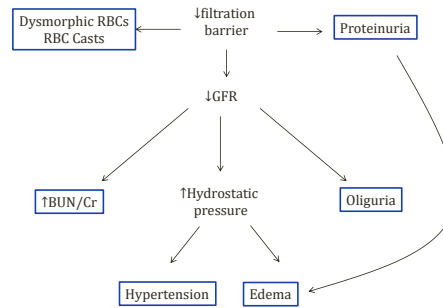
Nephritic Syndrome

Nephritic Syndrome

Jason Ryan, MD, MPH



Nephritic Syndrome



Nephritic Syndrome

- Classic presentation
 - Dark urine (RBCs)
 - Swelling/edema
 - Fatigue (uremia)
 - Proteinuria (<3.5g/day)



Nephritic/Nephrotic

Sites of Glomerular Injury

- Major determinant of whether a disease process leads to nephritic or nephrotic syndrome is the **site of glomerular injury**



Nephritic/Nephrotic

Sites of Glomerular Injury

- Podocyte injury → protein loss only → nephrotic
- **Endothelial and mesangial cells**
 - Exposed to blood elements
 - Injury lead to inflammation (nephritis)
 - Loss of red blood cells and protein in urine
- Most causes of nephritic syndrome related to **endothelial/mesangial injury** with influx of inflammatory cells



Nephritic Syndrome

Major Causes

1. Post-streptococcal
2. Berger's (IgA) nephropathy
3. Diffuse proliferative glomerulonephritis
4. Rapidly progressive glomerulonephritis (RPGN)
5. Alport syndrome
6. Membranoproliferative glomerulonephritis



Post-streptococcal GN

- Follows **group A β -hemolytic strep infection**
 - Impetigo (skin)
 - Pharyngitis
- Nephritogenic strains
 - Carry specific subtypes of M protein virulence factor

Post-streptococcal GN

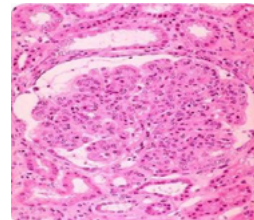
- Immune complexes deposit in kidney
 - Circulating antigen-antibodies complexes
 - In situ formation in kidney
- Fix complement
- Attract PMNs
- **Hypocomplementemia** (also lupus, MPGN)

Post-streptococcal GN

- Common in children (can also occur in adults)
- Classic case
 - Child
 - 2-3 weeks following strep throat infection
 - Nephritic syndrome

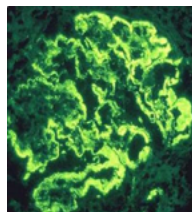
Post-streptococcal GN

- Glomeruli: Enlarged, hypercellular



Post-streptococcal GN

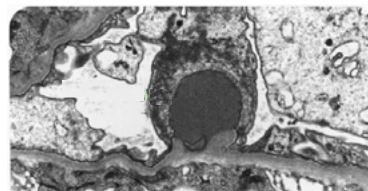
- Subendothelial antibodies/complexes
- Granular IF (IgG, C3)



Images courtesy of bilalbanday

Post-streptococcal GN

- Electron microscopy: **Subepithelial** "humps"
 - Immune complexes



Post-streptococcal GN

- Good prognosis in children
 - 95% recover completely
- Adults have worse prognosis
 - About 60% recover
 - Many develop renal insufficiency
 - Can be late: 10 to 40 years after initial illness
 - Can develop RPGN

Post-streptococcal GN

- No specific therapy (supportive)
- Spontaneous resolution

IgA Nephropathy

Berger's Disease

- Most common form glomerulonephritis worldwide
- Repeated episodes of hematuria (nephritic)
- Over time leads to ESRD and HD (50% patients)

IgA Nephropathy

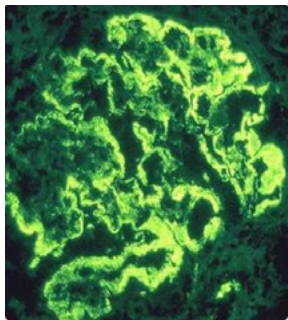
Berger's Disease

- Overactive immune system
- ↑IgA synthesis in response to triggers
 - Respiratory infection
 - GI infection
- IgA immune complexes → **mesangium**
- Activate complement
 - Alternative and lectin pathways
 - No hypocomplementemia
- Glomerular injury occurs

IgA Nephropathy

Berger's Disease

- Granular IF
- Stained for IgA



Images courtesy of bilalbanday

IgA Nephropathy

Berger's Disease

- Classic case
 - Recurrent episodes hematuria since childhood
 - Episodes follow URI or diarrheal illness
 - Slowly worsening renal function (BUN/Cr) over time
 - Possible progression to ESRD and HD (20yrs+)
- Don't confuse with other glomerular disorders
 - Post-strep GN: weeks after infection
 - IgA GN: days after infection
 - Minimal change: nephrotic syndrome after URI

Henoch-Schonlein Purpura

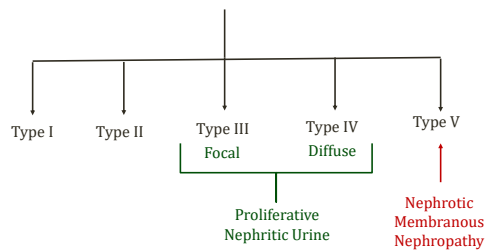
- IgA nephropathy with extra-renal involvement
- Most common **childhood** systemic vasculitis
- Skin: palpable purpura on buttocks/legs
- GI: abdominal pain, melena
- Joint pains
- **Diffuse IgA deposition**
- Tissue biopsy: demonstrates IgA

DPGN

Diffuse proliferative glomerulonephritis

- Systemic lupus erythematosus (SLE)
 - Most common subtype of SLE renal disease
 - "Type IV Lupus Nephritis"
 - Often presents with other SLE features: fever, rash, arthritis
- **Immune complex deposition** in glomeruli
 - IC → inflammatory response

Lupus Nephritis



DPGN

Diffuse proliferative glomerulonephritis

- Diffuse: More than 50% glomeruli affected
- Proliferative:
 - Increase in cellularity of glomeruli
 - Mesangial cells
 - Endothelial cells
 - Monocyte/neutrophil infiltration

DPGN

Diffuse proliferative glomerulonephritis

- **Subendothelial deposits** drive immune response
 - Anti-dsDNA
 - Hypocomplementemia (also post-strep, MPGN)
- Classic finding: **capillary loops thickened**
 - "Wire looping"

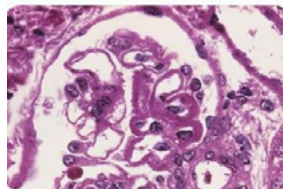


Image courtesy of bilalbanday

DPGN

Diffuse proliferative glomerulonephritis

- **Granular IF**
- "Full house" immunofluorescence
 - IgG, IgA, IgM, C3, C1q

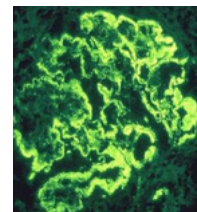


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DPGN

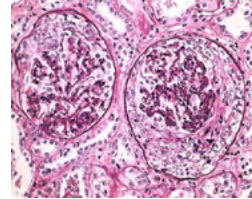
Diffuse proliferative glomerulonephritis

- Mixed clinical presentation
 - Proteinuria (sometimes nephrotic)
 - Hematuria
 - Reduced GFR
- Severe, often leads to ESRD and HD

RPGN

Rapidly progressive glomerulonephritis

- Also called “crescentic” glomerulonephritis
- Pathologic description: Many causes
 - Many diseases lead to this condition

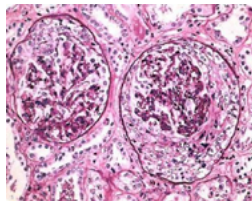


Images courtesy of bilalbanday

RPGN

Rapidly progressive glomerulonephritis

- Crescents formed by **inflammation**:
 - Monocytes/macrophages
 - Fibrin



Images courtesy of bilalbanday

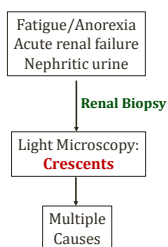
RPGN

Rapidly progressive glomerulonephritis

- Severe form of glomerulonephritis
- Progressive loss of renal function
- Rapid onset
- Often presents as acute renal failure
- Generalized symptoms: fatigue, anorexia

RPGN

Rapidly progressive glomerulonephritis



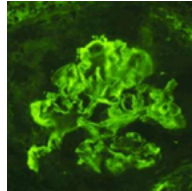
RPGN

Rapidly progressive glomerulonephritis

- Causes distinguished based on immunofluorescence
- Type I: Linear IF
- Type II: Granular IF
- Type III: Negative IF

RPGN Type I

- Anti-glomerular basement membrane antibodies
 - "Anti-GBM antibodies"
- Antibodies against GBM antigens
 - Unknown stimulus
 - Type II hypersensitivity
- **Linear IF**
 - IgG antibodies
 - Linear pattern



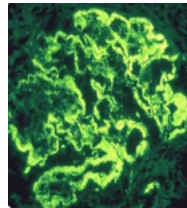
Images courtesy of bilalbanday

Goodpasture's Syndrome

- Antibody to collagen
- Antibodies to **alpha-3 chain** of type IV collagen
 - Found in GBM and alveoli
- Hemoptysis and nephritic syndrome
- Classic case
 - Young adult
 - Male
 - Hemoptysis
 - Hematuria

RPGN Type II

- Immune complex deposition
 - Type III hypersensitivity
- Granular IF



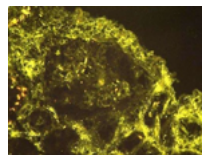
Images courtesy of bilalbanday

RPGN Type II

- **Post-streptococcal glomerulonephritis**
 - Can progress to RPGN
 - Most common cause RPGN
- **Systemic lupus erythematosus (SLE)**
 - Diffuse proliferative glomerulonephritis
 - Can progress to RPGN

RPGN Type III

- Negative IF
 - No staining for IgG, IgA, etc.
- "Pauci-immune"
- Most patients **ANCA positive**
 - c-ANCA or p-ANCA
- Most patients have a **vasculitis syndrome**



Images courtesy of bilalbanday

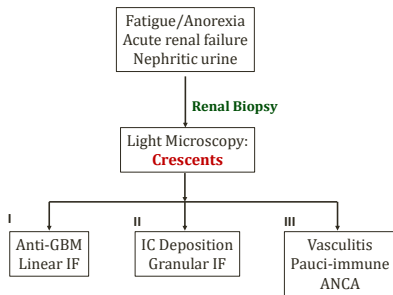
ANCA Diseases

Anti-neutrophil cytoplasmic antibodies

- Wegener's Granulomatosis (c-ANCA)
- Microscopic Polyangiitis (p-ANCA)
- Churg-Strauss syndrome (p-ANCA)
- All can lead to pauci-immune nephritis

RPGN

Rapidly progressive glomerulonephritis



Alport Syndrome

Hereditary Nephritis

- Genetic **type IV collagen** defect
 - Mutations in **alpha-3, alpha-4, or alpha-5** chains
 - Chains found in basement membranes kidney, eye, ear
- Inherited: X-linked
- Classic triad:
 - Hematuria
 - Hearing loss
 - Ocular disturbances
- Look for child with triad and family history

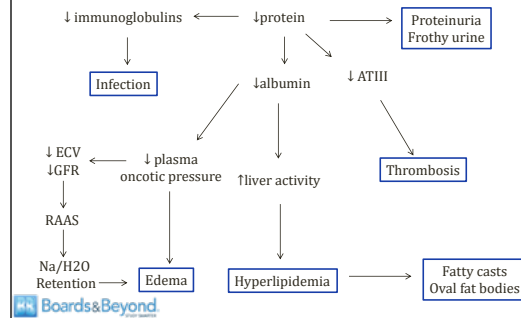
Nephrotic Syndrome

Nephrotic Syndrome

Jason Ryan, MD, MPH

Boards&Beyond

Nephrotic Syndrome



Boards&Beyond

Nephrotic Syndrome

- Classic presentation
 - Frothy urine
 - Swelling of ankles
 - Swelling around eyes (periorbital)
 - Serum total cholesterol >300mg/dl
 - Proteinuria (>3.5g/day)

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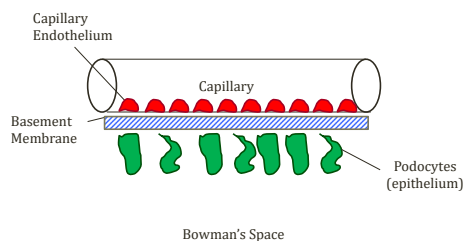
Nephritic/Nephrotic

Sites of Glomerular Injury

- Major determinant of whether a disease process leads to nephritic or nephrotic syndrome is the **site of glomerular injury**
- **Podocytes**
 - Separated from blood by GBM
 - Injury does not lead to inflammation
 - Damage → loss of filtration barrier to protein only
- Most causes of nephrotic syndrome related to injury of podocytes or epithelial side of GBM

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Glomerular Filtration Barrier



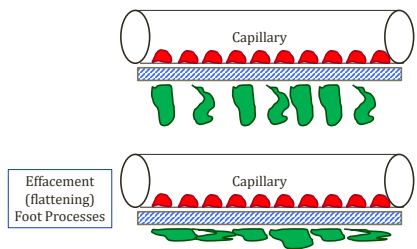
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Nephrotic Syndrome Causes

1. Minimal change disease
2. Focal segmental glomerulosclerosis (FSGS)
3. Membranous nephropathy
4. Diabetic
5. Amyloidosis
6. Membranoproliferative glomerulonephritis

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Minimal Change Disease



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Minimal Change Disease

Pathology

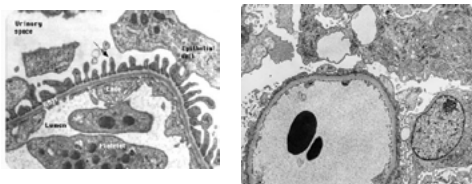
- Caused by **effacement of foot processes**
- Loss of anion (-) charge barrier GBM
- Triggered by **cytokines** → damage to podocytes
- Usually idiopathic
- Associated with Hodgkin Lymphoma

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Minimal Change Disease

Renal Biopsy

- Normal light microscopy
- No important findings IF
- Only finding is effacement foot processes EM



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Images courtesy of bilalbanday

Minimal Change Disease

Other Features

- Sometimes has immunological trigger (days before)
 - **Viral infection (URI)**
 - Allergic reaction (bee sting)
 - Recent immunization
- **"Selective"** proteinuria
 - Only albumin in urine (not immunoglobulin)
 - Contrast with other glomerular disease "non-selective"
- Most common cause nephrotic syndrome in children
 - Classic presentation is a child with recent URI

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Minimal Change Disease

Prognosis and Treatment

- Favorable prognosis
- Responds very well to **steroids**
 - Unique among nephrotic syndrome causes

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FSGS

Focal segmental glomerulosclerosis

- Glomerulosclerosis
 - Pink/dense deposition of collagen in glomerulus
- Segmental
 - Only portion of glomerulus involved
- Focal
 - Only some glomeruli involved

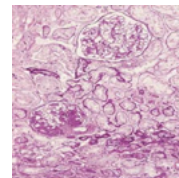


Image courtesy of bilalbanday

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FSGS

Pathology

- Sclerotic segments
 - Collapse of basement membranes
 - Hyaline deposition ("hyalinosis")
- Effacement of foot processes
 - Seen on electron microscopy

FSGS

Renal Biopsy

- Light microscopy: focal, segmental lesions
- Electron microscopy: effacement of foot processes
- Immunofluorescence
 - Usually negative (no immune complexes)
 - Sometimes IgM, C3, C1 (nonspecific finding)

FSGS

Focal segmental glomerulosclerosis

- Caused by **podocyte injury**
- Unknown cause
- Often progresses to chronic renal failure
 - 40-60% within 10 to 20 years
 - Does not respond to steroids
 - Severe version of minimal change disease

FSGS

Epidemiology

- **African Americans**
 - Most common cause nephrotic syndrome

Nephrotic Syndrome Causes

	African-American	Caucasian
FSGS	57%	23%
Membranous	17%	36%
Minimal Change	14%	20%

FSGS

Focal segmental glomerulosclerosis

- Usually idiopathic (primary)
- Many secondary causes

FSGS

Focal segmental glomerulosclerosis

- HIV
- Sickle cell patients
- Heroin users



Psychonaught/Wikipedia

FSGS

Other Associations

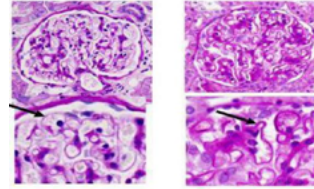
- Massive obesity
- Interferon treatment
 - Used to treat HCV and HBV
 - Some leukemias and lymphomas, melanoma
- Loss of nephrons
 - Single kidney (congenital)
 - Surgical kidney removal



Tibor Vigh

Membranous Nephropathy

- Thick glomerular basement membrane
 - "Membranous"
- Absence of hypercellularity



Normal

Membranous

Membranous Nephropathy

- Membrane thick from **immune complex deposition**
 - IF microscopy very useful
 - "Granular" deposits of IgG and C3 staining

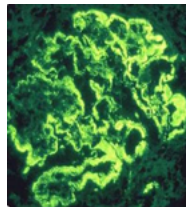
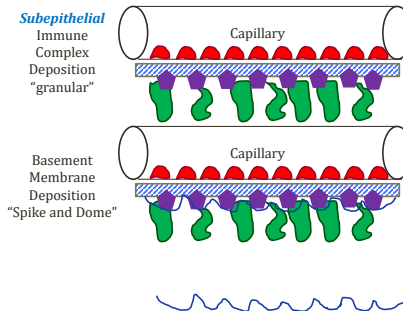


Image courtesy of bilalbanday

Membranous Nephropathy

Pathophysiology



Subepithelial Deposits

"Electron dense deposits"

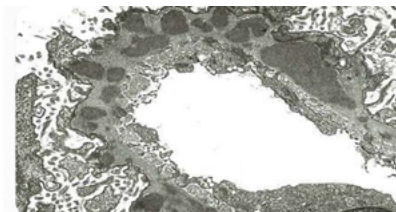
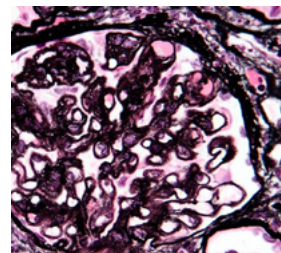


Image courtesy of bilalbanday

Spike and Dome



Nephron/Wikipedia

Membranous Nephropathy

Renal Biopsy

- Light microscopy: capillary/BM thickening
- Electron microscopy: subepithelial deposits
- Immunofluorescence: granular IgG/C3

Membranous Nephropathy

- Often idiopathic
- **Autoantibodies**
- Antigen: **phospholipase A2 receptor (PLA2R)**
- Expressed on podocytes

Membranous Nephropathy

Secondary Causes

- Systemic lupus erythematosus (SLE)
- Most lupus renal disease in nephritic
- Diffuse proliferative glomerulonephritis
- If nephrotic, this is cause (10-15%)



Wikipedia/Public Domain

Membranous Nephropathy

Secondary Causes

- Solid tumors
 - Colon cancer, lung cancer, melanoma
- Infections
 - Hep B, Hep C
- Drugs
 - Penicillamine, gold, NSAIDs
 - All used to treat **rheumatoid arthritis**

**Tumor
Hepatitis
Rheumatoid Arthritis**

Membranous Nephropathy

Other Features

- Most common cause nephrotic syndrome in adults
- Excellent prognosis in children
- Some adults develop ESRD

Nephrotic Syndrome Causes

	African-American	Caucasian
FSGS	57%	23%
Membranous	17%	36%
Minimal Change	14%	20%

Autoantibodies

- Most antibody disorders are nephritic
- IC → inflammation → nephritis → nephritic syndrome
- Membranous is nephrotic
- Subepithelial deposits → nephrotic syndrome



Diabetic Nephropathy

- Non-enzymatic glycosylation
- Basement membranes: leakage of protein
- Long term effect: sclerosis of glomerulus
- Proteinuria
- Can develop nephrotic syndrome

Amyloidosis

- Extracellular buildup of amyloid proteins
- Classic biopsy findings
 - Apple-green birefringence
 - Congo red stain
- Kidney is **most commonly involved organ**

Nephrotic Syndrome Causes

1. Minimal change disease ← Cytokines
2. FSGS ← Podocyte Damage
3. Membranous ← Immune Complexes
4. Diabetic ← Glucose
5. Amyloidosis ← Amyloid
6. Membranoproliferative glomerulonephritis

MPGN

MPGN

Jason Ryan, MD, MPH

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MPGN

Membranoproliferative Glomerulonephritis

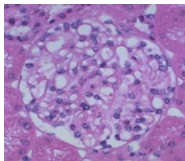
- Rare glomerular disorders
- Can cause nephritic or nephrotic syndrome
- Varying degrees of renal dysfunction
- Renal failure (↑BUN/Cr)
- Hematuria
- Proteinuria (+/- nephrotic range)

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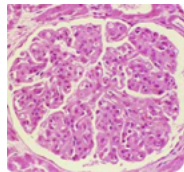
MPGN

Membranoproliferative Glomerulonephritis

- Membrano
 - Thick basement membrane
- Proliferative
 - Proliferation of mesangial cells, mesangial matrix



Normal



MPGN: Hypercellular, Thick walls

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Images courtesy of bilalbanday

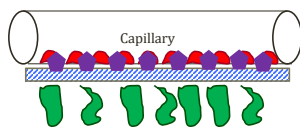
MPGN

Membranoproliferative Glomerulonephritis

- Two major types
- Type I much more common
- Type II (dense deposit disease) rare

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MPGN Type I

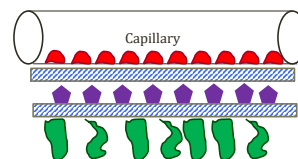


Type I

Subendothelial immune complex deposition
IgG → complement activation

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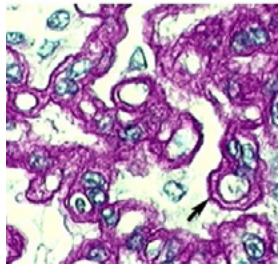
MPGN Type I



IC deposits trigger **mesangial ingrowth**
Splits basement membrane
"Tram track" appearance on light microscopy
Common (80%) in Type I

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MPGN Type I



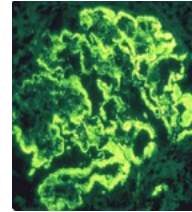
Type I: Tram Tracks

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Image courtesy of Shashidhar Venkatesh Murthy

MPGN Type I

- Subendothelial antibodies/complexes
- Granular IF for IgG and C3



Images courtesy of bilalbanday

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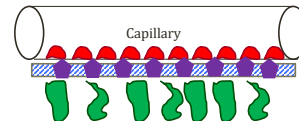
MPGN Type I

- May be idiopathic
- Associated with **hepatitis B and C infection**

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MPGN Type II

Dense Deposit Disease



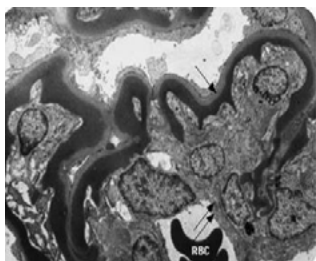
Type II

Basement Membrane
"Electron dense" deposits
Mediated by **complement**
IgG usually absent

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MPGN Type II

Dense Deposit Disease



Type II: Dense Deposits
IF shows C3 but not IgG

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Image courtesy of bilalbanday

C3 Nephritic Factor

C3 Convertase Stabilizing Antibody

- Found in >80% patients with MPGN II
- C3 convertase activates **alternative pathway**
- Stabilized by C3 nephritic factor
- Over activation of complement system
- Hypocomplementemia ($\downarrow$ C3)

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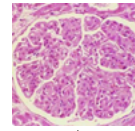
MPGN Type II

Dense Deposit Disease

- Mostly a disease of **children**
- Usually 5 to 15 years old
- 50% develop ESRD within ten years

MPGN

Membranoproliferative Glomerulonephritis



	Type I	Type II
Pathology	Immune Complex	Complement (C3)
Location	Subendothelial	Basement Membrane
Microscopy	LM: Tram Tracks	EM: Dense Deposits
Associations	Hepatitis	Children

Tubulointerstitial Disease

Tubulointerstitial Disorders

Jason Ryan, MD, MPH



Acute Renal Injury

- Acute fall in GFR
- Rise in serum BUN and Creatinine
- Usually reversible
- Common problem in hospitalized patients
- Two major causes:
 - Poor renal perfusion
 - Acute tubular necrosis



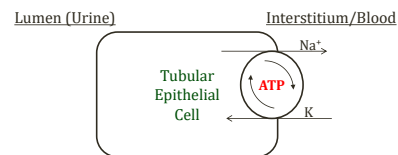
Acute Tubular Necrosis

- Sudden damage to **tubular epithelial cells**
 - Ischemia (ANY cause severe ↓ blood flow)
 - Drugs
 - Toxins



Ischemic ATN

- Ischemia → vasoconstriction → ↓ GFR
- Loss of tubular cell polarity
- Na-K-ATPase moves to luminal side
- More sodium in urine
- Macula densa → vasoconstriction



Ischemic Causes

- Hypovolemia
- Cardiogenic shock
- Massive bleeding
- Any cause of decreased perfusion



Toxin/Drug Causes

- Aminoglycosides
- Contrast dye
- Uric acid (tumor lysis syndrome)
- Myoglobin (rhabdomyolysis)
- Lead
- Cisplatin
- Ethylene glycol (antifreeze)



Acute Tubular Necrosis

- Tubular epithelial cells die, slough off into urine
- Obstructs urine flow → **intrinsic renal failure**
 - ↓ GFR
 - ↑ BUN and Cr
- Epithelial cells form casts in tubules
 - Granular casts
 - **"Muddy brown"**



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Image courtesy of Anwar Siddiqui

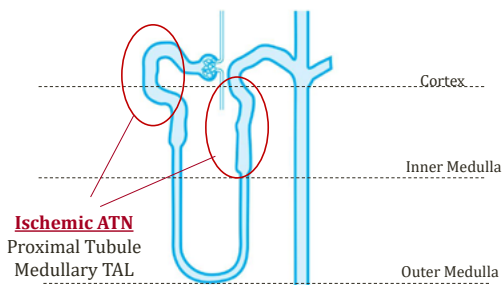
Acute Tubular Necrosis

Pathology

- Patchy, focal necrosis of nephron
- Large skip areas of normal nephron
- Occlusion of tubular lumen by casts
- Ischemic injury
 - Proximal tubule
 - Thick ascending limb
- Toxic injury
 - Proximal tubule

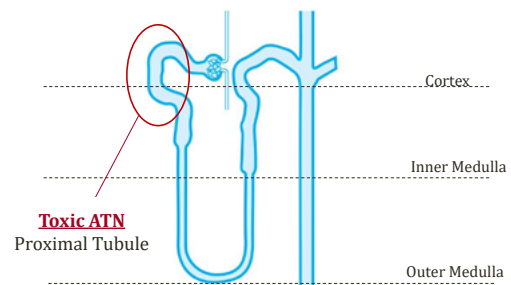
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Acute Tubular Necrosis



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Acute Tubular Necrosis



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ATN Phases

- Phase 1: Injury occurs
 - Slight decline urine output
- Phase 2: Maintenance
 - Oliguria
 - Rising BUN/Cr
 - **Hyperkalemia**
 - AG metabolic acidosis
 - May last weeks
- Phase 3: Recovery
 - **Polyuria**
 - Risk of **hypokalemia**

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Prognosis

- Typical course is kidney recovery
- Tubular cells capable of regeneration
 - "Tubular re-epithelialization"
- Some patients require temporary dialysis

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Acute Interstitial Nephritis

Tubulointerstitial Nephritis

- Inflammation of renal tubules and “interstitium”
 - Space between cells
- Hypersensitivity (allergic) reaction
 - Mediated by eosinophils and neutrophils
- Usually triggered by drugs
- Sometimes infections or autoimmune disease
- Usually resolves with stopping offending agent

Acute Interstitial Nephritis

Tubulointerstitial Nephritis

- Main clinical feature is **renal failure**
- Absence of nephritic/nephrotic syndrome
 - Considered a glomerular disease when occur together

Acute Interstitial Nephritis

Tubulointerstitial Nephritis

- **Drugs – 75% of cases**
 - Sulfonamides (TMP-SMX)
 - Rifampin
 - Penicillin
 - Diuretics (furosemide, bumetanide, thiazides)
 - NSAIDs
- Drugs often act as **haptens**
 - Haptens = illicit immune response when bound
 - Attach to basement membrane/epithelial cells
 - Illicit immune response

Acute Interstitial Nephritis

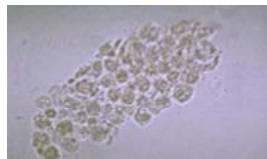
Tubulointerstitial Nephritis

- Infection – 5-10% of cases
 - Multiple organisms reported
 - Legionella, Leptospira, CMV, TB
- Systemic diseases – 5-10% of cases
 - Sarcoidosis
 - Sjögren's syndrome
 - SLE

Acute Interstitial Nephritis

Tubulointerstitial Nephritis

- Classic findings:
 - Exposure to a trigger
 - Fever, rash, malaise
 - Acute renal failure (↑ BUN/Cr)
 - WBC casts (**without** symptoms of cystitis)
 - “Sterile pyuria”
 - Peripheral eosinophilia
 - **Urine eosinophils**



Anwar Siddiqui

Acute Interstitial Nephritis

- Usually resolves with stopping offending agent
- Rarely progresses to **papillary necrosis**

Chronic Interstitial Nephritis

- **Mononuclear cell infiltration**
- **Fibrosis and atrophy of tubules**
- Seen with **longstanding use of NSAIDs**
- Mild elevation of BUN/Cr
- Resolves with stoppage of drugs
- Classic case:
 - Patient on NSAIDs for chronic pain
 - Mild increase BUN/Cr
 - Renal function improves with stoppage of drug

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NSAIDs

- Acute interstitial nephritis (fever, renal failure)
- Chronic interstitial nephritis (renal failure)
- Acute tubular necrosis
 - Ischemia
 - Block PG vasodilation of afferent arteriole
- Membranous glomerulonephritis
 - Nephrotic syndrome
- Papillary necrosis

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Papillary Necrosis

- Coagulative necrosis of renal papillae
- Sloughing of tissue
- **Gross hematuria**
- **Often painless**
- In isolation
 - No intrinsic renal failure
 - No WBC casts
- May obstruct urine flow

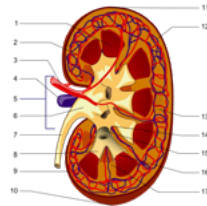


Image courtesy of Piotr Michal Jaworski

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Papillary Necrosis

Classic Causes

- Chronic phenacetin use
- Diabetes
- Acute pyelonephritis
- **Sickle cell anemia**

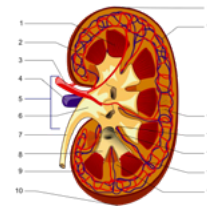


Image courtesy of Piotr Michal Jaworski

Boards&Beyond

Papillary Necrosis

- Typical presentation
 - Patient with typical trigger
 - **Gross hematuria**
 - **Painless**
- Normal renal function

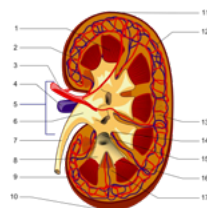


Image courtesy of Piotr Michal Jaworski

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Cortical Necrosis

- Caused by **ischemia of renal cortex**
- Acute onset severe renal failure
- Seen in very sick patients
 - Septic shock
 - Obstetric catastrophes (abruptio placentae, fetal demise)
- Acute renal failure
- Oliguria → anuria

Boards&Beyond

Renal Failure

Renal Failure

Jason Ryan, MD, MPH



Terms

- Acute renal failure
 - Decrease in Cr clearance over days
 - Often associated with symptoms
 - Many causes
- Chronic renal failure (chronic kidney disease)
 - Slow, steady deterioration of renal function (years)
 - Usually due to diabetes, hypertension
 - Symptoms only in most severe stages



Terms

- Azotemia
 - Insufficient filtering of blood by kidneys
- Uremia
 - Azotemia + "uremic" symptoms



Uremic Symptoms

- Anorexia
- Nausea, vomiting
- Platelet dysfunction (bleeding)
- Pericarditis
- Asterixis
- Encephalopathy



Acute Renal Failure

1. Insufficient blood flow to kidneys (pre-renal)
 - Dehydration
 - Shock
 - Heart failure
2. Obstruction of urine outflow (post-renal)
 - Need bilateral obstruction
 - Kidney stones, BPH, tumors, congenital anomalies
3. Renal dysfunction (intrinsic)
 - Acute tubular necrosis
 - Glomerulonephritis



Key Labs

- Creatinine
 - Similar to inulin
 - Freely filtered
 - Small amount of secretion
- Blood urea nitrogen
 - Freely filtered
 - Reabsorbed when kidney reabsorbs water



Key Labs

- In acute renal failure both rise (less filtered)
- In acute renal failure from dehydration:
 - BUN rises more (less filtered, more reabsorbed)

Real Life Acute Renal Failure

- Routine labs on outpatient or inpatient
- BUN/Cr elevated
- Work up:
 - Urinalysis (protein, blood, casts)
 - Ultrasound (hydronephrosis)
 - Careful history (meds, co-morbidities, hydration)
 - Physical exam (low blood pressure, dehydration, CHF, etc)
 - Limited use of blood, urine chemistries

USMLE Acute Renal Failure

- Determine cause based on blood, urine testing
 - BUN (rises in ARF)
 - Cr (rises in ARF)
 - BUN/Cr ratio (normal ~20:1)
 - U_{Na}
 - Fe_{Na}
 - Uosm

Renal Failure Measurements

- Urinary sodium (U_{Na})
 - Varies based on intake of sodium and water
 - Very low when kidney retaining salt/water
 - <20 mEq/L is very low
- Fractional excretion of Na (Fe_{Na})
 - Amount of filtered Na that is excreted
 - Very low when kidney retaining salt/water
 - $<1\%$ is low
- Urinary osmolality (Uosm)
 - Measure of concentrating ability of kidney
 - Very high when kidney retaining water
 - >550 mOsm/kg is high

Pre-Renal Failure

BUN/Cr

- Decreased blood flow to kidneys
- Less BUN/Cr filtered
- Rising BUN/Cr in blood
- More resorption H₂O
- BUN resorbed with H₂O
- BUN rises $\gg$ Cr rises
- Result
 - $\uparrow\uparrow$ BUN
 - $\uparrow$ Cr
 - $\uparrow$ BUN/Cr ratio

Pre-Renal Failure

Urinary Findings

- Lots of H₂O resorbed
- Concentrated urine
- $\uparrow$ Uosm
- Lots of Na resorbed
- $\downarrow U_{Na}$
- $\downarrow Fe_{Na}$

Pre Renal Failure

	Normal	Pre-Renal Failure
BUN (mg/dl)	20	60
Cr (mg/dl)	1.0	2.0
BUN:Cr	20:1	> 20:1
UNa (mEq/L)	variable	<20
FeNa (%)	variable	<1
Uosm (mOsm/kg)	variable	>550

Intrinsic Renal Failure

BUN/Cr

- Kidney cannot filter blood
- Less BUN/Cr filtered
- Rising BUN/Cr in blood
- No extra rise in BUN from ↑resorption
- Normal ratio (20:1)

Intrinsic Renal Failure

Urinary Findings

- Urine: kidney cannot resorb water
- Uosm low (can't concentrate urine)
- U_{Na} high (can't resorb Na)
- Fe_{Na} high (can't resorb Na)

Intrinsic Renal Failure

	Normal	Intrinsic Failure
BUN (mg/dl)	20	40
Cr (mg/dl)	1.0	2.0
BUN:Cr	20:1	20:1
UNa (mEq/L)	variable	>20
FeNa (%)	variable	>2
Uosm (mOsm/kg)	variable	<350

Post Renal Failure

- Obstruction to outflow
- Urine backs up
- High pressure in tubules
- Kidney cannot filter blood
- Kidney's resorptive mechanisms damaged/destroyed

Post Renal Failure

- Diagnosis rarely made by plasma/urinalysis
- Key features:
 - Anuria
 - Hydronephrosis
- Renal ultrasound is test of choice
- Shows enlarged, dilated kidneys

Post Renal Failure

- Plasma/urine findings similar to intrinsic renal
- High pressure in tubules prevents filtration
- Only exception is BUN/Cr ratio
- BUN may rise like pre-renal
- High pressure in tubules → forces BUN out
- BUN rises more than Cr
- ↑BUN/Cr ratio similar to pre-renal

Post Renal Failure

- Lots of variation in lab values based on tubules
- Early post renal → tubular function okay
- Late → high pressure disrupts tubular resorption
- Urine chemistries variable

Post Renal Failure

	Normal	Post Renal Failure
BUN (mg/dl)	20	60
Cr (mg/dl)	1.0	2.0
BUN:Cr	20:1	> 20:1
UNa (mEq/L)	variable	>20
FeNa (%)	variable	>2
Uosm (mOsm/kg)	variable	<350

Pre, Intrinsic, Post Problems

- Diseases often cross boundaries
 - Pre-renal → ATN
- Diuretics obscure urine findings
- Pre-existing chronic renal disease

Fractional Excretion Na

- Pre-renal
 - $Fe_{Na} < 1\%$
 - $U_{Na} < 20$
- Intrinsic renal
 - $Fe_{Na} > 1\%$
 - $U_{Na} > 40$

$$Fe_{Na} = \frac{P_{Cr} * U_{Na}}{P_{Na} * U_{Cr}}$$

Chronic Kidney Disease

- Slow, steady fall in creatinine clearance
 - Blood tests show ↑BUN/Cr
- Eventually progresses to dialysis for many patients
- Most common causes diabetes and hypertension
 - Hypertensive nephrosclerosis
 - Diabetes nephropathy

Stages of Kidney Disease

- Stage 1 → GFR >90
- Stage 2 → GFR 60-89
- Stage 3 → GFR 30-59
- Stage 4 → GFR 15-29 (approaching dialysis)
- Stage 5 → GFR <15 (usually on dialysis)

Indications for Dialysis

- Acidemia
- Electrolytes (hyperkalemia)
- Intoxication (overdose dialyzable substance)
- Overload of fluid (CHF)
- Uremic symptoms

Dialyzable Substances

- Salicylates (aspirin)
- Lithium
- Isopropyl alcohol
- Magnesium laxatives
- Ethylene glycol

Dialysis Methods



- Hemodialysis
 - Requires vascular access
 - Blood pumped from body → filter → back to body
 - Done in "sessions" of few hours at a time
- Peritoneal dialysis
 - Fluid cycled through peritoneal cavity
 - Peritoneum used as dialysis membrane
- Hemofiltration
 - Constant filtering of blood
 - Usually done at bedside for critically ill patients

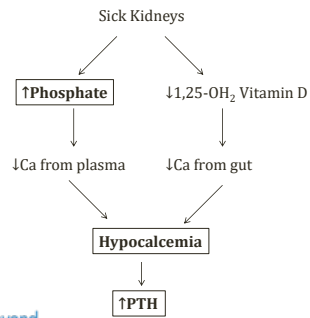
Vascular Access

- For acute dialysis, central line can be placed
- Ideal method is fistula
 - Connection between artery and vein
 - Placed surgically, usually in arm
 - Lowest rates thrombosis, infection
- Fistula must "mature" for use
- Ideally placed several months before dialysis

Complications CKD

- Anemia (loss of EPO)
- Dyslipidemia
 - Mostly triglycerides
 - Protein loss in urine → stimulation of liver synthesis
 - Impaired clearance of chylomicrons and VLDL
- Growth failure in children
- Renal osteodystrophy

Calcium-Phosphate in Renal Failure



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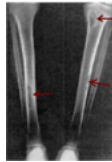
Calcium-Phosphate in Renal Failure

- Secondary hyperparathyroidism
 - Parathyroid stimulation in renal failure
- Tertiary hyperparathyroidism
 - Parathyroid becomes autonomous from constant stimulation
 - VERY high PTH levels
 - Calcium becomes elevated
 - Often requires parathyroidectomy

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Calcium-Phosphate in Renal Failure

- Untreated hyperparathyroidism leads to renal osteodystrophy
 - Bone pain (predominant symptom)
 - Fracture (weak bones 2° chronic high PTH levels)
- Osteitis fibrosa cystica
 - Untreated, severe high PTH levels
 - Bone cysts
 - Brown tumors (osteoclasts w/fibrous tissue)



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Phosphate Binders

- Bind phosphate in GI tract
- Calcium carbonate
- Calcium acetate (Phoslo)
- Sevelamer (Renagel)
- Lanthanum

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Drugs and Renal Function

- Many drugs worsen renal function
- Decrease GFR
- Associated with ↑BUN/Cr
- Loop, Thiazide, and K sparing diuretics
- ACE inhibitors
- NSAIDs

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Urinary Tract Infections

Urinary Infections

Jason Ryan, MD, MPH



Urinary Infections

- Cystitis
 - Infection of bladder
 - "Lower" urinary tract
- Pyelonephritis
 - Infection of kidneys
 - "Upper" urinary tract



Urinary Infections

- Most infections "ascend"
- Urethra → Cystitis → Pyelonephritis



Etiology

- Escherichia coli (75-95%)
- Proteus mirabilis
 - Urease producing bacteria
 - Struvite kidney stones
- Klebsiella pneumoniae
- Staphylococcus saprophyticus
- Enterococcus faecalis



Symptoms

- Cystitis
 - Dysuria (pain with urination)
 - Frequency (going a lot)
 - Urgency (always feel like you have to go)
 - Suprapubic pain
 - No systemic symptoms
 - Usually normal plasma WBC count



Symptoms

- Pyelonephritis
 - Systemic symptoms (fever, chills)
 - Flank pain
 - CVA tenderness
 - Hematuria
 - WBC casts

WBC Cast

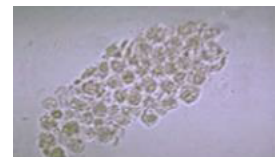


Image courtesy of Anwar Siddiqui .



Diagnosis

- Urinalysis
 - Cloudy urine
 - Leukocyte esterase
 - Produced by WBCs in urine
 - Nitrites
 - 90% UTI bugs convert nitrates to nitrites
 - Some that don't: *enterococcus*, *staph saprophyticus*
 - Best for detecting aerobic gram-negative rods (*E. Coli*)
 - >10WBC/hpf
- Culture
 - >100,000 CFUs



Risk Factors

- Women
 - 10x more likely than men to get UTIs
 - Shorter urethra, closer to fecal flora
- Sexual activity
- Urinary catheterization
- Diabetes
- Pregnancy

Risk Factors

- Infants with vesicoureteral reflux
 - Ureters insert abnormally into bladder
 - Chronic reflux of urine back into ureters
- Urinary obstruction
 - Anatomic abnormalities in children
 - Bladder tumors in adults
 - Enlarged prostate in older males

Treatment

- Fluoroquinolones
 - Ciprofloxacin, levofloxacin, ofloxacin
 - Usually 3 day course
- Nitrofurantoin (Macrobid)
 - Used in pregnancy
- Trimethoprim-sulfamethoxazole (TMP-SMX)

Sterile Pyuria

- Some women with chlamydia/gonorrhea complain of urinary tract symptoms
- Urinalysis shows pyuria but no bacterial growth
- Majority women are asymptomatic with chlamydia or gonorrhea

Chronic Pyelonephritis

- Consequence of recurrent pyelonephritis
- Vesicoureteral reflux in children
- Recurrent stones in adults
- Scarring of kidneys
- Corticomedullary scarring
- Blunted calyx
- "Thyroidization of kidney"
 - Tubules contain eosinophilic material
 - Looks like thyroid tissue on microscopy

Cystic Kidney Disease

Cystic Kidney Disease

Jason Ryan, MD, MPH



Cystic Kidney Diseases

1. Multicystic Dysplastic Kidney
2. Autosomal Recessive Polycystic Kidney Disease
3. Autosomal Dominant Polycystic Kidney Disease
4. Medullary Cystic Kidney Disease



Multicystic Dysplastic Kidney

- Abnormal ureteric bud-mesenchyme interaction
- Kidney replaced with cysts
- No/little functioning renal tissue
- Absent ureter
- Often detected in utero by ultrasound



Multicystic Dysplastic Kidney

- If unilateral → remaining kidney hypertrophies
- If bilateral → Potter's syndrome
 - Oligohydramnios
 - Failure of lung maturation
 - Compressed face/limbs
 - Not compatible with life



Multicystic Dysplastic Kidney

- Spontaneous
 - Non-inherited
 - Different from other cystic disorders
 - Subsequent pregnancies often okay



Polycystic Kidney Disease

- Autosomal recessive (infants)
- Autosomal dominant (young adults)



ARPKD

- Old name: "juvenile" PKD
- Occurs in infants
- Can occur with Potter's syndrome
- Renal failure
- High blood pressure
- Key associations:
 - Liver disease (fibrosis/cysts)
 - Can cause portal hypertension (ascites)

ADPKD

- Occurs in adults
- Microscopic cysts present at birth
 - Too small to visualize with ultrasound
 - Kidneys appear normal at birth
- Cysts develop over many years
- Inherited mutation of PKD1 or PKD2 genes

ADPKD

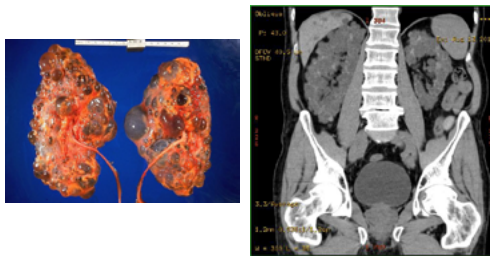


Image courtesy of Hg6996

ADPKD

- Key associations
 - Berry aneurysm (subarachnoid hemorrhage)
 - Liver cysts
 - Mitral valve prolapse

ADPKD

- Classic presentation
 - Young adult
 - High blood pressure (↑RAAS system)
 - Hematuria
 - Renal failure
 - Family history of sudden death (aneurysm)

Medullary Cystic Kidney Disease

- Autosomal dominant
- Cysts in collecting ducts of medulla
 - Name is misnomer
 - Most patients DO NOT have cysts
- Kidney fibrosis occurs → small, shrunken kidneys
 - Contrast with ADPKD (enlarged kidneys)
- Often have early onset (adolescent) gout
- Renal failure

Cystic Kidney Diseases

Disease	Kidney Involvement	Key Features
Multicystic Dysplastic	Unilateral	Detected in utero; cysts with no renal tissue; not hereditary
ARPKD	Bilateral; large	Children; large kidneys with cysts; renal failure; HTN; Liver (cysts, ascites)
ADPKD	Bilateral; large	Adults; large kidneys with cysts; hematuria; renal failure; berry aneurysms
Medullary Cystic	Bilateral; small	Cysts in collecting ducts; small shrunk kidneys; early gout; renal failure

Diuretics

Diuretics

Jason Ryan, MD, MPH

Boards&Beyond

Diuretics

Drugs that increase urine output

1. Carbonic Anhydrase Inhibitors
2. Osmotic Diuretics
3. Loop Diuretics
4. Thiazide Diuretics
5. K⁺ Sparing Diuretics

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Sodium

11
Na

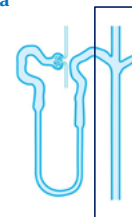
- Normal plasma [Na] = 140 meq/L
- [Na] tightly regulated
 - Renin-angiotensin-aldosterone
 - Antidiuretic hormone (ADH)
- Sodium intake → H₂O retention → [Na] = 140 meq/L
- **Sodium loss** → H₂O excretion → [Na] = 140 meq/L
- Any drug that ↑ Na excretion → volume loss
- **Many diuretics work by ↑ Na excretion**

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Potassium

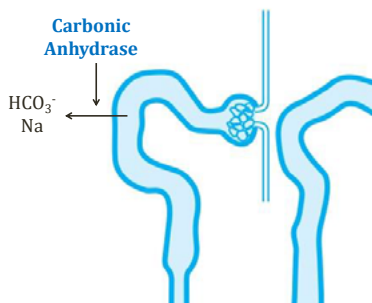
- Secreted by **distal tubule and collecting duct**
- Varies with Na/H₂O delivery to distal nephron
- More urine flow → more secretion of potassium
- Most diuretics lead to **hypokalemia**

19
K



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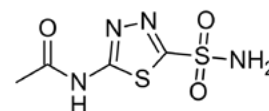
Carbonic Anhydrase Inhibitors



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Carbonic Anhydrase Inhibitors

- **Acetazolamide**
- Weak diuretic effect
 - Block some Na resorption
- Causes a non-AG **metabolic acidosis**
 - Increased elimination of HCO₃⁻



Acetazolamide

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Carbonic Anhydrase Inhibitors

Clinical Uses

- Severe metabolic alkalosis
- Glaucoma
 - Blocks formation of aqueous humor

Carbonic Anhydrase Inhibitors

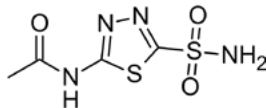
Clinical Uses

- Pseudotumor cerebri
 - Reduced rate of CSF formation
- Prevention of high altitude sickness
 - Low pO₂ at high altitude → hyperventilation
 - Low CO₂ → respiratory alkalosis
 - Acetazolamide → acidosis → reverses alkalosis

Carbonic Anhydrase Inhibitors

Side Effects

- Metabolic acidosis
- Paresthesias ("tingling" in extremities)
- Sulfa allergy

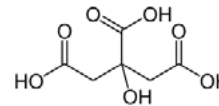


Acetazolamide

Carbonic Anhydrase Inhibitors

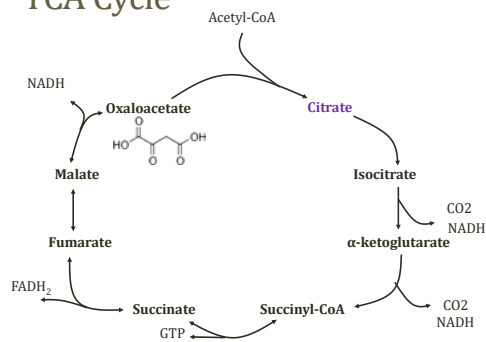
Side Effects

- Cause kidney stones
 - Reduce urinary **citrate** excretion
 - Citrate inhibits calcium stone formation



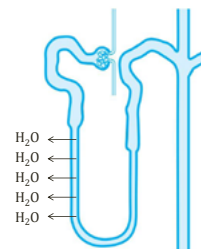
Citrate
(Citric Acid)

TCA Cycle

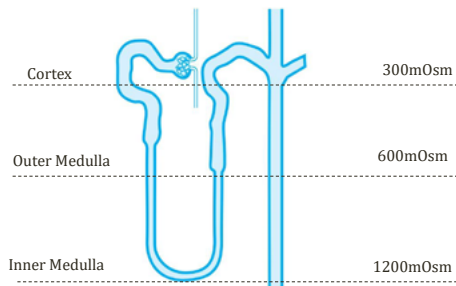


Osmotic Diuretics

- **Thin descending limb**
- Concentrates urine
- Absorbs water
- Impermeable to NaCl
- Water leaves urine
- Drawn out by hypertonicity in medulla



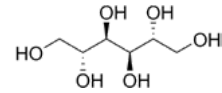
Osmotic Diuretics



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Mannitol

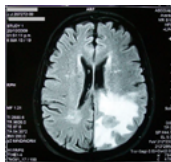
- Sugar alcohol
- Freely filtered by glomerulus
- No tubular reabsorption
- Raises osmolality
- Reduces water reabsorption
- Increases urine output



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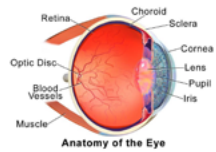
Mannitol

- Main use is in cerebral edema, glaucoma
- Goal is to create a **HYPER**osmolar state
- "Osmotherapy"
- Draws fluid out (brain, eye)



Bobigalindo/Wikipedia

Boards&Beyond



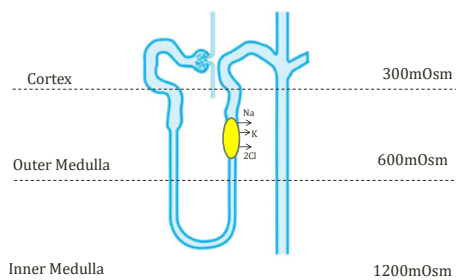
BruceBlais

Mannitol

- Cannot use in **heart failure** patients
 - Draws fluid out of tissues
 - Expands intravascular volume
 - Can cause pulmonary edema
- Can't use with severe renal disease
 - High doses cause acute anuric renal failure
 - Mannitol can cause renal vasoconstriction → anuria

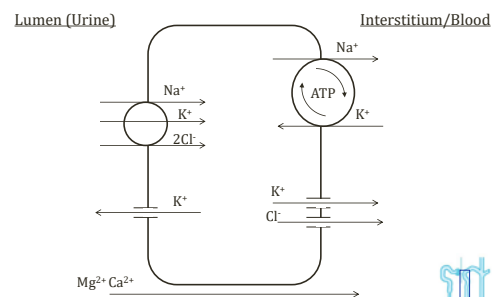
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Loop Diuretics



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Loop Diuretics

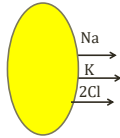


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Loop Diuretics

Furosemide, bumetanide, torsemide, ethacrynic acid

- Inhibit Na-K-2Cl pump
- **Strong diuretic effect**
- Two mechanisms that promote diuresis
 - ↑ Na excretion
 - ↓ **medullary osmotic gradients**
- Used for edematous states
 - Heart failure, cirrhosis

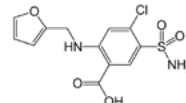


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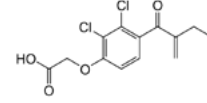
Loop Diuretics

Furosemide, bumetanide, torsemide, ethacrynic acid

- Hypokalemia
- Hypocalcemia
- Hypomagnesemia
- Most are sulfa drugs
- Exception: Ethacrynic acid (used in allergic patients)



Furosemide



Ethacrynic Acid

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Loop Diuretics

Furosemide, bumetanide, torsemide, ethacrynic acid

- **Ototoxicity**
 - Very high doses or given with other ototoxic agents
 - Tinnitus, loss of hearing (usually reversible)
- Acute interstitial nephritis
 - ↑ BUN/Cr
 - White blood cell casts
 - Urine eosinophils
- Gout



Pixabay/Public Domain

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Uric Acid

- Complex mechanism of renal handling
- Thiazides, loop diuretics ↑ uric acid reabsorption
- **Gout promoted by diuretics**



James Heilman, MD/Wikipedia

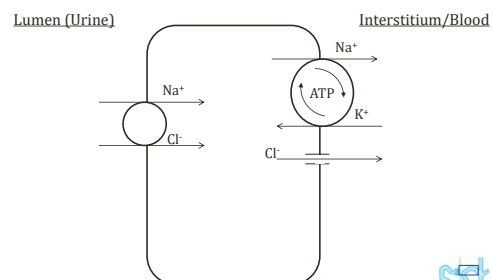
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Metabolic Alkalosis

- pH > 7.45
- ↑ HCO_3^-
- Diuretics → ↑ urine output → ↓ ECV
- Renin-Angiotensin-Aldosterone activation
- ↑ H^+ secretion → metabolic alkalosis
- "Contraction alkalosis"
- Seen with loop diuretics and thiazides

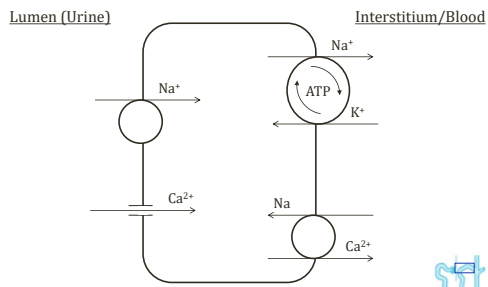
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Thiazide Diuretics



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Thiazides: Hypercalcemia

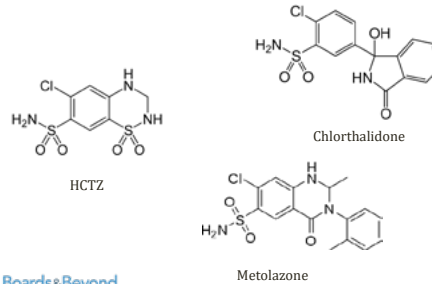


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Thiazide Diuretics

Hydrochlorothiazide; chlorthalidone; metolazone

- Sulfa Drugs (allergy)



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Thiazide Diuretics

Hydrochlorothiazide; chlorthalidone; metolazone

- Elevates blood levels
 - Glucose
 - Lipids
 - Uric acid
 - Calcium
 - **HyperGLUC**
- Caution: diabetes, $\uparrow$ lipids, gout, hypercalcemia

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Thiazide Diuretics

Hydrochlorothiazide; chlorthalidone; metolazone

- **Hyponatremia**
 - Drugs promote Na loss
 - H_2O resorption intact (normal medullary gradients)
 - High H_2O intake $\rightarrow$ hyponatremia
- Hypokalemia
- Metabolic alkalosis

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Thiazide Diuretics

Hydrochlorothiazide; chlorthalidone; metolazone

- Clinical uses
 - Hypertension
 - Recurrent calcium kidney stones
 - Osteoporosis
 - Diabetes insipidus

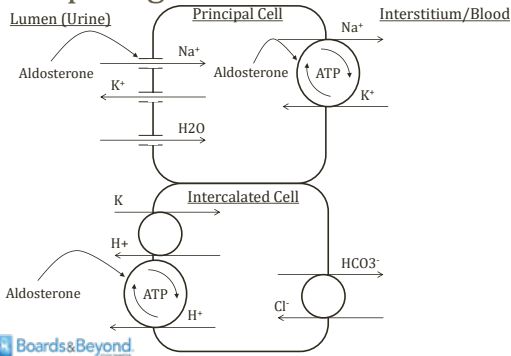
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K-Sparing Diuretics

- Spironolactone/eplerenone
 - Block aldosterone receptor site
- Triamterene/amiloride
 - Block aldosterone Na channel
- Good choice for patients with low K
 - Often from other diuretics

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K Sparing Diuretics



K Sparing Diuretics

Spironolactone, Eplerenone, Triamterene, Amiloride

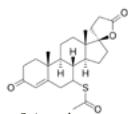
- All $\uparrow$ Na/ H_2O excretion (diuretics)
- All "spare" potassium
 - Unlike other diuretics, do not increase K^+ excretion
- HYPERkalemia is side effect

Spironolactone

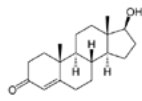
- Similar structure to testosterone
 - Blocks testosterone effects
 - **Gynecomastia** in men
 - Eplerenone: No gynecomastia
- Derivative of progesterone
 - Activates progesterone receptors
 - **Amenorrhea** in women



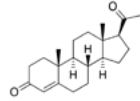
Image courtesy Dr. Mordcai Blau/Wikipedia



Spironolactone



Testosterone



Progesterone

Renal Failure

- All diuretics can cause renal failure
- $\downarrow$ ECV $\rightarrow$ $\downarrow$ GFR
- BUN/Cr may rise in the plasma

RAA System

Renin-Angiotensin-Aldosterone

- Diuretics result in volume loss
- Activates renin-angiotensin-aldosterone system
- $\uparrow$ RAAS $\rightarrow$ $\uparrow$ Na/ H_2O reabsorption
- Some adaptation to diuretic effect over time

Rules of Thumb

- All diuretics except K sparing: $\uparrow$ K excretion
- CA inhibitors and K sparing cause acidosis ($\downarrow$ pH)
 - CA Inhibitors: HCO_3^- excretion
 - K sparing: $\downarrow$ aldosterone; hyperkalemia (H^+/K^+ exchanger)
 - Others cause contraction alkalosis
- Loops and Thiazides have opposite effects on Ca
 - Loops $\rightarrow$ hypocalcemia
 - Thiazides $\rightarrow$ hypercalcemia

Kidney Stones

Kidney Stones

Jason Ryan, MD, MPH

Boards&Beyond

Kidney Stones

Nephrolithiasis

1. Calcium
2. Struvite
3. Urate
4. Cystine

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Symptoms

- Flank pain (side between the ribs and the hip)
- Colicky (waxes and wanes in severity)
- Hematuria

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Risk Factors

- High amount of stone substance in blood
 - Hypercalcemia
 - Hyperuricemia
- Low urine volume
 - Usually from dehydration
 - Increases concentration of urine substances
- In general, hydration lowers risk of stones

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Calcium Stones

- Calcium oxalate (most common)
- Calcium phosphate
- Most common type of kidney stone (80%)
- Key risk factors
 - Hypercalcemia
 - High oxalate levels in blood
- Radiopaque
 - Seen on x-ray and CT scan



Image courtesy of James Heilman, MD

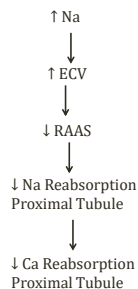
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Risk Factors

- Most common etiology: idiopathic hypercalciuria
- Hypercalcemia (hyperparathyroidism)
- High oxalate levels
 - Crohn's disease: Fat malabsorption → Fat binds to calcium, leaving oxalate free to be absorbed in the gut
 - Gastric bypass patients
- Ethylene glycol (antifreeze)
 - Formation of oxalate
 - Increases oxalate concentration in urine
- Vitamin C abuse
 - Oxalate generated from metabolism of vitamin C

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Dietary Sodium



More Na = More Ca Urine
High Na diet = Stone formation
Low Na diet = Treatment stones

Calcium Stones

- Classic case
 - Patient drinking less water
 - Flank pain, hematuria
 - Calcium stone on imaging
 - Normal Ca level in plasma
 - Increased calcium level in urine

Treatment

- Most stones pass on their own
- Large stones that do not pass require surgery
- Recurrent stone formers may take medication
- Thiazides
 - Decrease Ca in urine
- Citrate (Potassium citrate)
 - Binds with calcium but remains dissolved
 - Lowers urinary Ca available for stones
 - Inhibits of stone formation

Struvite Stones

- Ammonium-Magnesium-Phosphate stones
- 2nd most common stone type (15%)
- Consequence of urinary tract infection
- Urease-positive bacteria
 - Proteus, Staphylococcus, Klebsiella
 - All hydrolyze urea to ammonia
 - Urine becomes alkaline

Struvite Stones

- Can form "staghorn calculi"
 - Stones form a cast of the renal pelvis and calices
 - Looks like horns of a stag
- Won't pass → surgery required
- Untreated → bacterial reservoir
 - Recurrent infection
- Radiopaque
 - Seen on x-ray and CT scan

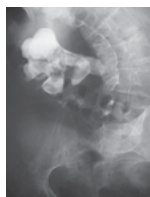


Image courtesy of Nevit Dilmen

Struvite Stones

- Classic presentation
 - UTI symptoms (dysuria, frequency)
 - Mild flank pain
 - Hematuria
 - Large, branching staghorn stone on imaging
- Treatment:
 - Surgery
 - Antibiotics

Uric Acid Stones

- Cause by high uric acid in urine or acidic urine
- $H^+ + Urate^- \leftrightarrow Uric\ acid$
- Radiolucent stones
 - Not visible on x-ray
 - Can see with CT scan
- Lowest pH is in the distal tubule/collecting duct

Risk Factors

- High uric acid levels
 - Gout
 - Leukemia, myeloproliferative disease
- Acidic urine (precipitates uric acid)
 - Chronic diarrhea
- More common in hot, arid climates
 - Low urine volume, acidic urine more common
 - 5-10% stones in US/Europe
 - 40% stones in other climates

Treatment

- Hydration
- Alkalinization of urine
 - Potassium bicarbonate
- Rarely allopurinol
 - Xanthine oxidase inhibitor
 - Reduces uric acid production
- Medically therapy often effective
- Usually does not require surgery

Uric Acid Stones

- Classic case
 - Flank pain, hematuria
 - No stone on x-ray
- Choose medical therapy, not surgery

Cystine Stone

- Rare type of stone
- Seen in children with cystinuria
- Tubular defect $\rightarrow$ cannot absorb cystine
- Also form staghorn calculi

Cystine Stone

- Classic case
 - Child
 - No history of UTI (contrast with Struvite)
 - Large, staghorn stone
- Treatment:
 - Hydration
 - Alkalinization of urine

Renal and Bladder Malignancy

Renal and Bladder Malignancies

Jason Ryan, MD, MPH



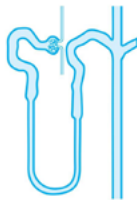
Renal and Bladder Malignancies

1. Renal Cell Carcinoma
2. Wilms' Tumor
3. Renal Angiomyolipoma
4. Transitional Cell Carcinoma
5. Squamous Cell Carcinoma
6. Adenocarcinoma



Renal Cell Carcinoma

- Most common kidney tumor
- Epithelial tumor
- Commonly arise from proximal tubule cells



Renal Cell Carcinoma

Risk Factors

- Males
- Age 50-70
- Cigarette smoking
- Obesity



Renal Cell Carcinoma

Symptoms

- Classic triad
 - Hematuria
 - Palpable abdominal mass
 - Flank pain
- Many patients have fever, weight loss
- Many patients asymptomatic until disease advanced
- At presentation ~25% have metastases/advanced disease



Renal Cell Carcinoma

Symptoms

- Invades renal vein
 - May cause thrombosis
 - Can block renal vein/IVC
 - Can block drainage of testicles
 - May cause a varicocele
- Spreads through venous system
- Common sites for metastasis:
 - Lung
 - Bone
- Can also spread to retroperitoneal lymph nodes



Renal Cell Carcinoma

Paraneoplastic syndromes

- Many paraneoplastic syndromes
- Polycythemia (↑Hct)
 - Increased EPO production by tumor
- Hypercalcemia
 - Tumor production of PTHrP
 - Increased Ca from bones

Renal Cell Carcinoma

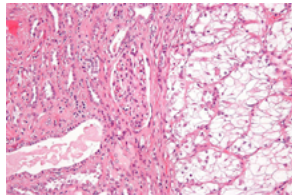
Paraneoplastic syndromes

- Hypertension
 - Renin production by tumor
- Cushing's Syndrome
 - ACTH production by tumor
 - Look for weight gain, hypertension, hyperglycemia

Renal Cell Carcinoma

Pathology

- Most common type is clear cell carcinoma
- Cells filled with glycogen and lipids



Renal Cell Carcinoma

Genetics

- Associated with gene deletion chromosome 3
- Von-Hippel-Lindau (VHL) gene
- Sporadic mutation
 - Single tumor
 - Older patient, usually smoker
- Inherited
 - Younger patient
 - Multiple, bilateral tumors

Von-Hippel-Lindau Disease

- Autosomal dominant
- Von-Hippel-Lindau (VHL) gene inactivation
- Many tumors
 - Renal cell carcinomas
 - Cerebellar hemangioblastoma
 - Retinal hemangioblastoma

Renal Cell Carcinoma

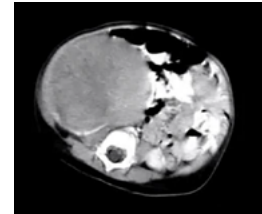
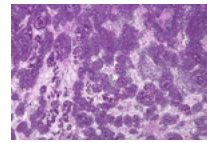
Treatment

- Surgical resection in early disease
- Poorly responsive to chemotherapy/radiation
- Recombinant cytokines used
 - Aldesleukin (interleukin-2)
 - Hypotension, fevers, chills are important side effects

Wilms' Tumor

- Most common renal malignancy of young children
- Proliferation of metanephric blastema
 - Embryonic glomerular structures
- Classic case
 - Young child (~3years old)
 - Huge, palpable flank mass
 - Hematuria
 - Hypertension (renin secretion)

Wilms' Tumor



Wilms' Tumor

- Associated with loss of function mutation
- WT1 tumor suppressor gene
- Chromosome 11
- May be sporadic
- Often part of a syndrome

WAGR Syndrome

- Wilms' tumor
- Aniridia
 - Absence of the iris
 - Visual problems
- Genital anomalies
 - Cryptorchidism, ambiguous genitalia
- Mental Retardation
- Deletion of WT1 gene chromosome 11



Beckwith Wiedemann Syndrome

- Pediatric overgrowth disorder
- Macrosomia
 - Height/weight often >97th percentile
- Hemihyperplasia
 - Muscles in one limb bigger than other
- Macroglossia
- Many embryonal tumors
 - Wilms' tumor
 - Neuroblastoma
 - Rhabdomyosarcoma

Renal Angiomyolipoma

- Benign tumor – young children
- Tumors of blood vessels, smooth muscle, fat
- Associated with Tuberous Sclerosis
 - Autosomal dominant condition
 - Cortical tubers in brain
 - Subependymal hamartomas in brain
 - Seizures, mental retardation
 - Cardiac rhabdomyomas
 - Leaf-like patches of skin with no pigment (ash-leaf patches)

Transitional Cell Carcinoma

- Most common tumor of urinary tract system
- Most common type of bladder cancer
- Locations:
 - Bladder (most common)
 - Also renal calyces, renal pelvis, ureters
- Often multifocal and recurrent
 - "Field defect"
 - Damage to entire urothelium

Transitional Cell Carcinoma

Risk Factors

- Smoking
- Cyclophosphamide
- Phenacetin
- Aniline dyes (hair coloring)
- Workplace exposures
 - Rubber, textiles, leather
 - Naphthalene (industrial solvent)
 - Painters, machinists, printers

Transitional Cell Carcinoma

- Classic case
 - Older, white male
 - Smoker
 - Painless hematuria
 - No casts in urine
- Test of choice: cystoscopy and biopsy

Transitional Cell Carcinoma

Treatment

- Surgical resection
- Radiation
- Chemotherapy
 - Combination chemotherapy with platinum-based regimens
 - Cisplatin, carboplatin

Squamous Cell Carcinoma

- Rare bladder cancer
- Need chronic inflammation of bladder
- Several key risk factors
 - Recurrent kidney stones or cystitis
 - UTI with *Schistosoma haematobium*

Schistosoma haematobium

- Trematode
- Found in Africa and Middle East (Egypt)
- Acquired from freshwater containing larvae
- Penetrate the skin
- Migrate to liver and mature to adults
- Infects bladder
- Usually causes hematuria
- Can result in bladder cancer

Adenocarcinoma

- Very rare bladder cancer
- Glandular proliferation in bladder
- Occurs in special circumstances
 - Urachal remnant
 - Long history of cystitis
 - Exstrophy: bladder protrusion through abdominal wall defect

Rhabdomyolysis

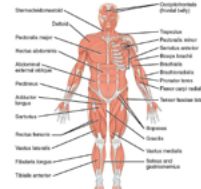
Rhabdomyolysis

Jason Ryan, MD, MPH



Rhabdomyolysis

- Syndrome caused by muscle necrosis
- Can lead to **renal failure** and death



OpenStax College



Rhabdomyolysis

Causes of Muscle Damage

- Intense **physical exercise**
 - Especially if dehydrated
- **Crush injuries** (trauma)
- **Drugs**
 - Statins
 - Fibrates



Muscle Contents

- **Creatine kinase**
 - Elevated levels are hallmark of rhabdomyolysis
- Aldolase, lactate dehydrogenase, AST/ALT



Muscle Contents

- **Potassium** and **phosphate**
 - Hyperkalemia/hyperphosphatemia in rhabdomyolysis

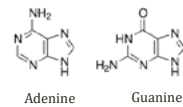
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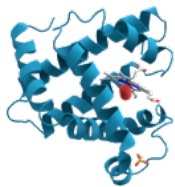
Muscle Contents

- Purines
 - Metabolized to uric acid in liver
 - Can lead to **hyperuricemia**

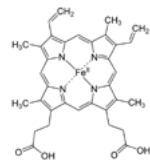


Myoglobin

- Protein monomer (NOT tetramer like Hgb)
- Contains **heme** (porphyrin plus iron)
- Binds oxygen for use by muscle tissue

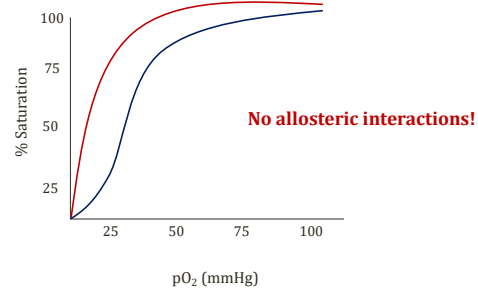


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Myoglobin



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Myoglobin

Renal Toxicity

- Obstructs tubules
- Toxic to **proximal tubular** cells
- Vasoconstriction
 - Especially in medulla
 - Leads to renal hypoxia

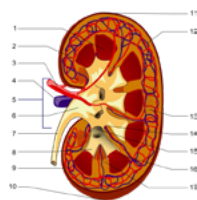


Image courtesy of Piotr Michal Jaworski

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Myoglobin

Renal Toxicity

- Made worse by **volume depletion** in rhabdomyolysis
 - Intravascular fluid influx into muscle tissue
- Feared outcome rhabdomyolysis: **renal failure/death**



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Rhabdomyolysis

Symptoms

- Muscle pain
- Weakness
- Dark urine (from myoglobin)



James Hellman, MD/Wikipedia

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Rhabdomyolysis

Diagnosis

- **Creatine kinase**
 - Usually very high
 - Normal < 250 IU/L
 - Rhabdomyolysis > 1000 IU/L
 - Sometimes up to 25,000 or more IU/L

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Rhabdomyolysis

Diagnosis

- Urinalysis for **heme**
 - Heme has peroxidase activity
 - Breaks down peroxide
 - Changes test strip color
- Positive dipstick = hemoglobin or myoglobin



Image courtesy of J3D3

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Rhabdomyolysis

Diagnosis

- Microscopy for red blood cells
- Classic finding rhabdomyolysis
 - Dark urine
 - Positive dipstick for heme
 - No evidence of red blood cells

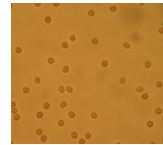


Image courtesy of Bobgalindo

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Rhabdomyolysis

Treatment

- Volume resuscitation
 - IV Fluids (usually isotonic saline)
 - Titrated to maintain good urine output
- Treatment of electrolyte abnormalities
- Dialysis

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Hypocalcemia

- Calcium deposits in damaged myocytes
- Initial phases rhabdomyolysis: **hypocalcemia**
- Recovery phase: release from myocytes
 - Levels return to normal
 - Can become elevated

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